
Towards Reliable Multimodal Financial Reasoning with Large Language Models



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Abstract

Financial reasoning with large language models requires more than fluent answer generation. In finance, reliable reasoning depends on identifying relevant evidence, integrating heterogeneous information, applying domain knowledge, performing valid numerical or logical operations, and producing conclusions that can be inspected and verified. This thesis advances reliable multimodal financial reasoning with large language models through three connected studies: capability evaluation, adaptive evidence acquisition, and cross-model error reflection.

First, the thesis reviews financial reasoning research and organizes the literature into five task families: financial numerical, tabular, document, multimodal, and conversational reasoning. It identifies three gaps in current research: limited evaluation of knowledge-intensive multimodal reasoning, insufficient adaptive evidence acquisition, and weak use of step-level error reflection. Second, the thesis introduces FinMR, a benchmark of 3,200 multimodal financial reasoning questions across 15 topics. FinMR evaluates mathematical reasoning, financial expertise reasoning, single-image reasoning, and multi-image reasoning, and reveals persistent limitations in current MLLMs. Third, the thesis introduces SEA, a step-wise multi-agent framework in which specialized agents dynamically acquire visual, textual, and financial knowledge evidence through query-based communication. Fourth, the thesis introduces FinErrorSet and CLER, a cross-MLLM error reflection framework that retrieves step-level errors from other models to guide reasoning revision.

Together, these studies show that reliable multimodal financial reasoning is an end-to-end property of evaluation, evidence acquisition, reasoning validity, reflection, and verification. The thesis contributes benchmarks, methods, error resources, and an integrated framework for building more reliable financial reasoning systems.

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Shuangyan Deng
Auckland, 2026

Declaration

I declare that this thesis is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person except where due acknowledgement is made. The thesis includes material arising from collaborative research manuscripts. My contributions to these works include conceptualization, methodology development, experiment design, analysis, and manuscript writing, as described in the publications statement.

Shuangyan Deng
2026

Publications Arising from This Thesis

The following publications and manuscript form the main technical basis of this thesis:

1. Shuangyan Deng et al. *FinMR: A Knowledge-Intensive Multimodal Benchmark for Advanced Financial Reasoning*. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*, pp. 168–176, 2025. doi: 10.1145/3768292.3770365.
2. Shuangyan Deng et al. *Multi-Agent SEA: Step-wise Evidence Acquisition for Multimodal Financial Reasoning*. *Information Fusion*, 136, Article 104550, 2026. doi: 10.1016/j.inffus.2026.104550.
3. Shuangyan Deng et al. *CLER: Improving Multimodal Financial Reasoning by Cross-MLLM Error Reflection*. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(36), pp. 30494–30502, 2026. doi: 10.1609/aaai.v40i36.40303.
4. Shuangyan Deng et al. *Financial Reasoning with Large Language Models: A Survey of Benchmarks, Methods, and Challenges*. Under review; this survey manuscript forms the literature foundation of this thesis.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
LLM	Large Language Model
MLLM	Multimodal Large Language Model
NLP	Natural Language Processing
QA	Question Answering
VQA	Visual Question Answering
FNR	Financial Numerical Reasoning
FTR	Financial Tabular Reasoning
FDR	Financial Document Reasoning
FMR	Financial Multimodal Reasoning
FCR	Financial Conversational Reasoning
CoT	Chain-of-Thought
PoT	Program-of-Thoughts
RAG	Retrieval-Augmented Generation
PRM	Process Reward Model
SEA	Step-wise Evidence Acquisition
CLER	Contrastive Learning for Error Retrieval
EM	Exact Match
API	Application Programming Interface
GPU	Graphics Processing Unit
SEC	U.S. Securities and Exchange Commission
EDGAR	Electronic Data Gathering, Analysis, and Retrieval
XBRL	eXtensible Business Reporting Language
CFA	Chartered Financial Analyst

FRM	Financial Risk Manager
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Chapter 1

Introduction

1.1 Background

Artificial Intelligence (AI) has moved through several phases of capability. Earlier systems were often designed around perception, classification, retrieval, information extraction, speech recognition, or prediction: they detected objects, labelled documents, extracted entities, recognized speech, or estimated future values from structured signals [129, 113, 16, 80, 62, 83]. The emergence of LLMs and MLLMs has changed these expectations [116, 55, 7]. Contemporary models are no longer used only to recognize patterns or generate fluent text. They are increasingly asked to interpret evidence, combine information from different sources, perform calculations, justify conclusions, and revise their answers when an error is detected, a shift reflected in work on chain-of-thought prompting, tool use, agentic reasoning, and reflection [156, 130, 175, 106]. In other words, AI is moving from perception and language generation toward complex reasoning.

Finance is a demanding domain in which to study this shift. Financial work depends on evidence that is heterogeneous, dense, and often highly structured. Relevant information may appear in annual reports, regulatory filings, spreadsheets, balance sheets, income statements, cash-flow tables, earnings presentations, chart images, analyst notes, screenshots, and conversational explanations. A single financial question may require a system to read a chart, identify the correct reporting period, align units, retrieve a formula, apply financial knowledge, perform arithmetic, and justify why one answer follows from the available evidence [25, 203, 67, 128]. This makes finance a natural but difficult testbed for reasoning with LLMs and MLLMs.

LLMs and MLLMs have already been applied to many financial tasks, including document question answering, financial report analysis, numerical reasoning, table interpretation, visual question answering, conversational assistance, and financial education [27, 165, 171, 44, 169]. These applications show the practical appeal of general-purpose models in finance: they can accept flexible natural-language questions, process long contexts, and interact with users in ways that are easier to use than many traditional financial analytics tools. However, fluency does not imply financial competence. A model may produce an answer that sounds plausible while relying on the wrong figure, applying an inappropriate formula, overlooking a chart annotation, or confusing financial concepts that have similar surface language.

Financial reasoning is therefore not merely financial text understanding. Traditional financial NLP has often focused on tasks such as sentiment classification, event extraction, document retrieval, named entity recognition, and price-related prediction [66, 158, 43, 45]. These tasks remain important, but they do not fully capture the reasoning demands of many financial questions. Financial reasoning requires a system to identify what evidence is relevant, understand how that evidence is structured, apply domain-specific knowledge, perform numerical or logical operations, and produce a conclusion that can be inspected. In this thesis, financial reasoning is understood as a domain-specific reasoning process in which an intelligent system integrates financial knowledge, heterogeneous evidence, numerical or logical operations, and contextual interpretation to derive justified and decision-relevant conclusions.

Multimodality makes this problem more difficult. Financial evidence is rarely presented as plain text alone. Tables encode row-column relations, reporting periods, units, and subtotals. Charts encode trends, comparisons, proportions, and visual annotations. Document images preserve layout, headings, footnotes, legends, and relationships that may be lost in text extraction. Professional financial questions may also combine a visual figure with a textual scenario and rely on external financial knowledge that is not explicitly stated in either modality. Recent multimodal financial benchmarks show that these evidence forms remain challenging for MLLMs [169, 103, 141]. A multimodal financial reasoning system must therefore align visual evidence, textual context, and financial knowledge rather than treating the image as a decorative supplement to text.

The stakes of financial reasoning are also high. Incorrect reasoning can lead to misleading risk assessment, unsupported investment interpretation, inaccurate understanding of disclosures, poor educational feedback, or flawed audit and compliance analysis. This thesis does not argue that LLMs or MLLMs should be used as au-

tonomous financial decision makers. Rather, it argues that systems used to support financial analysis must be evaluated and improved with reliability in mind. **In this thesis, reliability is defined as the ability of a financial reasoning system to consistently produce correct, evidence-grounded, robust, and verifiable conclusions, consistent with recent emphasis on evidence-grounded QA, process supervision, and verifiable reasoning [67, 93, 166, 197].** Correctness concerns the final answer; evidence grounding concerns whether the answer is supported by relevant information; robustness concerns stability under noisy or perturbed inputs; and verifiability concerns whether a human can inspect the reasoning process.

These dimensions of reliability are related but not interchangeable. A model can produce a correct answer for the wrong reason, cite relevant evidence while misusing it, or generate a convincing explanation that cannot be checked. Conversely, a model can retrieve the correct evidence but fail at the calculation or synthesis step. This thesis therefore treats reliable multimodal financial reasoning as an end-to-end process. Reliability must be considered from evaluation, to evidence acquisition, to reasoning, to reflection and verification.

1.2 Research Context

This thesis focuses on multimodal financial reasoning with LLMs. The focus on financial reasoning is motivated by the nature of financial questions: they are not only about recalling facts or summarizing text, but about making justified inferences from evidence. A financial reasoning task may ask for a ratio, a trend interpretation, a comparison between reporting periods, the selection of a financial concept, or the explanation of a risk-related implication. These tasks require knowledge, calculation, and interpretation to work together.

The focus on multimodality is motivated by the form of financial evidence. In finance, important information is distributed across text, tables, charts, and document layouts. Text may describe a business situation, a chart may provide values, a table may contain the required line item, and domain knowledge may determine which formula is appropriate. A model that can read only text is therefore limited when the decisive evidence is visual or structural. A model that can describe an image is also insufficient if it cannot connect the visual content to a financial concept or operation. Multimodal financial reasoning requires cross-modal alignment between what is seen, what is stated, what is known, and what must be inferred.

The thesis adopts the following conceptual reasoning pipeline:

Evidence $\rightarrow$ Understanding $\rightarrow$ Reasoning $\rightarrow$ Answer $\rightarrow$ Reflection and Verification.

Evidence refers to the visual, textual, numerical, and knowledge sources that can legitimately support an answer. Understanding refers to the interpretation of these sources, such as locating a value, recognizing a trend, identifying a relevant entity, or determining which financial topic is involved. Reasoning refers to the operations performed over the interpreted evidence, including arithmetic, comparison, financial formula application, logical inference, and professional judgement. Answer refers to the final response in the required format. Reflection and verification refer to the process of checking whether the evidence, intermediate steps, and final conclusion are consistent.

This pipeline explains why benchmarks are needed. Without appropriate benchmarks, it is difficult to determine whether an MLLM genuinely reasons over multi-modal financial evidence or merely exploits surface cues. Existing financial and multi-modal benchmarks provide important starting points, but they differ in their coverage of numerical reasoning, document grounding, multimodal perception, and domain expertise [25, 203, 181, 169, 102]. A benchmark must expose the capabilities that matter for financial reasoning: mathematical operations, financial expertise, multimodal evidence use, topic coverage, and fine-grained error patterns. Evaluation is therefore the first technical step in the thesis. It provides a way to make the reliability problem visible.

The pipeline also explains why multi-agent evidence acquisition is needed. Many models receive all available context at once and then produce a single answer. This approach gives the model little explicit opportunity to ask what evidence is missing, which modality should be inspected, or whether the current evidence is sufficient. Retrieval-augmented and agentic methods show the value of external evidence and tool-mediated reasoning, but fixed retrieval or fixed role decomposition does not guarantee task-specific evidence sufficiency [87, 175, 157, 201]. In financial reasoning, the necessary evidence can change as reasoning proceeds. A partial solution may reveal that the model needs a specific chart value, a textual condition, or a financial definition. A multi-agent framework can decompose these needs into specialized roles, allowing visual evidence, textual context, financial knowledge, planning, and reasoning to be coordinated step by step.

Finally, the pipeline explains why error reflection is needed. Even when relevant evidence is available, a model can still produce incorrect intermediate reasoning. It

may select the wrong formula, misread a sign, confuse a unit, make an arithmetic error, or draw a conclusion that does not follow from its own steps. Self-reflection and iterative refinement are useful but limited because a model may repeat its own bias [106, 131]. This motivates cross-model error reflection, where step-level errors produced by different MLLMs are treated as resources that can help another model diagnose and revise its reasoning, drawing on broader work on learning from negative examples and mistakes [13, 146, 145, 6].

The scope of the thesis is deliberately bounded. It does not study autonomous trading, portfolio optimization, price forecasting, or automated investment advice. These areas are important, but they involve additional issues such as market feedback, investor preferences, regulatory constraints, and real-time deployment risk. The thesis instead studies reasoning over financial questions and multimodal evidence, especially in settings where answers and reasoning steps can be evaluated through benchmark labels, expert judgement, or controlled comparison. This scope enables a focused investigation of reliability while avoiding a drift into broader financial prediction or decision automation.

1.3 Problem Statement

This thesis addresses three connected problems: evaluation, evidence acquisition, and error correction. Together, these problems define the main reliability challenge for multimodal financial reasoning with LLMs and MLLMs

1.3.1 Problem 1: Evaluation

Existing benchmarks do not yet provide a sufficiently comprehensive evaluation of knowledge-intensive multimodal financial reasoning. Many financial reasoning benchmarks emphasize text, tables, or numerical operations [25, 203, 191], while many general multimodal benchmarks emphasize visual perception or broad-domain question answering [181, 101]. As a result, current evaluations often provide limited evidence about whether MLLMs can perform financial reasoning when the task requires domain knowledge, multimodal understanding, and precise calculation at the same time.

Several limitations are especially important. First, existing benchmarks often provide insufficient coverage of financial knowledge-intensive reasoning, where the answer depends on concepts such as risk, valuation, financial reporting, accounting relationships, or professional financial practice. Second, mathematical reasoning and financial

expertise reasoning are not always separated, making it difficult to distinguish failures caused by incorrect operations from failures caused by missing domain knowledge. Third, multi-image and multimodal inputs remain underrepresented, even though real financial documents often require information to be compared across several figures or aligned between image and text. Fourth, topic coverage can be narrow or uneven, which limits the diagnostic value of reported performance. Fifth, many evaluations report aggregate accuracy without enough fine-grained analysis of model capabilities and errors.

The evaluation problem is therefore not simply a problem of dataset size. A large benchmark can still fail to evaluate the capabilities that matter if most examples are answerable through shallow pattern matching or if the evidence source is obvious. A useful benchmark for this thesis must diagnose how models use visual evidence, textual context, financial knowledge, mathematical operations, and answer synthesis. It should also reveal how performance changes across topics, reasoning types, and input formats. Chapter 3 addresses this problem by introducing FinMR, a benchmark designed to evaluate knowledge-intensive multimodal financial reasoning.

1.3.2 Problem 2: Evidence Acquisition

Existing models often passively receive all available context and attempt to answer in a single generation. This one-shot setting can work when the context is short, complete, and easy to interpret. However, multimodal financial reasoning frequently requires a model to determine what evidence is missing, which modality should be inspected, and how visual, textual, and financial knowledge evidence should be coordinated. A fixed pipeline may provide captions, retrieved passages, or a preselected context, but it does not necessarily adapt to the specific evidence gaps revealed during reasoning.

This matters because financial evidence is often scattered across sources. A question may define the task in text, provide the relevant numbers in a chart, and require an external financial concept to choose the correct operation. If the model treats all context as equally relevant, it may overlook the decisive evidence or combine incompatible pieces of information. If the model uses a fixed sequence of retrieval, captioning, and reasoning, it may continue even when evidence is insufficient or stop before a crucial source has been inspected. In finance, this can lead to period mismatch, unit confusion, incorrect formula application, or unsupported answer synthesis.

The evidence acquisition problem is therefore adaptive. A reliable system should be able to ask what information is missing at a given step, route the question to a suitable evidence source, integrate the returned evidence, and decide whether more evidence

is required before answering. This motivates the multi-agent study in Chapter 4. The SEA framework uses a planner, an image evidence agent, a context evidence agent, a knowledge evidence agent, and a reasoning agent. Its central idea is not simply to use multiple agents, but to make evidence acquisition query-based, step-wise, and sensitive to evidence sufficiency.

1.3.3 Problem 3: Error Correction

Even if a model acquires relevant evidence, it may still reason incorrectly. Errors can occur at many points in the reasoning trace. A model may misinterpret an image, overlook a visual label, use an inappropriate financial formula, make a numerical computation error, produce an invalid intermediate step, or synthesize an answer that does not follow from its own reasoning. These failures show that evidence acquisition is necessary but not sufficient for reliable reasoning.

Error correction is difficult because final-answer feedback is often too coarse. If a model only learns that its answer is wrong, it may not know whether the failure occurred during visual perception, evidence selection, cross-modal alignment, financial knowledge use, operation selection, numerical computation, logical inference, or answer synthesis. Self-reflection can sometimes improve a response, but it can also reproduce the same model-specific bias that caused the original error. A model that selected the wrong formula may confidently defend that formula; a model that misread a chart may not discover the visual mistake by reflecting over its own generated text.

This thesis therefore treats step-level errors as useful evidence for reflection. If reasoning traces are decomposed into intermediate steps, errors can be represented more precisely and retrieved when a new reasoning problem contains a similar failure pattern. Moreover, errors from different MLLMs can be complementary: one model’s failure may reveal a trap that another model should avoid. Chapter 5 addresses this problem through FinErrorSet and CLER, a cross-MLLM error reflection framework that retrieves relevant step-level errors and uses them to guide reasoning revision.

1.4 Research Aim

The aim of this thesis is to advance reliable multimodal financial reasoning with large language models through comprehensive capability evaluation, adaptive multi-agent evidence acquisition, and cross-model error reflection.

This aim has two implications. First, the thesis is concerned with reliability across the reasoning process rather than only final-answer Accuracy. Second, the thesis treats evaluation, evidence acquisition, and error reflection as complementary components. Evaluation identifies what current systems can and cannot do; evidence acquisition improves the information available to the reasoner; and error reflection improves the system’s ability to diagnose and correct flawed intermediate reasoning.

1.5 Research Questions

The overall research question of the thesis is:

How can multimodal financial reasoning systems be made more reliable through evaluation, evidence acquisition, and error reflection?

This overall question is answered through three technical research questions:

- RQ1.* How can the knowledge-intensive multimodal financial reasoning capabilities of large language models be comprehensively evaluated?
- RQ2.* How can specialized agents dynamically acquire and integrate heterogeneous evidence to improve multimodal financial reasoning?
- RQ3.* How can step-level errors produced by different MLLMs be exploited to improve financial reasoning?

RQ1 is addressed in Chapter 3 through FinMR. RQ2 is addressed in Chapter 4 through the SEA framework. RQ3 is addressed in Chapter 5 through FinErrorSet and CLER. Chapter 6 returns to the overall research question and integrates the findings into a thesis-level account of reliable multimodal financial reasoning. Chapter 2 provides the conceptual foundation for these studies, but it is not framed as a separate original technical research question.

1.6 Research Objectives

The objectives of this thesis are to:

1. systematically synthesize the definitions, tasks, benchmarks, methods, and evaluation practices of financial reasoning with LLMs and MLLMs;

2. construct a knowledge-intensive multimodal financial reasoning benchmark that covers mathematical reasoning, financial expertise reasoning, multiple topics, and single-image and multi-image inputs;
3. systematically evaluate mainstream MLLMs and analyse their fine-grained capabilities and error patterns;
4. design a step-wise multi-agent evidence acquisition framework for multimodal financial reasoning;
5. compare agent communication paradigms and analyse evidence quality, evidence perturbation, cost, **latency**, and statistical significance;
6. construct a step-level cross-model financial error resource;
7. develop a cross-MLLM error reflection framework that retrieves relevant errors and supports reasoning revision;
8. integrate the findings into an account of the components required for reliable multimodal financial reasoning.

These objectives move from conceptual synthesis to empirical evaluation, method development, error correction, and thesis-level integration. They preserve a consistent research object across the thesis: reliable multimodal financial reasoning with LLM.

1.7 Contributions

This thesis makes five contributions.

1.7.1 Contribution 1: Literature Foundation

The thesis provides a structured synthesis of the financial reasoning literature that establishes the conceptual foundation for the thesis. It organizes existing work into five task families: financial numerical reasoning, financial tabular reasoning, financial document reasoning, financial multimodal reasoning, and financial conversational reasoning. It also organizes methods into four broad paradigms: prompting and program-based methods, retrieval-augmented and evidence-grounded methods, agent-based and interactive methods, and training-, reflection-, and verification-based methods. This synthesis identifies recurring evaluation and reliability issues and motivates the three technical studies that follow.

The boundary of this contribution is important. Chapter 2 is not presented as an additional technical system or benchmark. Its contribution is to clarify the field, define the thesis’s research object, and identify the evaluation, evidence acquisition, and error reflection gaps addressed in Chapters 3–5.

1.7.2 Contribution 2: FinMR

The thesis introduces FinMR, a benchmark for knowledge-intensive multimodal financial reasoning. FinMR contains 3,200 multimodal financial reasoning questions and 3,764 images across 15 financial topics. It distinguishes mathematical reasoning from financial expertise reasoning and includes both single-image and multi-image questions. The benchmark supports systematic baseline evaluation, fine-grained topic and modality analysis, and error analysis using a unified taxonomy. FinMR contributes an evaluation instrument for diagnosing what current MLLMs can and cannot do in multimodal financial reasoning.

1.7.3 Contribution 3: SEA

The thesis develops SEA, a step-wise multi-agent evidence acquisition framework for multimodal financial reasoning. SEA includes a planner, an image evidence agent, a context evidence agent, a knowledge evidence agent, and a reasoning agent. Its query-based communication protocol enables the system to identify evidence gaps, acquire visual, textual, and financial knowledge evidence, assess evidence sufficiency, and decide when to terminate the evidence acquisition process. The study also compares communication paradigms, evaluates perturbation effects, and analyses cost, latency, and statistical significance.

1.7.4 Contribution 4: CLER

The thesis introduces FinErrorSet and CLER for cross-MLLM error reflection. FinErrorSet is a step-level financial reasoning error resource constructed from reasoning traces produced by multiple MLLMs. CLER represents errors, retrieves relevant cross-model errors, uses contrastive learning to improve retrieval, and guides reflection and revision of current reasoning steps. The study includes human validation and cross-benchmark experiments, showing how step-level errors from diverse models can support more reliable financial reasoning.

1.7.5 Contribution 5: Integrated Insight

The thesis develops an integrated understanding of reliable multimodal financial reasoning:

$$\begin{aligned} \text{Reliable Reasoning} = & \text{Appropriate Evaluation} + \text{Relevant Evidence} \\ & + \text{Correct Reasoning} + \text{Effective Reflection}. \end{aligned}$$

This formulation explains why the three technical studies belong together. FinMR contributes appropriate evaluation, SEA contributes relevant evidence acquisition, and CLER contributes error reflection for correcting remaining reasoning failures. The integrated insight is not that any single component guarantees reliability, but that reliability depends on the interaction among evaluation, evidence, reasoning validity, and reflection.

1.8 Thesis Structure

The remainder of the thesis is organized as follows.

Chapter 2 reviews financial reasoning with LLMs. It defines the scope of financial reasoning, distinguishes financial reasoning from broader financial NLP, organizes the literature into five task families and four method paradigms, reviews evaluation practices, and identifies the research gaps that motivate the thesis.

Chapter 3 introduces FinMR, a benchmark for knowledge-intensive multimodal financial reasoning. It presents the task definition, benchmark design, data construction process, dataset statistics, experimental setup, baseline results, fine-grained analysis, and error analysis. The chapter answers RQ1 by showing how MLLM capabilities can be evaluated more comprehensively.

Chapter 4 presents SEA, a step-wise multi-agent evidence acquisition framework. It formulates the evidence acquisition problem, describes the specialized agent roles, introduces query-based communication, evaluates the framework across benchmarks, compares communication paradigms, and analyses ablations, perturbations, cost, latency, and statistical significance. The chapter answers RQ2 by showing how specialized agents can dynamically acquire and integrate heterogeneous evidence.

Chapter 5 presents FinErrorSet and CLER. It defines the step-level error reflection problem, describes the construction of FinErrorSet, introduces cross-model error retrieval and contrastive learning, and evaluates reflection and revision across bench-

marks and reasoners. The chapter answers RQ3 by showing how errors produced by different MLLMs can be exploited to improve financial reasoning.

Chapter 6 synthesizes the findings from the three technical studies. It answers the overall research question, presents an integrated framework for reliable financial reasoning, discusses cross-study findings, and considers theoretical, methodological, and practical implications, as well as overall limitations and future research directions.

Chapter 7 concludes the thesis. It summarizes the research problem, aim, contributions, and final thesis-level answer: reliable financial intelligence requires systems that can identify appropriate evidence, integrate heterogeneous information, perform valid reasoning, recognize errors, and support conclusions that can be inspected and verified.

Literature Review: Financial Reasoning with LLMs

2.1 Chapter Overview

This chapter reviews and synthesizes research on financial reasoning with LLMs and MLLMs. Its purpose is to establish the conceptual foundation for the three technical studies in this thesis rather than to present financial NLP as a broad application area. The chapter defines financial reasoning, explains why it differs from general language understanding, reviews major task families and benchmarks, analyses methodological directions, examines evaluation practice, and identifies the research gaps addressed in Chapters 3, 4, and 5.

The review is organized around the central thesis object: reliable multimodal financial reasoning with LLMs. Chapter 1 defined reliability as the ability of a financial reasoning system to consistently produce correct, evidence-grounded, robust, and verifiable conclusions. This chapter shows why that form of reliability is difficult. Financial reasoning requires a model to select evidence, interpret financial context, align numbers and modalities, apply financial knowledge, perform valid operations, and justify the answer. These requirements cannot be reduced to sentiment analysis, information extraction, price forecasting, or general financial text generation.

The chapter adopts a capability-oriented taxonomy with five task categories: financial numerical reasoning, financial tabular reasoning, financial document reasoning, financial multimodal reasoning, and financial conversational reasoning. This taxonomy is intentionally broader than the later technical chapters. Chapters 3–5 focus increasingly on multimodal financial reasoning, while this review explains how multi-

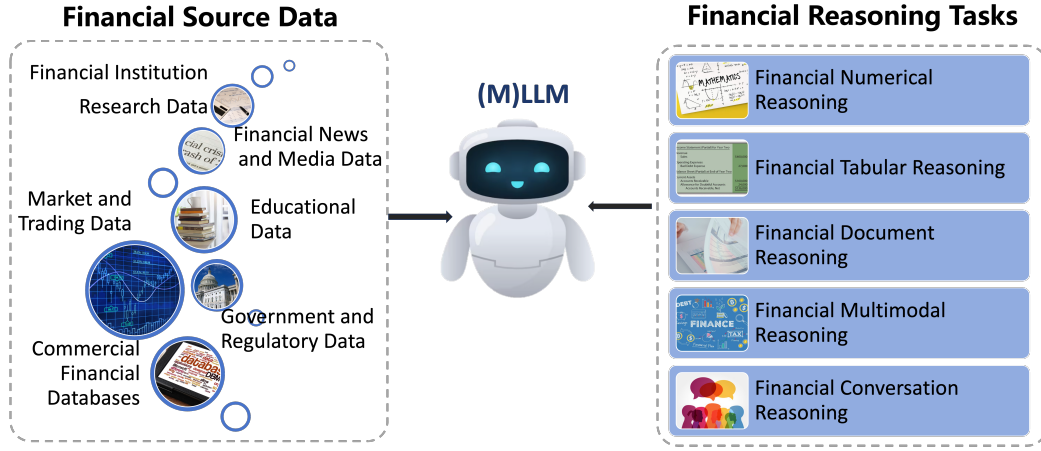


Figure 2.1: Scope of financial reasoning with LLMs and MLLMs. Heterogeneous financial data support numerical, tabular, document, multimodal, and conversational reasoning tasks.

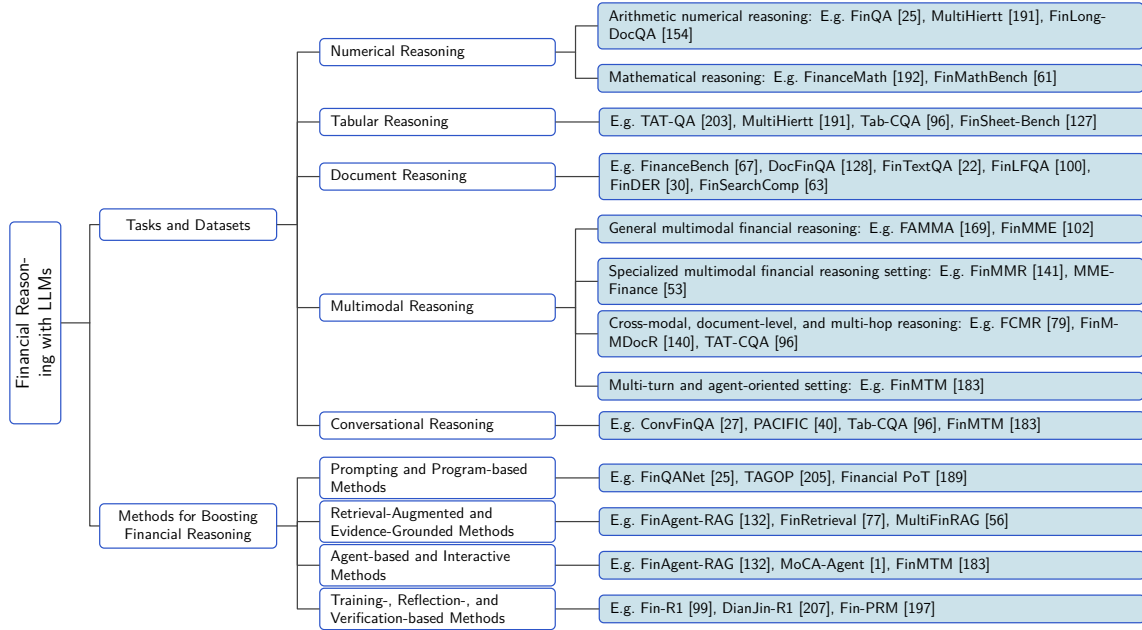


Figure 2.2: Taxonomy of tasks, datasets, and methods for financial reasoning with LLMs.

modal financial reasoning inherits challenges from numerical, tabular, document-level, and conversational financial reasoning.

The chapter proceeds as follows. Section 2.2 describes the review scope and literature selection process. Section 2.3 introduces foundations of financial reasoning, including general LLMs reasoning techniques, financial language models, the working definition used in this thesis, and the distinctive properties of financial reasoning.

Section 2.4 reviews tasks and benchmarks. Section 2.5 reviews methods. Section 2.6 analyses evaluation. Section 2.7 synthesizes the literature into three broad findings. Section 2.8 states the research gaps that motivate FinMR, SEA, and CLER. Section 2.9 concludes the chapter.

2.2 Review Scope and Literature Selection

The review covers work on financial reasoning tasks in which models must derive a justified answer from financial evidence rather than merely classify text or predict a market variable. The literature was selected to support the thesis aim: advancing reliable multimodal financial reasoning through evaluation, evidence acquisition, and error reflection. The review therefore emphasizes benchmarks, methods, and evaluation protocols that expose how LLMs and MLLMs reason over financial evidence.

The search scope included financial NLP, financial question answering, financial numerical reasoning, financial table reasoning, financial document question answering, multimodal financial reasoning, conversational financial reasoning, financial agents, retrieval-augmented financial QA, program-based financial reasoning, reflection, verification, and process-level evaluation. The sources considered included major NLP, AI, multimodal learning, and finance-oriented AI venues; public preprint repositories; and benchmark papers referenced by the source studies used in this thesis. The review gives priority to work from the recent LLMs and MLLMs period because the thesis studies large generative models, but earlier financial NLP and reasoning work is included when it establishes task definitions, benchmark formats, or evaluation conventions.

The keyword strategy combined domain terms, model terms, task terms, and reliability terms. Domain terms included finance, financial report, accounting, investment, risk, valuation, audit, filing, and financial statement. Model terms included LLMs, MLLMs, financial language model, vision-language model, and agent. Task terms included question answering, numerical reasoning, table reasoning, document reasoning, chart reasoning, multimodal reasoning, multi-turn reasoning, and conversational reasoning. Reliability terms included evidence, grounding, attribution, verification, reflection, error analysis, process supervision, robustness, perturbation, calibration, and human evaluation.

Studies were included when they satisfied at least one of the following criteria: they introduced a benchmark for financial reasoning; evaluated LLMs or MLLMs on financial reasoning; proposed prompting, program-based, retrieval-augmented, agentic,

training-based, reflection-based, or verification-based methods for financial reasoning; or provided evidence about model errors and reliability in financial analytical tasks. The review specifically includes financial numerical reasoning, financial table reasoning, financial document QA, multimodal financial reasoning, conversational financial reasoning, financial agents, and reflection and verification.

Studies were excluded when the main task did not require substantive reasoning. Excluded areas included pure sentiment analysis, stock price prediction as a standalone forecasting problem, portfolio optimization, generic information extraction, entity classification, and other financial classification tasks where the central objective is label prediction rather than evidence-grounded inference. These areas are important to financial AI, but they do not directly address the thesis object. For example, sentiment classification can support financial analysis, yet it usually does not require a model to align evidence, select formulas, perform calculations, or verify a reasoning trace.

The reviewed studies were classified through an iterative coding process. First, each benchmark or method was coded by modality, evidence type, answer format, reasoning requirement, supervision type, and evaluation protocol. Second, benchmark studies were grouped into five task categories: financial numerical, tabular, document, multimodal, and conversational reasoning. Third, methodological studies were grouped into four paradigms: prompting and program-based methods; retrieval-augmented and evidence-grounded methods; agent-based and interactive methods; and training-, reflection-, and verification-based methods. Finally, cross-cutting limitations were synthesized into three thesis-level gaps: evaluation, evidence acquisition, and error reflection.

The time frame of the review reflects the development of the field. Earlier financial NLP and financial QA studies are included when they establish durable benchmark formats, such as financial report QA, table-text reasoning, numerical operation annotation, or document-grounded evidence. The main emphasis, however, is on the period in which LLMs and MLLMs became the dominant experimental models. This emphasis is appropriate because the thesis studies the reliability of large generative systems rather than task-specific classifiers. It also means that benchmark design, prompting, retrieval, agentic workflows, and reflection are reviewed as parts of a shared reasoning pipeline rather than as isolated techniques.

The classification process also deliberately separates task families from methodological paradigms. A benchmark such as FinQA may be used to evaluate prompting, program generation, retrieval, or training; it is therefore coded as a task resource

rather than as a method. Conversely, a method such as retrieval-augmented generation can be applied to document reasoning, table reasoning, or multimodal reasoning. This distinction is useful for the thesis because the three technical studies contribute at different levels: FinMR contributes an evaluation resource, SEA contributes an evidence acquisition method, and CLER contributes an error reflection method.

2.3 Foundations of Financial Reasoning

2.3.1 Reasoning with LLMs

Reasoning is the process through which a system draws conclusions from premises, evidence, background knowledge, and intermediate inferences [147, 82, 72, 114, 108]. In logic-oriented accounts, reasoning connects premises to conclusions through explicit or implicit inference rules [17]. In practical AI systems, reasoning is broader: it includes selecting relevant information, decomposing a problem, applying knowledge, using tools, checking intermediate steps, and producing a conclusion that follows from the available evidence. This broader view is important for financial reasoning because financial questions rarely require only one isolated operation.

LLMs have made reasoning a central evaluation problem [167]. CoT prompting encourages models to generate intermediate reasoning steps before the final answer [156]. Self-consistency samples multiple reasoning paths and aggregates answers to improve robustness [152]. Program-of-Thought and program-aided methods ask models to externalize parts of reasoning as executable code, which is useful when exact numerical computation matters [23, 54]. Tool-use methods allow models to call calculators, search engines, retrievers, or other external functions when internal generation is insufficient [130]. Agentic methods, including ReAct, interleave reasoning with actions such as search or tool calls [175]. More deliberative approaches search over alternative reasoning paths or use multiple agents to critique candidate answers [174, 47, 157].

Reflection and verification extend this progression. Self-Refine and Reflexion use model-generated feedback or memory to revise outputs over multiple attempts [106, 131]. Process supervision evaluates intermediate reasoning steps rather than only final answers, showing that step-level feedback can be more informative than answer-level supervision [93]. These techniques are relevant to finance because an answer may be numerically correct by coincidence, while the reasoning path may still contain an unsupported assumption, an incorrect formula, or a mismatched reporting period.

However, general LLMs reasoning techniques do not automatically solve financial reasoning. A chain-of-thought can be fluent but ungrounded. A program can execute correctly while implementing the wrong financial operation. A retriever can return relevant passages that are misinterpreted. A critic agent can miss the same error made by the generator. Financial reasoning therefore requires domain-specific attention to evidence, units, temporal context, formulas, accounting relationships, and verifiability.

This observation motivates a process view of reasoning. In many LLMs evaluations, reasoning is treated as a property of the final response or of the generated rationale. In financial reasoning, the process must be decomposed more carefully. A system may first need to recognize what the question is asking, then identify which evidence source contains the required information, then extract values or concepts, then apply an operation, then check whether the answer is supported. Each stage can fail in a different way. A model that is strong at generic mathematical reasoning may still fail at financial reasoning because it selected the wrong line item. A model that is strong at visual perception may still fail because it does not know which financial formula is relevant.

General reasoning methods also differ in how inspectable they make the reasoning process. Chain-of-thought produces natural-language rationales that are easy for users to read but difficult to verify mechanically. Program-of-Thought produces executable traces that are easier to check but only when the task can be expressed as code. Retrieval-augmented generation exposes external sources but does not automatically expose how evidence is transformed into an answer. Agents expose interaction traces but introduce questions about role design, communication cost, and error propagation. Reflection exposes revision attempts but may not provide independent evidence. This thesis draws on all of these traditions while focusing on the specific reliability demands of financial reasoning.

The link to multimodal reasoning is especially important. In multimodal settings, the reasoning process starts before language generation. The model must perceive the image, identify the relevant visual elements, map them to financial variables, and align them with textual context. A reasoning trace that begins only after visual information has been converted into text may hide perception and alignment errors. This is why the later technical chapters treat visual perception errors, evidence selection errors, cross-modal alignment errors, financial knowledge errors, formula or operation selection errors, numerical computation errors, logical inference errors, and answer synthesis errors as distinct error sources.

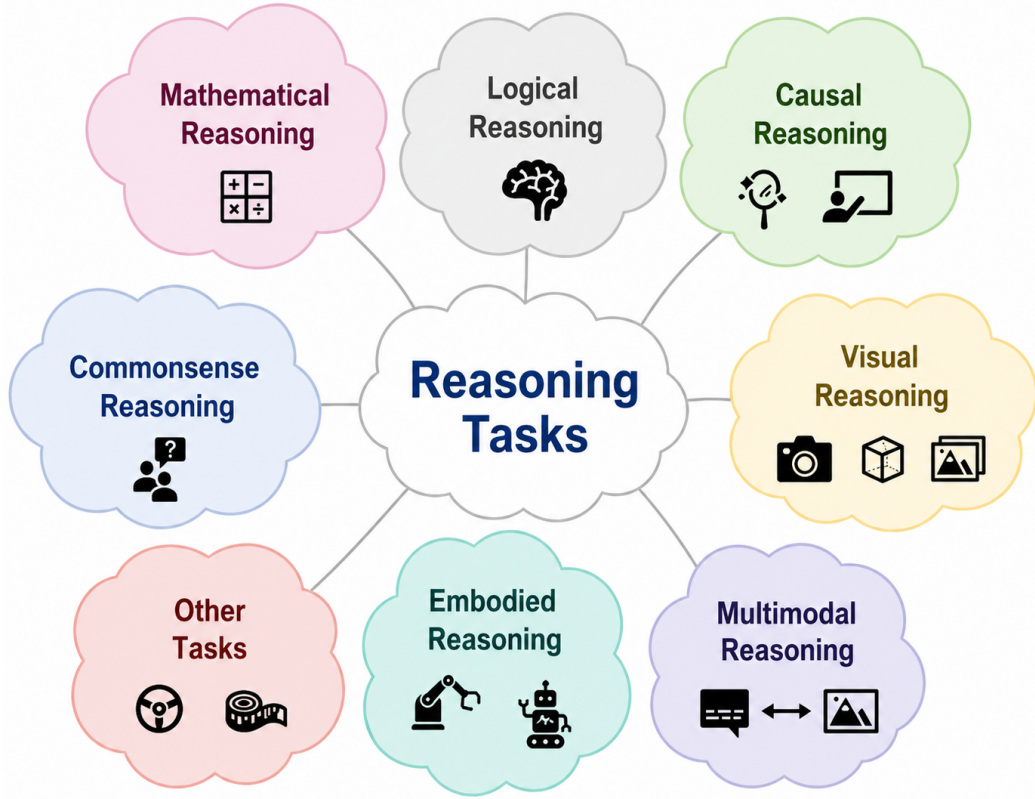


Figure 2.3: Reasoning tasks.

2.3.2 Financial NLP and Financial Language Models

Financial NLP provides an important foundation for financial reasoning, but financial language competence does not necessarily imply financial reasoning competence. Domain-adapted models such as FinBERT, BloombergGPT, FinGPT, and PIXIU/FinMA show that financial corpora, financial instruction data, and task-specific training can improve financial text understanding, sentiment analysis, information extraction, and financial QA performance [66, 158, 171, 165]. Surveys and benchmarks of financial language modeling further show the breadth of financial NLP, from sentiment and document understanding to domain-specific evaluation [84, 44, 164, 142, 149]. These models help address the vocabulary, style, and document conventions of finance.

Yet financial reasoning requires more than recognizing financial language. A model may understand the term "operating margin" but still fail to compute it from a table. It may identify a paragraph about liquidity but ignore that the question asks for a particular fiscal year. It may recognize a chart as financial but misread the legend or extract the wrong bar. It may retrieve a passage from a filing but draw a conclusion

beyond what the evidence supports. For this reason, the thesis distinguishes financial language competence from financial reasoning competence. The former concerns domain fluency and representation; the latter concerns evidence-grounded inference.

This distinction also explains why the literature review is not organized around financial NLP tasks such as sentiment classification, named entity recognition, or event extraction. Such tasks can support reasoning systems, but they do not by themselves evaluate whether a model can integrate evidence, apply financial knowledge, compute accurately, and justify a conclusion. The benchmarks and methods reviewed in this chapter are therefore selected because they make reasoning visible.

The distinction is also methodological. Domain-adapted language models can provide better priors for financial terminology, document style, and task instructions, but reasoning reliability often depends on external structure. A financial report may contain hundreds of pages, multiple tables, repeated entities, and similar line items across several reporting periods. No amount of language fluency removes the need to locate the right evidence. Similarly, knowledge of a formula is useful only if the model can select the correct variables and units. The literature therefore suggests that financial language models are best understood as foundations for reasoning systems, not as complete solutions to financial reasoning.

2.3.3 Definition of Financial Reasoning

This thesis uses the following definition:

Financial reasoning is a domain-specific reasoning process in which an intelligent system integrates financial knowledge, heterogeneous evidence, numerical or logical operations, and contextual interpretation to derive justified and decision-relevant conclusions.

This definition has four implications. First, financial reasoning is domain-specific. The same arithmetic operation can mean different things depending on whether the problem concerns liquidity, profitability, valuation, risk, or accounting recognition. Second, financial reasoning is evidence-grounded. Conclusions should be supported by financial documents, tables, charts, textual contexts, or relevant knowledge. Third, financial reasoning is operational. It often requires calculations, comparisons, formula selection, logical inference, or multi-step synthesis. Fourth, financial reasoning is decision-relevant. The answer should not only be syntactically valid; it should make sense in the financial context.

The definition covers numerical, tabular, document, multimodal, and conversational settings. In numerical reasoning, the evidence may be values and formulas. In tabular reasoning, the evidence may be cells, headers, periods, and row hierarchies. In document reasoning, the evidence may be long filings, report sections, or attributed sources. In multimodal reasoning, the evidence may be charts, report figures, visual tables, or document images. In conversational reasoning, the evidence may include dialogue history and previously computed intermediate results. These settings differ in representation, but they share the same underlying requirement: justified inference from financial evidence.

The definition also clarifies what is outside the scope of this thesis. A system that predicts tomorrow’s price movement from historical prices is not necessarily performing financial reasoning in the sense used here, unless it produces a justified inference from evidence that can be inspected. A system that classifies a headline as positive or negative may support financial reasoning, but it is not itself a reasoning system if it only maps input text to a label. A system that generates a persuasive financial explanation is also insufficient if the explanation cannot be grounded in evidence or checked for numerical validity. The thesis therefore focuses on tasks where evidence, knowledge, operations, and justification interact.

Finally, the definition makes reliability a process-level property. A financial reasoning system can be wrong because the evidence is missing, because the evidence is misread, because the wrong operation is selected, because the calculation is wrong, or because the final answer is poorly synthesized. These failure modes motivate the unified error taxonomy used in later chapters. They also explain why this chapter reviews not only benchmarks and methods, but also evaluation protocols and research gaps.

2.3.4 What Makes Financial Reasoning Distinct?

Financial reasoning is distinct from general reasoning in several ways. First, it requires domain-specific knowledge. Financial problems involve concepts such as revenue, operating margin, cash flow, leverage, liquidity, volatility, valuation, default risk, and financial reporting. These concepts are not merely words; they are tied to accounting principles, market mechanisms, and professional financial practice. A model that lacks domain knowledge may apply a superficially plausible operation that is financially invalid.

Second, financial reasoning depends on heterogeneous evidence. Information may appear in narrative disclosures, structured tables, report figures, charts, screenshots,

document images, analyst commentary, and external knowledge sources. Benchmarks such as FinQA and TAT-QA show that financial numerical reasoning often requires both textual and tabular evidence [25, 203]. FinanceBench and DocFinQA show that document-level evidence grounding is central to financial QA [67, 128]. FAMMA, FinMMR, FinMME, and MME-Finance show that visual and multimodal evidence are increasingly important for evaluating MLLMs in finance [169, 141, 102, 53].

Third, financial reasoning requires numerical and structural precision. Small mistakes in row selection, unit alignment, period interpretation, or sign direction can change the answer. A model may perform arithmetic correctly while using the wrong operands. It may identify the correct line item but choose the wrong reporting period. It may compute a percentage change while ignoring that the table reports values in millions. Such errors are especially common when tables, charts, and text must be aligned.

Fourth, financial reasoning is temporally and contextually sensitive. Financial evidence is time-dependent. A ratio, cash-flow item, or risk indicator must usually be interpreted in relation to a reporting period, market condition, accounting context, or firm-specific event. A statement that is correct for one year may be wrong for another. A trend that appears positive in isolation may be misleading when interpreted with cost structure, debt pressure, or industry conditions.

Fifth, financial reasoning is decision-oriented. It often supports judgements about performance, risk, valuation, compliance, investment education, or business conditions. This does not mean that LLMs should make autonomous financial decisions. It means that outputs should be grounded, inspectable, and appropriately cautious because users may rely on them to support analysis.

Sixth, financial reasoning has high reliability requirements. In finance, a plausible but unsupported answer can be harmful. Reliability therefore requires more than final-answer Accuracy. It requires evidence grounding, process validity, robustness, and verification. These requirements motivate the three technical studies in this thesis: FinMR evaluates capabilities and errors; SEA improves evidence acquisition; and CLER improves step-level error reflection.

The six characteristics are tightly connected. Domain-specific knowledge determines which evidence is relevant and which operation is valid. Heterogeneous evidence requires structural and cross-modal alignment. Numerical precision depends on temporal and unit consistency. Decision orientation increases the cost of unsupported conclusions. Reliability requirements force the evaluation to examine not only answers,

but also evidence and reasoning processes. This interconnectedness is why financial reasoning is difficult to benchmark with a single metric or a single task format.

These characteristics also explain why multimodal financial reasoning is a natural focal point for the thesis. Multimodal tasks combine several sources of difficulty at once. A chart may require visual perception, unit interpretation, temporal comparison, and financial knowledge. A report image may require layout understanding, table reasoning, and document grounding. A multi-image question may require a model to determine which figure contains the relevant evidence and how the figures relate. Such tasks expose weaknesses that may remain hidden in text-only or table-only settings.

2.4 Tasks and Benchmarks

Financial reasoning benchmarks can be organized into five task families: financial numerical reasoning, financial tabular reasoning, financial document reasoning, financial multimodal reasoning, and financial conversational reasoning. The categories are not mutually exclusive. FinQA and TAT-QA involve both numerical and tabular reasoning; FinanceBench and DocFinQA involve document grounding and may require numerical reasoning; FAMMA and FinMMR involve multimodal reasoning and often require financial knowledge; ConvFinQA extends numerical reasoning into dialogue [25, 203, 67, 128, 169, 141, 27]. The taxonomy is therefore best understood as a way to identify dominant evidence forms and capability requirements.

2.4.1 Financial Numerical Reasoning

Financial numerical reasoning evaluates whether a model can derive numerical or quantitative answers from financial evidence. It includes arithmetic reasoning, financial mathematical reasoning, formula selection, calculation, and interpretation. The task is not limited to arithmetic. A model must identify the correct quantities, align periods and units, choose the relevant operation, execute the calculation, and interpret the result in context.

FinQA is a foundational benchmark for numerical reasoning over financial reports [25]. It pairs questions with financial report evidence and provides gold reasoning programs. This design makes the reasoning process partially observable: a model can be evaluated not only on the final answer, but also on whether it generates a plausible operation sequence. TAT-QA extends this setting to hybrid tabular and textual contexts and introduces questions requiring arithmetic and aggregation over

table-text evidence [203]. MultiHiertt increases structural complexity by introducing multi-hierarchical financial tables and longer contexts [191].

FinanceMath and FinanceReasoning shift the emphasis toward knowledge-intensive financial mathematical reasoning [192, 139]. These benchmarks require models to combine mathematical reasoning with finance-domain concepts and formula knowledge. They are important because they reveal a limitation of report-grounded numerical QA: a model may perform table arithmetic but fail when the problem requires selecting a financial formula or applying professional knowledge.

Financial numerical reasoning is central to this thesis because many multimodal financial questions require numerical operations over visual or textual evidence. However, numerical correctness alone is insufficient. A correct calculation over the wrong values is still wrong. Conversely, a model may fail a numerical question because of visual perception or evidence selection rather than arithmetic. This motivates the error taxonomy used throughout the thesis.

The typical reasoning chain for financial numerical reasoning can be described as evidence identification, variable grounding, operation selection, execution, and interpretation. Evidence identification determines which table, sentence, chart element, or formula is relevant. Variable grounding maps words in the question to concrete quantities. Operation selection decides whether the task requires addition, subtraction, percentage change, ratio computation, present-value reasoning, or another operation. Execution produces the numerical result. Interpretation explains what the result means in financial terms. Each stage requires different capabilities, and each stage creates different error opportunities.

Financial numerical reasoning benchmarks vary in how much of this chain they annotate. FinQA-style program supervision makes operations explicit, but does not necessarily cover visual evidence or broader financial expertise. FinanceMath-style solution references provide richer expert reasoning, but they may be less tied to real financial documents. Benchmarks that include only final answers cannot reveal whether the model selected the correct formula or merely guessed. This is why Chapter 3 distinguishes mathematical reasoning from financial expertise reasoning: the two overlap but are not the same capability.

2.4.2 Financial Tabular Reasoning

Financial tabular reasoning evaluates whether a model can reason over tables such as financial statements, spreadsheet-like data, or report tables. It requires row-column localization, period and unit alignment, multi-table reasoning, and table-text integra-

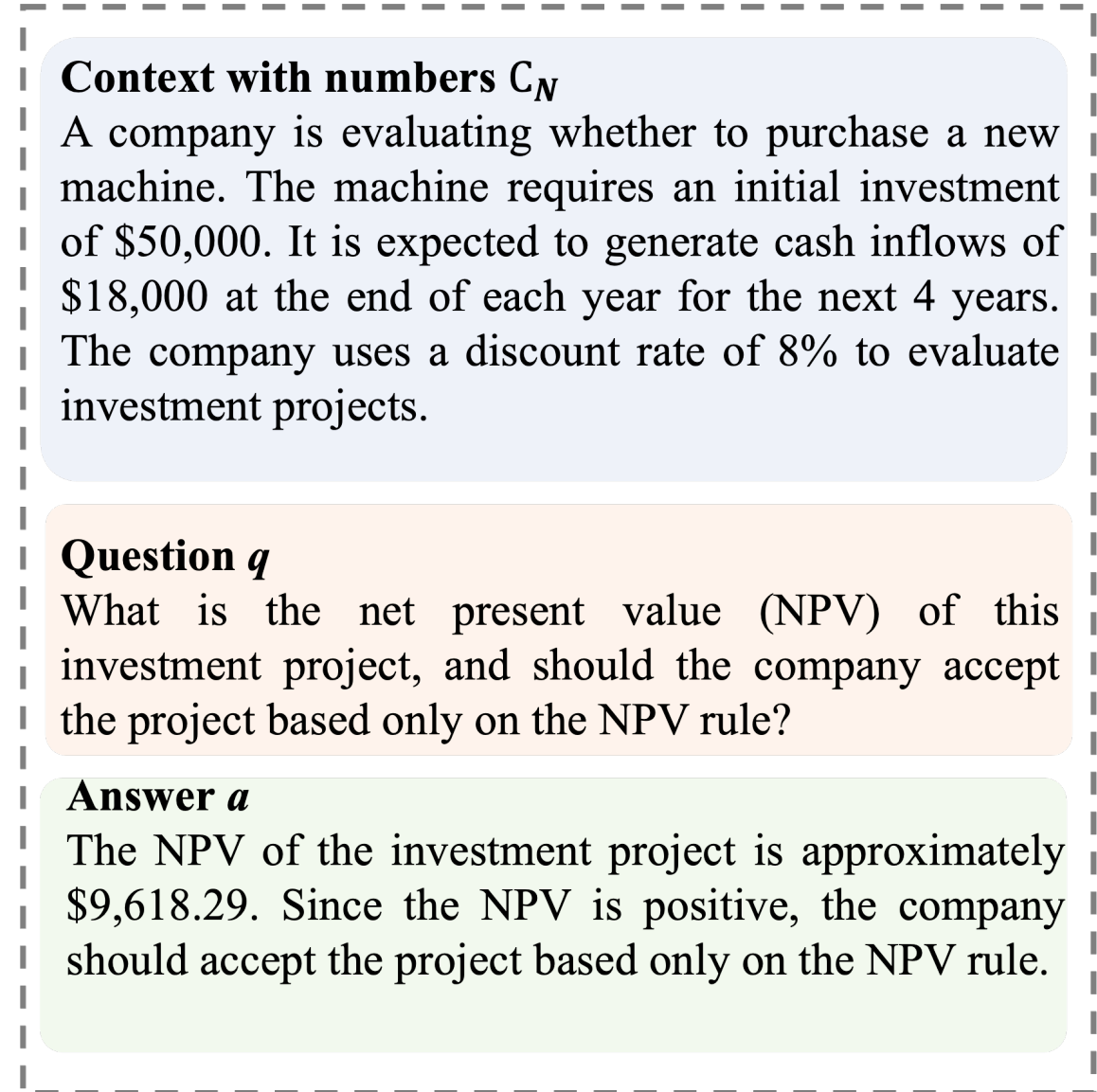


Figure 2.4: A financial numerical reasoning question-answering example.

tion. Tables are compact but difficult: the meaning of a cell depends on headers, row labels, units, footnotes, hierarchy, and surrounding text.

TAT-QA is a key benchmark for hybrid tabular-textual financial reasoning [203]. It requires models to identify supporting table cells and text spans before applying symbolic operations. MultiHiertt further introduces hierarchical structures, multiple tables, and longer textual contexts [191]. These benchmarks show that financial table reasoning is not merely table lookup. It often requires understanding accounting hierarchies, subtotal relationships, and cross-period comparisons.

Tabular reasoning also overlaps with document reasoning. In financial reports, tables are embedded in long documents and may be separated from explanatory notes. A question may require values from one table and contextual constraints from another section. This makes table-text integration essential. A model that extracts a value from a cell without understanding the reporting period, unit, or footnote may produce an answer that is numerically precise but financially invalid.

The main challenge in financial tabular reasoning is structural grounding. Financial tables are not generic grids. They often contain hierarchical row labels, multi-level headers, negative values expressed with parentheses, omitted repeated units, and subtotals that depend on accounting conventions. A model must understand that a value under a year column belongs to a specific period, that a subtotal may include several child rows, and that a footnote may change interpretation. These structural requirements explain why table reasoning errors often appear as calculation errors even when the underlying problem is evidence alignment.

Another challenge is table-text interaction. Management discussion, accounting notes, or surrounding captions may specify the business event, segment, currency, or accounting policy that gives the table its meaning. A model that treats the table as self-contained may miss these constraints. Conversely, a model that relies on text alone may ignore the exact numbers in the table. Reliable financial tabular reasoning therefore requires both structural precision and contextual interpretation.

2.4.3 Financial Document Reasoning

Financial document reasoning evaluates whether models can answer questions grounded in long financial reports, filings, disclosures, or other documents. It involves long-context understanding, retrieval, cross-section reasoning, attribution, and document-grounded generation. The central challenge is that evidence may be buried in long, complex, and heterogeneous documents.

FinanceBench evaluates open-book financial QA over public company documents and provides evidence strings for answers [67]. It is influential because it emphasizes realistic document-grounded financial questions and reveals that strong models can still generate unsupported answers. DocFinQA extends financial reasoning to full-document contexts, forcing models to locate relevant evidence before performing FinQA-style reasoning [128]. FinTextQA and FinLFQA extend the setting toward attributed long-form financial QA, where answers must be explanatory and grounded rather than only short or numerical [22, 100].

Document:
(... abbreviate...)
The following table presents product and service sales and operating expenses by segment (dollar in millions):

hierarchical row headers		hierarchical column headers			
		Year Ended December 31			
		2018		2017	
Segment		Sales	Expenses	Sales	Expenses
Innovation Systems					
Product		2,894	2,582	—	—
Service		382	351	—	—
Aerospace Systems					
Product		11,087	9,889	10,064	8,988
Service		2,009	1,796	2,067	1,854
Mission Systems					
Product		7,329	6,335	7,012	6,088
Service		4,380	3,854	4,458	3,940
Technology Service					
Product		485	450	391	360
Service		3,812	3,404	4,296	3,878

Product sales for 2018 increased \$4.3 billion, or 25 percent, as compared with 2017. The increase was primarily due to the addition of \$2.9 billion of product sales from Innovation Systems and higher restricted and F-35 volume at Aerospace Systems.
(... abbreviate...)

The table below reconciles funds provided to each segment (dollar in millions):

Segment	2018	2017	% Change
	Funded	Funded	
Innovation Systems	5,928	—	—
Aerospace Systems	11,448	9,560	19.7 %
Mission Systems	9,676	9,277	4.3 %
Technology Services	2,883	2,792	3.3 %

Approximately \$26.6 billion of the \$53.5 billion total at December 31, 2018 is expected to be converted into sales in 2019. (... abbreviate...)

Question: In 2018, what was the total sales increase in the segment with most funds in 2017?

Answer: 965

Numerical expression: $(11087 - 10064) + (2009 - 2067)$

Figure 2.5: An example of financial tabular reasoning from MultiHiertt [191].

Retrieval-centric benchmarks further highlight the importance of evidence acquisition. FinDER evaluates financial QA and retrieval-augmented generation using query-evidence-answer triplets from real filings [30]. FinSearchComp and FinAgentBench move toward realistic financial search and agentic retrieval, where models must decide where to search and which evidence to use [63, 31]. These benchmarks show that document reasoning is not only about reading long contexts; it is about finding, grounding, and using the right evidence.

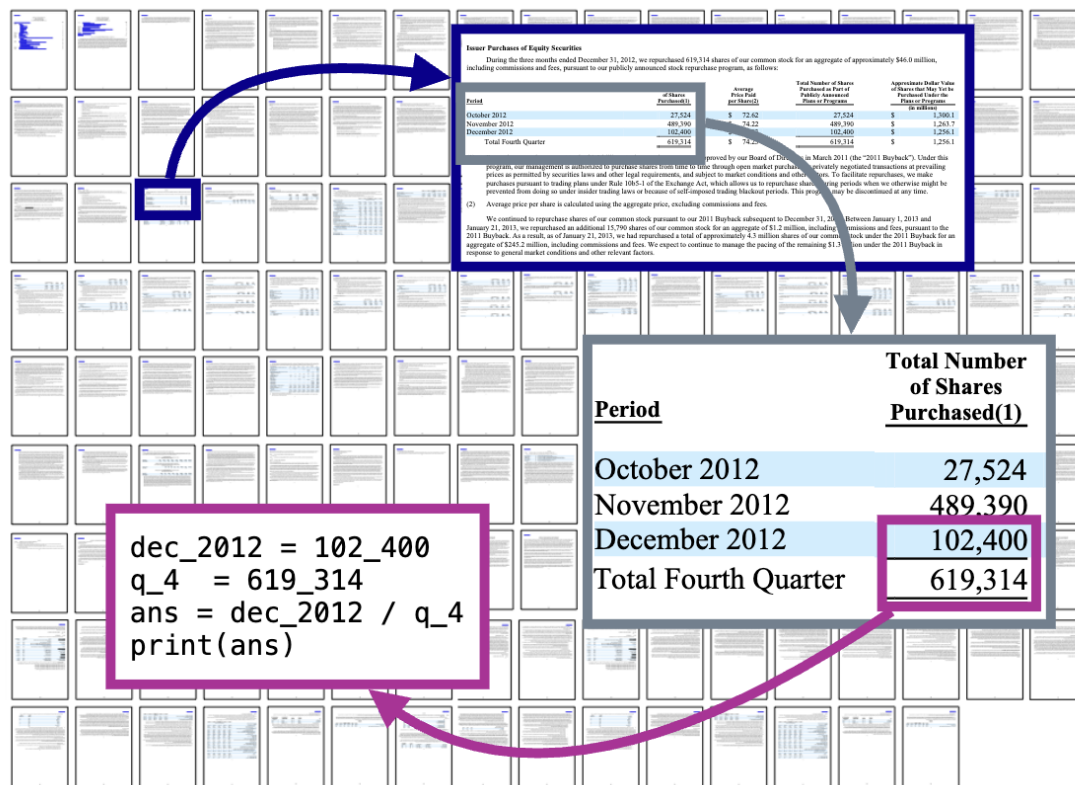


Figure 2.6: An example of document reasoning from DocFinQA [128].

Financial document reasoning also raises the problem of attribution. In high-stakes settings, an answer should be traceable to the document region that supports it. A generated explanation that cites an irrelevant passage is unreliable even if the final answer is plausible. Attribution is especially difficult when the answer requires combining information from several sections, such as a filing note, a financial statement table, and a management discussion paragraph. The model must retrieve evidence, preserve its context, and avoid overstating what the source says.

Long-context models do not remove the need for evidence acquisition. Even if an entire filing fits into a context window, the model must still identify the relevant parts and ignore distractors. Long context can also increase the risk of confusion when similar line items, years, or entities appear repeatedly. This is why retrieval and document reasoning should be evaluated together with reasoning-step correctness and evidence faithfulness.

2.4.4 Financial Multimodal Reasoning

Financial multimodal reasoning evaluates whether models can integrate financial text with charts, report figures, visual tables, screenshots, document images, and other visual evidence. It includes chart interpretation, report-figure understanding, visual table reasoning, document-image reasoning, multi-image reasoning, and cross-modal alignment.

FAMMA is an early multilingual multimodal benchmark for financial domain question answering [169]. It includes financial images and evaluates whether models can answer questions that require visual understanding and financial knowledge. FinMMR focuses on multimodal financial numerical reasoning, requiring models to extract quantitative information from visual inputs and perform financial calculations [141]. FinMME and MME-Finance extend financial multimodal evaluation to broader visual forms, including charts, market visualizations, diagrams, and expert-level financial images [102, 53]. General expert multimodal benchmarks such as MMMU include finance-related categories, but they are not designed specifically to cover the breadth of financial reasoning [181].

Multimodal financial reasoning is especially important for this thesis. Many financial artifacts are created for human visual inspection. Charts encode trends and comparisons; visual tables encode structure; document images preserve layout; screenshots and figures contain labels and annotations. A model must not only describe what is visible, but also identify which visual elements are relevant, align them with the question and textual context, and apply financial knowledge. This is why Chapter 3 focuses on knowledge-intensive multimodal financial reasoning.

Multimodal financial reasoning differs from generic visual question answering in the density of visual information. A chart is not a natural image; it is a compact encoding of quantities, axes, labels, legends, units, and relationships. A document image is not only a picture of text; it preserves layout, section hierarchy, and table structure. A financial diagram may require conceptual interpretation rather than object recognition. These properties make visual perception only the first step. The model must transform visual elements into financial variables and then reason over them.

Multi-image reasoning adds another layer of difficulty. A question may require comparing two charts, using one figure to identify a trend and another to identify a value, or reconciling evidence across several pages. In this setting, the model must decide which image to inspect, whether evidence from one image is sufficient, and how evidence from multiple images should be integrated. Existing benchmarks provide

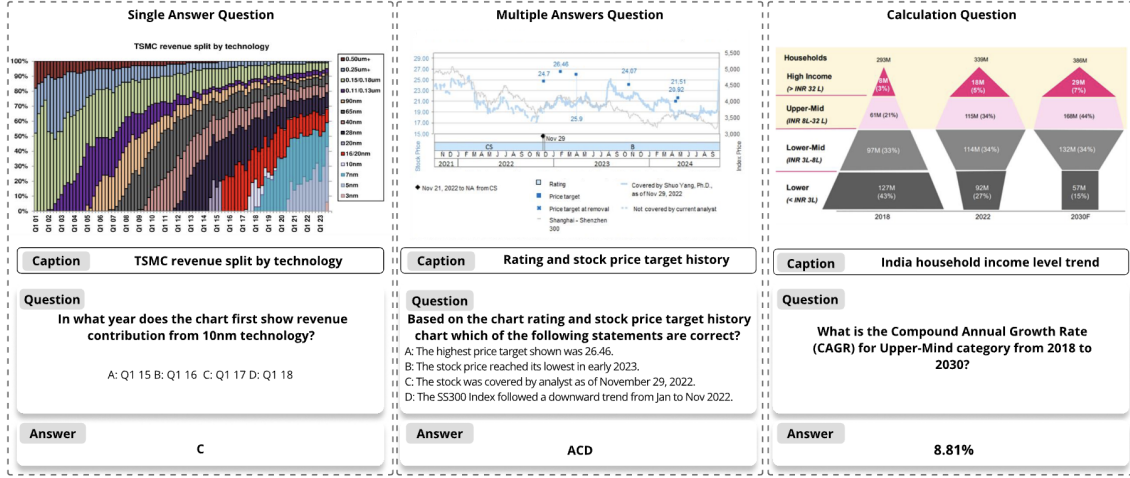


Figure 2.7: Representative examples of different question types from the FINMME dataset [102].

limited coverage of this capability, which directly motivates the FinMR benchmark design.

2.4.5 Financial Conversational Reasoning

Financial conversational reasoning evaluates whether models can answer financial questions across multiple turns. It requires dialogue history, reference resolution, intermediate-result reuse, clarification, and multi-turn financial analysis. The challenge is not only remembering previous turns, but also maintaining consistency among evidence, assumptions, and calculations.

ConvFinQA extends financial numerical reasoning into a conversational setting over financial reports [27]. Later questions may depend on previous answers, so the model must resolve references such as "it", "that number", or "the previous year" and reuse intermediate calculations. PACIFIC and Tab-CQA extend financial conversational QA toward proactive clarification and tabular financial conversations [40, 96]. Emerging multi-turn multimodal settings further connect conversation with visual evidence and agentic reasoning [183].

Conversational reasoning is relevant to reliable financial reasoning because real users often ask follow-up questions, refine information needs, and request justification. A model that answers each turn independently may lose context or contradict earlier computations. Reliable systems should therefore support clarification, reuse of intermediate results, and inspection of the evidence behind each turn.

Paper Abbrev.	Year	Venue Abbrev.	Language	Size	Task Label	Evaluation Method Used
FinQA [25]	2021	EMNLP 2021	English	8,281	FNR, FTR	PoT
TAT-QA [203]	2021	ACL 2021	English	16,552	FTR, FNR	PoT; Training
ConvFinQA [27]	2022	EMNLP 2022	English	14,115	FCR, FNR, FTR	Prompt; PoT
MultiHiertt [191]	2022	ACL 2022	English	10,440	FTR, FNR, FDR	PoT; RAG; Training
PACIFIC [40]	2022	EMNLP 2022	English	19,008	FCR, FTR, FDR, FNR	PoT; Training
FinanceBench [67]	2023	arXiv	English	10,231	FDR	Prompt; RAG
NUFR [9]	2023	arXiv	English	8,281	FNR, FTR, FDR	RAG; Training/Finetuning
Tab-CQA [96]	2023	ACL Industry 2023	Chinese	109,089	FCR, FTR, FNR	Training
PIXIU [165]	2023	NeurIPS 2023	English	–	FDR, FNR	Prompt; Training/Finetuning
DocFinQA [128]	2024	ACL 2024	English	7,437	FDR, FTR, FNR	Prompt; PoT; RAG; Training/Finetuning
FAMMA [169]	2024	arXiv	English, Chinese, French	1,935	FMR, FNR	Prompt; PoT; RAG; Agent; Training/Finetuning; Verification
FinanceMath [192]	2024	ACL 2024	English	1,200	FNR	Prompt; PoT; RAG
EEDP [135]	2024	ACL Findings 2024	English	–	FNR, FDR, FTR	Prompt
BizBench [81]	2024	ACL 2024	English	–	FNR	Prompt; PoT
FinTextQA [22]	2024	ACL 2024	English	1,262	FDR	RAG; Verification
FinMMR [141]	2025	ICCV 2025	English, Chinese	4,300	FMR, FNR	Prompt; PoT; RAG
FinanceReasoning [139]	2025	ACL 2025	English	2,238	FNR	Prompt; PoT
FinLFQA [100]	2025	EMNLP Findings 2025	English	1,008	FDR	RAG; Reflection; Verification
FinDER [30]	2025	ICAIF 2025	English	5,703	FDR	RAG
FinSearchComp [63]	2025	arXiv	English, Chinese	635	FDR, FNR	RAG; Agent
FinAgentBench [31]	2025	ICAIF 2025	English	3,429	FDR	RAG; Agent; Training/Finetuning
FinMME [102]	2025	ACL 2025	English, Chinese	11,099	FMR, FNR	Prompt; Verification
MME-Finance [53]	2025	ACM MM 2025	English	4,751	FMR, FNR	Prompt; Verification
FCMR [79]	2025	ACL 2025	English	2,199	FMR, FDR, FNR	Prompt; RAG
VisFinEval [98]	2025	EMNLP 2025	English	15,848	FMR, FNR	Prompt
FinAuditing [153]	2025	arXiv	English	1,102	FDR, FTR, FNR	Prompt; RAG; Verification
FinMMDocR [140]	2026	AAAI 2026	English, Chinese	1,200	FMR, FDR, FNR	Prompt; PoT; RAG
FinMTM [183]	2026	arXiv	English, Chinese	11,133	FCR, FMR, FNR	RAG; Agent
FinMathBench [61]	2026	AAAI 2026	English	946	FNR	Prompt
TaxReasoning [64]	2026	AAAI 2026	English	940	FNR, FDR	Prompt; RAG
FinLongDocQA [154]	2026	arXiv	English	7,527	FDR, FTR, FNR	PoT; RAG; Agent; Verification
BankMathBench [85]	2026	arXiv	Korean, English	13,839	FNR	PoT; Training/Finetuning
FinChain [166]	2026	ACL 2026	English	–	FNR	PoT; Verification
FinRule-Bench [107]	2026	arXiv	English	–	FTR, FNR	Prompt; Verification
Financial-PoT [189]	2026	arXiv	Chinese	–	FNR	PoT; Reflection; Verification
FinSheet-Bench [127]	2026	arXiv	English	–	FTR, FNR	Prompt
FinRetrieval [77]	2026	arXiv	English	500	FDR	RAG; Agent
FIND [34]	2026	arXiv	English	18,900	FMR, FNR	Training/Finetuning; Verification
FinDocMRE [206]	2026	arXiv	English	12,207	FMR, FDR, FNR	Prompt; Verification
FIRE [185]	2026	arXiv	Mainly Chinese	17,000	FDR, FNR	Prompt
FinTradeBench [3]	2026	arXiv	English	1,400	FNR, FDR	Prompt; RAG; Reflection; Verification
Hedge-Bench [29]	2026	arXiv	English	102	FDR, FNR	Agent; Verification
FinBoardBench [65]	2026	arXiv	English	–	FCR, FNR	Agent
UniFinEval [173]	2026	arXiv	Chinese, English	3,767	FMR, FDR	Prompt
EDINET-Bench [136]	2026	ICLR 2026	Japanese	3,485	FDR, FTR, FNR	Prompt

Table 2.1: Comparison of representative financial reasoning benchmarks. Task labels include financial numerical reasoning (FNR), financial tabular reasoning (FTR), financial document reasoning (FDR), financial conversational reasoning (FCR), and financial multimodal reasoning (FMR).

Conversation also changes the error profile. An error in an early turn may propagate into later turns, especially when a later question reuses an earlier result. A model may answer a follow-up question using an inconsistent denominator, assume that a pronoun refers to the wrong financial item, or fail to revise an earlier assumption when new information appears. These problems are process-level rather than answer-level. They show why reliability in financial reasoning must include traceability across turns and not only correctness of isolated answers.

2.4.6 Comparative Analysis of Benchmarks

Table 2.1 restores the benchmark-oriented comparison from the source survey. It summarizes representative resources by publication year, venue, language, size, task label, and evaluation method used. The table is not intended to be exhaustive. Its purpose is to show how benchmark design choices shape what can be evaluated.

The comparison reveals three patterns. First, financial reasoning benchmarks have expanded from text and tables toward documents, visual evidence, and interaction.

Financial report:
... the total income tax benefit recognized for share-based compensation in the accompanying statements of income is also presented.

	2010	2009	2008
share-based compensation cost	\$18.10	\$14.60	\$13.80
income tax benefit	-\$6.30	-\$5.20	-\$4.90

Conversational QA:

Q1: In the year of 2010, what was the share-based compensation cost?
A1: 18.1

Q2: and what was the income tax benefit?
A2: -6.3

Q3: what was, then, the sum of both?
A3: $\text{add}(18.1, -6.3) = 11.8$

Q4: and what was that sum in 2009?
A4: $\text{add}(14.6, -5.2) = 9.4$

Q5: what, then, was the change in the sum of those amounts from 2009 to 2010?
A5: $\text{add}(18.1, -6.3), \text{add}(14.6, -5.2), \text{subtract}(\#0, \#1) = 2.4$

Figure 2.8: A conversational reasoning example from ConvFinQA [27]: each question may depend on previous questions to answer.

Second, evidence annotation is uneven: some benchmarks provide programs or supporting facts, while others mainly provide final answers. Third, current benchmarks often specialize in one aspect of reasoning, such as table arithmetic, document retrieval, or visual QA. This specialization supports controlled evaluation but makes it difficult to diagnose end-to-end reliability.

The comparison also shows why benchmark size is not a sufficient indicator of evaluation quality. A large dataset can still have narrow modality coverage, weak financial knowledge demands, or limited process annotation. Conversely, a smaller expert-designed benchmark may be valuable if it isolates a difficult capability, such as knowledge-intensive financial mathematics or evidence-grounded document QA. For the purposes of this thesis, the most important benchmark properties are whether the dataset exposes multimodal evidence use, financial expertise, mathematical reasoning, topic coverage, and fine-grained error patterns.

Another important dimension is interaction. Most benchmarks remain single-turn and static. Conversational and agentic benchmarks introduce dialogue history, search actions, or tool use, but these are still less common than conventional QA formats. This matters because evidence acquisition in real financial analysis is often iterative. The system may not know in advance which document, figure, or formula is needed. Benchmarks that provide all context upfront may therefore underestimate the difficulty of adaptive evidence acquisition.

2.4.7 Limitations of Existing Benchmarks

Existing benchmarks provide only partial coverage of the capabilities needed for reliable multimodal financial reasoning. The most direct limitation for this thesis is that multimodal financial benchmarks remain relatively limited compared with text, table, and document benchmarks. Many datasets evaluate financial reasoning without visual evidence, while many multimodal datasets include financial examples only as a small subset. As a result, current evaluation does not sufficiently test how MLLMs align charts, report figures, visual tables, document images, textual context, and financial knowledge.

Multi-image reasoning is also underrepresented. Real financial analysis often requires comparing several figures, using a chart together with a table, or integrating visual evidence across document pages. Many existing multimodal benchmarks center on a single image or a localized visual context. This leaves open whether models can reason across multiple financial images and decide which visual source contains the decisive evidence.

Financial expertise coverage is another limitation. Some benchmarks emphasize arithmetic or table lookup, while professional financial reasoning often requires concepts from valuation, risk management, corporate finance, accounting, investment analysis, and financial reporting. Mathematical reasoning and financial expertise reasoning are not always separated, making it difficult to determine whether a model fails

because it cannot compute or because it lacks the relevant financial concept. Topic coverage is also uneven, which limits fine-grained diagnosis.

Finally, existing benchmarks often provide limited process-level annotation and quality control for reliability analysis. Many datasets evaluate final answers without sufficient labels for evidence selection, formula validity, reasoning-step correctness, or error category. Contamination risk and annotation quality are also concerns for public benchmarks [159]. These limitations motivate FinMR in Chapter 3, which is designed to evaluate knowledge-intensive, multi-topic, single-image and multi-image financial reasoning with fine-grained analysis of mathematical and financial expertise capabilities.

These limitations should not be read as shortcomings of individual datasets alone. Each benchmark makes design trade-offs. A dataset focused on executable numerical programs may not include visual evidence. A document benchmark may prioritize retrieval realism over step-level reasoning labels. A multimodal benchmark may emphasize broad visual coverage over detailed financial formulas. The gap is at the field level: existing resources do not yet provide a unified view of knowledge-intensive multimodal financial reasoning. FinMR is positioned as a response to this field-level gap, not as a replacement for all prior benchmarks.

2.5 Methods for Financial Reasoning

Methods for financial reasoning can be organized into four paradigms: prompting and program-based methods; retrieval-augmented and evidence-grounded methods; agent-based and interactive methods; and training-, reflection-, and verification-based methods. These paradigms are not mutually exclusive. A financial reasoning system may use retrieval to select evidence, a program to perform calculation, an agent to plan evidence acquisition, and a verifier to check the result. The categories are useful because they reveal which part of the reliability pipeline a method tries to improve.

Table 2.2 restores the method-oriented comparison from the source survey and summarizes representative approaches according to benchmark usage, task label, and method category.

2.5.1 Prompting and Program-Based Methods

Prompting methods attempt to elicit reasoning from a model through instructions, examples, or structured output formats. Chain-of-Thought prompting asks the model

Paper Abbrev.	Year	Venue Abbrev.	Representative Benchmark(s) Used	Task Label	Method Category
RegHNT [86]	2022	NAACL 2022	TAT-QA	FNR, FTR	PoT; Training
Zero-shot FinQA [121]	2023	arXiv	FinQA; ConvFinQA; TAT-QA	FDR, FTR, FNR	Prompt; PoT; Verification
PIXIU [165]	2023	NeurIPS 2023	FinQA; ConvFinQA	FDR, FNR	Prompt; Training/Finetuning
FSLLMs [122]	2024	EMNLP Findings 2024	FinQA; ConvFinQA; TAT-QA	FDR	PoT; Training/Finetuning
FinQAPT [133]	2024	ICAIF 2024	FinQA	FDR	Prompt; RAG
Multi-Agent QA Framework [49]	2024	ICAIF 2024	FinQA; ConvFinQA; TAT-QA	FDR	Agent; Reflection
TAT-LLMs [205]	2024	ICAIF 2024	FinQA; TAT-QA; TAT-DQA	FTR, FNR	Prompt; PoT; Training/Finetuning
BRS [73]	2025	EMNLP Industry 2025	IndicFinNLP Task 2	FDR, FNR	Training/Finetuning
Data-Driven Prompt Tuning [177]	2025	arXiv	Synthetic financial tables and documents	FTR, FDR, FNR	Prompt; Reflection; Verification
MultiFinRAG [56]	2025	arXiv	Custom SEC EDGAR multimodal corpus	FMR, FDR, FNR	Prompt; RAG
FinSage [151]	2025	CIKM 2025	FinanceBench; Custom 75-question financial-filings QA set	FDR	RAG; Training/Finetuning
Optimizing Retrieval Strategies for FinQA [78]	2025	arXiv	FinDER; FinQA-Bench; FinanceBench; TAT-QA; FinQA	FDR, FTR, FNR	RAG; Training/Finetuning
Metadara-RAG [33]	2025	arXiv	FinanceBench	FDR	Prompt; RAG
Role-Aware Multi-Agent Framework [201]	2025	ICAIF 2025	Study.com Finance QA; Six-finance-textbook corpus	FDR	Prompt; RAG; Agent; Reflection
David vs. Goliath [94]	2025	EMNLP Findings 2025	CodeFinQA; CodeTAT-QA; FinanceMATH; ConvFinQA(E); TAT-QA(E)	FDR	Agent
Fin-R1 [99]	2025	arXiv	Fin-R1-Data; FinQA; ConvFinQA	FNR	Training/Finetuning
DianJin-R1 [207]	2025	arXiv	FinQA	FNR, FDR	Training/Finetuning; Verification
FEVO [119]	2025	arXiv	FEVO	FNR, FDR	Training/Finetuning; Verification
FinPoT [76]	2025	EMNLP 2025	FinQA; ConvFinQA	FNR, FTR, FDR	Prompt; PoT; RAG
Financial-PoT [189]	2026	arXiv	Cognitive Complexity Benchmark	FNR	PoT; Reflection; Verification
Structure First, Reason Next [111]	2026	arXiv	FinQA	FNR, FTR, FDR	RAG
MoCA-Agent [1]	2026	arXiv	FinQA; FinanceMath; MultiHiort	FNR	PoT; Agent; Reflection; Verification
FinAgent-RAG [132]	2026	arXiv	FinQA; ConvFinQA; TAT-QA	FDR	PoT; RAG; Agent; Training; Verification
Fin-PRM [197]	2026	IJCAI 2026	CFLUE; FinQA; Math500	FNR	Training/Finetuning; Verification
FinAcumen [58]	2026	arXiv	BizBench; FinMMR; FinTMMBench; FinMME	FMR, FNR	RAG; Agent; Reflection; Verification
Online Financial QA [20]	2026	KDD 2026	FinQA; ConvFinQA	FNR, FDR	PoT; RAG; Agent; Training; Verification
Neuro-Symbolic Financial Reasoning [2]	2026	arXiv	FinQA; TAT-QA; FinanceBench	FNR, FDR	PoT; RAG; Agent; Training; Reflection; Verification
TaxReasoning [64]	2026	AAAI 2026	TaxReasoning	FNR, FDR	Prompt; RAG
FinLongDocQA [154]	2026	arXiv	FinLongDocQA	FDR, FTR, FNR	PoT; RAG; Agent; Verification
BankMathBench [85]	2026	arXiv	BankMathBench	FNR	PoT; Training/Finetuning
FinChain [166]	2026	ACL 2026	FinChain	FNR	PoT; Verification
FinRule-Bench [107]	2026	arXiv	FinRule-Bench	FTR, FNR	Prompt; Verification
FinSheet-Bench [127]	2026	arXiv	FinSheet-Bench	FTR, FNR	Prompt
FinRetrieval [77]	2026	arXiv	FinRetrieval	FDR	RAG; Agent
FIND [34]	2026	arXiv	FinVQA	FMR, FNR	Training/Finetuning; Verification
FinDocMRE [206]	2026	arXiv	FinDocMRE	FMR, FDR, FNR	Prompt; Verification
FIRE [185]	2026	arXiv	FIRE	FDR, FNR	Prompt
FinTradeBench [3]	2026	arXiv	FinTradeBench	FNR, FDR	Prompt; RAG; Reflection; Verification
Hedge-Bench [29]	2026	arXiv	Hedge-Bench	FDR, FNR	Agent; Verification
FinBoardBench [65]	2026	arXiv	FinBoardBench	FCR, FNR	Agent
UniFinEval [173]	2026	arXiv	UniFinEval	FMR, FDR	Prompt
EDINET-Bench [136]	2026	ICLR 2026	EDINET-Corpus; EDINET-Bench	FDR, FTR, FNR	Prompt

Table 2.2: Comparison of representative methods for financial reasoning.

to generate intermediate reasoning steps [156]. In financial reasoning, this can help expose how the model identifies values, selects operations, and reaches an answer. However, free-form rationales are not guarantees of correctness. A fluent explanation can hide an incorrect formula, a wrong value, or an arithmetic mistake.

Program-based methods address numerical precision by separating reasoning from execution. Program-of-Thought and program-aided methods ask the model to generate executable code or symbolic operations, then delegate calculation to an external interpreter [23, 54]. FinQA and TAT-QA established this direction in financial QA by using programs, evidence tagging, or symbolic operators to represent the reasoning process [25, 203]. FinanceMath and FinanceReasoning further show the value of solution references and executable reasoning for knowledge-intensive financial mathematical problems [192, 139].

The strength of this paradigm is executability. If the program is correct, numerical execution can be exact, reproducible, and easier to inspect than a free-form rationale. This is important for financial reasoning because users need to know not only the final value but also which values and operations produced it. Program supervision can also support process-level evaluation by comparing generated operations with gold solutions.

The limitation is that correct execution does not imply correct financial logic. A model may generate executable code that uses the wrong variables, the wrong reporting period, or a financially inappropriate formula. The program may faithfully compute an answer that is invalid because the evidence was misread or the domain concept was misunderstood. Prompting and program-based methods therefore improve numerical reliability but do not fully solve evidence selection, cross-modal alignment, or financial expertise errors.

Prompting methods also face reproducibility and controllability issues. Results can depend on prompt wording, examples, output formatting, model version, temperature, and parsing rules. In finance, small formatting differences can change whether a model reports a raw number, a percentage, or a rounded value. Program-based methods reduce some ambiguity, but they introduce their own failure modes, including invalid code, unsupported assumptions, and mismatches between natural-language reasoning and executable operations. These limitations indicate that prompting and programs should be combined with evidence grounding and verification rather than treated as complete solutions.

2.5.2 Retrieval-Augmented and Evidence-Grounded Methods

Retrieval-augmented and evidence-grounded methods improve reasoning by providing external evidence. This paradigm builds on open-domain retrieval and retrieval-augmented generation methods such as Dense Passage Retrieval, REALM, Retrieval-Augmented Generation, and Fusion-in-Decoder [75, 60, 87, 68]. In finance, this is essential because relevant information is often document-specific, company-specific, time-sensitive, or too long to fit reliably in a model’s internal memory. Retrieval can provide report passages, table snippets, filing sections, definitions, or supporting financial knowledge. Evidence grounding can also make answers more inspectable.

FinanceBench, DocFinQA, FinDER, FinSearchComp, and FinAgentBench illustrate the movement from fixed contexts toward retrieval-intensive financial reasoning [67, 128, 30, 63, 31]. These benchmarks show that retrieval is not merely preprocessing. It is part of reasoning because the model must determine what evidence is needed, where it may be located, and whether the retrieved information is sufficient. FinTextQA and FinLFQA further emphasize long-form answer generation with attribution, where source grounding and citation faithfulness become central [22, 100].

Document and multimodal retrieval introduce additional challenges. A retrieved passage may be relevant but incomplete. A document image may contain the correct table, but the model must still locate the correct row, period, and unit. A chart may

include labels or legends that are easy to overlook. A financial report may describe several periods or entities in adjacent sections. Reporting-period mismatch is therefore a recurring risk: the model retrieves a plausible source but applies it to the wrong year, quarter, company, or line item.

The main limitation of this paradigm is that relevant retrieval does not equal valid reasoning. A system may retrieve the right document but misunderstand the evidence. It may cite a passage that supports only part of the answer. It may combine evidence from incompatible periods or sources. It may produce citations that look faithful but do not actually support the generated claim. Retrieval-augmented financial reasoning therefore requires evidence precision, grounding, citation correctness, and reasoning-step validation, not only high retrieval recall.

Retrieval-augmented systems also face the problem of evidence sufficiency. A retrieved passage may be relevant but not enough to answer the question. For example, a document section may identify the relevant line item but not the denominator needed for a ratio. A chart may show a trend but not the exact value needed for calculation. A formula may be retrieved without the required variables. A reliable system should be able to recognize such insufficiency and acquire additional evidence. This requirement directly motivates the step-wise evidence acquisition problem addressed in Chapter 4.

2.5.3 Agent-Based and Interactive Methods

Agent-based and interactive methods decompose reasoning into roles, tools, and iterative interactions. General agent frameworks such as ReAct, Toolformer, Tree-of-Thoughts, CAMEL, AutoGen, generative agents, and multi-agent debate show that LLMs can interleave reasoning, actions, tool calls, search, and critique [175, 130, 174, 88, 157, 120, 47]. In finance, this paradigm is attractive because complex tasks may require a planner, retriever, calculator, visual interpreter, financial expert, critic, and verifier.

Financial agent systems and benchmarks increasingly reflect this direction. Multi-agent reflection can use critic agents to inspect financial QA outputs [49]. Role-aware systems can separate generation, retrieval, and expert review [201]. FinAgent-RAG combines iterative retrieval, reasoning, program modules, and self-verification for financial document QA [132]. MoCA-Agent combines multi-agent interaction with code-based reasoning for financial and numerical QA [1]. FinAgentBench evaluates agentic retrieval, making document selection and evidence localization explicit [31].

The value of agents is not simply that there are multiple prompts. Their potential value lies in specialization and adaptive communication. A planner can decide what

evidence is missing; an image agent can inspect visual evidence; a context agent can retrieve textual evidence; a knowledge agent can supply financial definitions or formulas; a reasoning agent can integrate the evidence. This is directly relevant to SEA in Chapter 4, where the focus is step-wise evidence acquisition.

However, role assignment does not automatically produce genuine specialization. A system may label agents as analyst, critic, or expert while all agents rely on similar prompts and make correlated errors. Fixed workflows may be inefficient or inappropriate when different questions require different evidence. Many systems lack step-wise evidence queries, explicit evidence sufficiency checks, and principled stopping conditions. Agent communication is also rarely evaluated systematically. Comparisons often focus on final-answer Accuracy, leaving uncertainty about whether gains come from better evidence use, more reasoning attempts, majority voting, or increased computation. These limitations motivate the query-based and evidence-driven design of SEA.

A further limitation is cost. Multi-agent systems may improve performance by increasing the number of model calls, retrieval operations, or tool executions. In high-stakes applications, this cost may be acceptable if it improves reliability; in routine analysis, it may be excessive. Evaluation should therefore report not only accuracy but also call counts, token cost, latency, and statistical significance. The SEA study includes these analyses because agentic improvement is meaningful only when the accuracy gain can be interpreted alongside computational cost and communication efficiency.

2.5.4 Training-, Reflection-, and Verification-Based Methods

Training-, reflection-, and verification-based methods aim to improve reasoning reliability through model adaptation, feedback, or checking. Training-based methods include continued pretraining, instruction tuning, supervised fine-tuning, reinforcement learning, and domain-specific reasoning supervision. Instruction tuning and reinforcement learning from human feedback show how general LLMs behavior can be aligned with user instructions before domain adaptation [118]. Financial LLMs such as FinGPT and PIXIU/FinMA show that domain adaptation can improve financial tasks [171, 165]. More specialized reasoning datasets and models aim to teach models financial formulas, numerical problem solving, and document-grounded QA.

Verification-based methods check whether reasoning steps, retrieved evidence, executable programs, or final answers are valid. Process reward models evaluate intermediate steps rather than only final answers, which is especially relevant for financial

reasoning because the same final answer can be produced by valid or invalid reasoning [93, 197]. Benchmarks such as FinChain further emphasize verifiable chain-of-thought financial reasoning with executable traces [166]. Verification can detect arithmetic inconsistency, unsupported evidence, and some formula errors.

Reflection-based methods ask a model or another component to diagnose and revise errors. Self-Refine and Reflexion illustrate the general idea of iterative self-feedback and memory [106, 131]. In financial reasoning, reflection can help identify wrong values, missing evidence, or invalid intermediate steps. However, self-reflection may reproduce the model’s own bias. A model that chose the wrong formula may rationalize it; a model that misread a chart may not discover the error by rereading its generated text.

The limitations of this paradigm motivate CLER in Chapter 5. First, final-answer feedback is too coarse for diagnosing financial reasoning failures. Second, high-quality step-level financial error resources are scarce. Third, self-reflection does not provide independent error signals when the same model generates and critiques the reasoning. Fourth, cross-model error complementarity has not been sufficiently exploited. CLER addresses these issues by retrieving structured step-level errors produced by diverse MLLMs and using them to guide reflection and revision.

Reflection and verification also differ in their assumptions. Verification asks whether a claim, step, program, or answer is valid according to evidence or rules. Reflection asks how the system should revise its reasoning after identifying a potential error. A verifier may detect that a calculation is inconsistent, but the reasoner must still decide what should be corrected. Conversely, a reflection prompt may encourage revision without strong evidence that a specific step is wrong. Financial reasoning benefits from connecting these two ideas: reflection should be informed by concrete error evidence, and verification should operate at the level where errors occur.

2.6 Evaluation of Financial Reasoning Systems

Evaluation determines what kinds of reliability can be observed. Financial reasoning systems are often evaluated at the answer level, but answer-level metrics alone cannot show whether a model used correct evidence, selected a valid formula, or followed a faithful reasoning process. A comprehensive evaluation should therefore consider answer-level, program-level, evidence-level, process-level, and reliability-level metrics.

Answer-level metrics include Accuracy, Exact Match, F1, and numerical tolerance. Accuracy is common for multiple-choice and classification-style tasks. Exact Match is

used when the answer must match a number, span, or short phrase. F1 is useful when answers are textual or span-based. Numerical tolerance is important because financial answers may involve rounding, percentages, currency units, or decimal precision. These metrics are easy to report and compare, but they provide limited diagnostic information.

Program-level metrics evaluate whether the generated reasoning program or executable solution is correct. Program accuracy measures whether the generated program matches a reference program or has the correct structure. Execution accuracy measures whether executing the program produces the correct answer. These metrics are useful in benchmarks such as FinQA, FinanceMath, FinanceReasoning, and FinChain [25, 192, 139, 166]. However, execution accuracy can still hide financial logic errors if the program uses wrong variables or assumptions.

Evidence-level metrics evaluate retrieval and grounding. Retrieval recall measures whether the system retrieves the gold evidence. Evidence precision measures whether the retrieved set is focused rather than noisy. Grounding measures whether the answer is supported by the retrieved or provided evidence. Citation correctness evaluates whether cited sources actually support the claims they accompany. These metrics are important for document and multimodal financial reasoning, where the answer should be traceable to reports, filings, tables, figures, or knowledge sources.

Process-level metrics evaluate the reasoning trace. They include reasoning-step correctness, error localization, formula validity, operation validity, evidence consistency, and cross-modal alignment. Process-level evaluation is more expensive than final-answer scoring because it often requires expert annotation or structured traces. Nevertheless, it is crucial for reliability. A model may obtain the right final answer through an invalid path, or it may fail because one intermediate step corrupted the rest of the reasoning.

Reliability-level metrics evaluate behavior under realistic constraints and perturbations. Robustness measures stability when irrelevant context, noisy evidence, or alternative visual forms are introduced. Perturbation tests examine whether removing or changing evidence affects the answer appropriately. Calibration evaluates whether confidence aligns with correctness. Cost and latency matter for agentic systems because additional calls and retrieval steps may improve Accuracy while reducing practical usability. Human evaluation assesses usefulness, faithfulness, and interpretability when automatic metrics are insufficient. Statistical significance tests, including bootstrap confidence intervals, McNemar’s test, Friedman test, and Wilcoxon signed-rank test, help determine whether performance differences are meaningful rather than incidental.

Table 2.3: Evaluation levels for financial reasoning systems.

Level	Typical Metrics	What It Reveals
Answer-level	Accuracy, Exact Match, F1, numerical tolerance	Whether the final output is correct.
Program-level	Program accuracy, execution accuracy	Whether executable reasoning is structurally or computationally valid.
Evidence-level	Retrieval recall, evidence precision, grounding, citation correctness	Whether the answer is supported by relevant evidence.
Process-level	Step correctness, error localization, formula validity, evidence consistency	Where reasoning succeeds or fails.
Reliability-level	Robustness, perturbation, calibration, cost, latency, human evaluation, statistical significance	Whether the system is stable, inspectable, and practically reliable.

The choice of metric should match the research question. If the question concerns model ranking, overall Accuracy and confidence intervals may be appropriate. If the question concerns numerical reasoning, numerical tolerance and execution accuracy are necessary. If the question concerns evidence acquisition, retrieval recall, evidence precision, grounding, and perturbation tests are more informative. If the question concerns error correction, step-level correctness and human validation become important. A mismatch between research question and metric can produce misleading conclusions. For example, an agentic system may improve answer Accuracy while using much more evidence and cost; a retrieval system may improve recall while reducing precision; a reflection system may change answers without improving reasoning validity.

Financial reasoning evaluation also needs consistent reporting conventions. Model names, versions, access dates, prompts, output parsing rules, dataset splits, and statistical tests should be reported clearly. This is especially important for commercial MLLMs, whose behavior may change over time. Without version information and reproducible prompts, performance comparisons across studies can be difficult to interpret. The later technical chapters therefore emphasize experimental setup, model selection, evaluation metrics, and statistical analysis.

The literature indicates that financial reasoning evaluation should move beyond final-answer Accuracy. Accuracy remains necessary, but it is not sufficient for high-stakes domains. Reliable financial reasoning requires evidence-grounded, process-aware, statistically supported, and human-inspectable evaluation. This motivates the

fine-grained evaluation in FinMR, the evidence perturbation and cost analysis in SEA, and the step-level human validation in CLER.

2.7 Synthesis of the Literature

2.7.1 From Answer Prediction to Evidence-Grounded Financial Intelligence

The first major finding is that financial reasoning is moving from answer prediction toward evidence-grounded financial intelligence. Early financial NLP and QA systems often emphasized predicting a label, extracting a span, or producing a short answer. Recent benchmarks increasingly require systems to identify the supporting evidence, retrieve relevant document regions, interpret tables and figures, and justify the answer. FinanceBench, DocFinQA, FinDER, FinTextQA, and FinLFQA exemplify this transition in document reasoning [67, 128, 30, 22, 100].

This transition changes the meaning of performance. A correct answer is not enough if the model cannot show where the evidence came from or if its explanation is unsupported. Evidence-grounded financial intelligence requires a system to connect sources, reasoning steps, and conclusions. It also requires evaluation protocols that can distinguish a lucky answer from a justified answer. This finding directly motivates FinMR’s emphasis on capability diagnosis and error analysis.

The transition also changes the role of benchmarks. If the objective is answer prediction, a benchmark can be designed mainly around question-answer pairs and final labels. If the objective is evidence-grounded intelligence, the benchmark must represent the evidence structure that makes the answer valid. This includes the modality of evidence, the location of evidence, the financial concept that links the evidence to the question, and the reasoning operation that transforms evidence into an answer. Benchmarks that omit these dimensions may still be useful for measuring surface performance, but they provide weaker support for diagnosing reliability.

Evidence grounding is especially important because financial information is often institutionally produced and auditable. Annual reports, financial statements, filings, and professional materials are designed to support accountability. When an LLMs produces a financial answer without traceable evidence, it breaks this accountability chain. For research purposes, this makes evidence grounding a methodological requirement. It allows model errors to be categorized as evidence selection errors, knowledge

errors, computation errors, or answer synthesis errors. Without evidence grounding, many of these errors collapse into a single incorrect-answer label.

This finding also reveals why financial reasoning differs from many open-domain QA tasks. In open-domain QA, a model may answer a factual question from general knowledge or a retrieved passage. In finance, the source, period, entity, and context are often part of the answer’s validity. The same formula or ratio may be interpreted differently depending on whether the evidence comes from a consolidated statement, segment disclosure, market chart, or analyst note. Evidence-grounded financial intelligence therefore requires both source-level grounding and financial-context grounding.

2.7.2 From Static Inputs to Dynamic Multimodal Interaction

The second major finding is that financial reasoning is moving from static inputs toward dynamic multimodal interaction. Traditional benchmarks often provide a fixed context and ask for a single answer. However, realistic financial reasoning is often iterative. A user asks a question, the system identifies missing evidence, retrieves a document, inspects a chart, applies a formula, and may need to revisit earlier assumptions. Conversational benchmarks such as ConvFinQA and PACIFIC show the importance of dialogue history and clarification [27, 40]. Agentic benchmarks and systems such as FinAgentBench and FinAgent-RAG show the importance of tool use and iterative retrieval [31, 132].

Multimodality strengthens this shift. When evidence is distributed across text, tables, charts, and document images, the model cannot assume that all relevant information is already in a convenient textual context. It must decide which modality to inspect and how to integrate the result. Existing systems often use fixed pipelines, but fixed pipelines may not fit questions with different evidence needs. This finding motivates SEA, which treats evidence acquisition as a step-wise and query-driven part of reasoning.

The shift from static inputs to dynamic interaction is not only a user-interface issue. It reflects a deeper reasoning problem: the evidence required to answer a question may be unknown until the model begins reasoning. A first step may reveal that a denominator is missing. A visual inspection may reveal that a chart uses a secondary axis. A retrieved formula may reveal that the model needs values from two different periods. Static contexts assume that all relevant evidence has already been selected, whereas dynamic interaction allows the system to discover evidence needs during reasoning.

This is particularly important for multimodal financial reasoning because modalities are not interchangeable. A textual context may state the financial concept, while the chart provides the values. A table may provide exact numbers, while the surrounding prose explains an accounting change. A document image may preserve layout that is lost in extracted text. A model that passively consumes a fixed concatenation of all inputs may not know which source to trust or inspect. Dynamic interaction provides a mechanism for targeted evidence acquisition and evidence sufficiency assessment.

The literature therefore suggests that reliable multimodal financial reasoning should be evaluated as an adaptive process. The question is not only whether a model can answer after receiving all context, but whether it can identify what evidence is missing, ask for the appropriate source, integrate the response, and stop when evidence is sufficient. This process view links benchmark design to agent design. FinMR exposes cases where multimodal evidence is required; SEA then investigates how evidence can be acquired and coordinated step by step.

2.7.3 From Local Verification to End-to-End Reliability

The third major finding is that financial reasoning reliability cannot be established through local verification alone. Program execution can verify arithmetic but not necessarily financial logic. Retrieval metrics can verify whether a relevant passage was found but not whether it was interpreted correctly. A critic model can identify some inconsistencies but may share the generator’s blind spots. Local components are useful, yet financial reasoning failures often arise from interactions among perception, evidence selection, knowledge, formula or operation selection, computation, inference, and synthesis.

End-to-end reliability therefore requires process-level evaluation and reflection. A model should be assessed on whether it uses the right evidence, applies the right financial concept, performs the correct operation, and produces a conclusion that can be verified. It should also be able to learn from errors at the step level. This finding motivates CLER, which treats errors produced by different MLLMs as reusable resources for reflection and revision.

Local verification remains useful, but its scope must be understood. Execution accuracy verifies that a generated program produces the expected output, but it does not verify that the program is financially appropriate. Retrieval recall verifies that evidence was found, but it does not verify that the model used it correctly. A citation may point to a relevant document while the generated claim overstates or misinterprets the source. A self-critic may detect inconsistencies in language but miss a domain-

specific assumption. Each local verifier therefore covers only part of the reliability problem.

End-to-end reliability requires connecting local checks across the reasoning pipeline. Visual perception should be checked against the image; evidence selection should be checked against the question; formula or operation selection should be checked against financial knowledge; numerical computation should be checked against the selected variables; and answer synthesis should be checked against the full reasoning trace. The challenge is that errors interact. A wrong value extracted from a chart can make a correct formula produce a wrong answer. A correct computation can support a misleading conclusion if the financial interpretation is wrong.

The literature also shows that reflection should be treated as evidence-sensitive rather than purely introspective. Self-reflection can improve style and sometimes catch obvious inconsistencies, but it may not introduce new information. For financial reasoning, useful reflection should point to concrete error patterns: a likely wrong variable, a mismatched period, an inappropriate formula, an unsupported citation, or a visual misreading. CLER builds on this insight by using step-level errors from other MLLMs as external reflection evidence.

Taken together, the three findings indicate that financial reasoning is a composite capability. It is not reducible to language understanding, retrieval, calculation, visual perception, or verification alone. A system can perform well on one component and still fail overall because the components depend on one another. This composite structure is the reason the thesis does not propose a single all-purpose solution. Instead, it studies evaluation, evidence acquisition, and error reflection as complementary layers of reliability.

The findings also establish a direction of increasing difficulty. Evidence-grounded intelligence asks whether systems can support answers with verifiable sources. Dynamic multimodal interaction asks whether systems can acquire the right evidence when it is not already organized for them. End-to-end reliability asks whether systems can detect and correct failures that occur across the whole process. This progression mirrors the thesis structure: the literature review defines the field, FinMR evaluates the capability, SEA improves evidence acquisition, and CLER improves error reflection.

2.8 Research Gaps

The literature points to three research gaps that structure the technical chapters of this thesis.

The gaps are interdependent. The evaluation gap concerns what can be measured; the evidence acquisition gap concerns what information the system can obtain and coordinate; and the error reflection gap concerns how the system can respond when reasoning goes wrong. Weakness in one area can hide or amplify weakness in another. If evaluation is too coarse, evidence acquisition errors may appear only as answer errors. If evidence acquisition is weak, reflection may operate over an already corrupted reasoning state. If reflection is weak, a system may acquire relevant evidence but still preserve invalid reasoning.

The three gaps also map onto different stages of the reasoning pipeline introduced in Chapter 1. Evaluation concerns whether the field can observe failures across evidence, understanding, reasoning, answer generation, and verification. Evidence acquisition concerns the transition from evidence to understanding. Error reflection concerns the transition from reasoning to verification and revision. The gaps therefore do not represent unrelated weaknesses in the literature; they are different manifestations of the same reliability problem.

For this reason, the thesis treats Chapter 2 as a foundation rather than as a fourth technical contribution. Its role is to make the conceptual structure of the problem explicit. Once the gaps are identified, the technical chapters can be read as targeted responses: FinMR makes capability limitations measurable; SEA makes evidence acquisition adaptive; and CLER makes step-level errors available for reflection.

2.8.1 Evaluation Gap

Existing benchmarks provide insufficient coverage of knowledge-intensive, multi-topic, and multi-image financial reasoning and offer limited fine-grained analysis of mathematical and financial expertise capabilities. This gap matters because strong performance on general multimodal benchmarks or financial text benchmarks does not necessarily demonstrate reliable financial reasoning. Chapter 3 addresses this gap by introducing FinMR, a benchmark designed to evaluate knowledge-intensive multimodal financial reasoning across mathematical and financial expertise questions, single-image and multi-image inputs, and multiple financial topics.

The evaluation gap has several dimensions. The first is modality coverage. Textual and tabular benchmarks have matured more quickly than multimodal financial

Table 2.4: Research gaps identified in the literature and thesis responses.

Gap	Literature	Limita- tion	Thesis Response	Reliability Component
Evaluation gap	Limited coverage of knowledge-intensive, multi-topic, and multi-image financial reasoning; limited fine-grained capability analysis.		FinMR evaluates mathematical and financial expertise reasoning across multimodal financial topics.	Appropriate evaluation
Evidence acquisition gap	Fixed contexts and pre-determined workflows do not adapt to missing modality-specific evidence.		SEA uses specialized agents and query-based communication for step-wise evidence acquisition.	Relevant evidence
Error reflection gap	Reflection is often self-based, answer-level, and weakly supported by structured financial error resources.		CLER retrieves cross-model step-level errors to guide reflection and reasoning revision.	Effective reflection

benchmarks, while general multimodal benchmarks only partially cover finance. As a result, current evaluations often understate the difficulty of interpreting charts, report figures, visual tables, and document images in financial contexts. The second dimension is image composition. Many multimodal tasks use one image at a time, whereas realistic financial analysis may require comparing several figures or combining a chart with a visual table. The third dimension is domain coverage. A benchmark may include financial-looking data but still provide limited coverage of valuation, risk, reporting, investment, derivatives, or other specialized topics.

The fourth dimension is reasoning-type coverage. Mathematical reasoning and financial expertise reasoning are related but not identical. A mathematical question may require calculation over well-specified values; an expertise question may require selecting a concept, interpreting a financial relationship, or applying professional knowledge. If a benchmark merges these categories without analysis, it becomes difficult to know whether a model’s weakness is arithmetic, formula or operation selection, financial knowledge, or evidence alignment. The fifth dimension is diagnostic granularity. Overall Accuracy cannot show whether a model fails because it misreads an image, retrieves the wrong evidence, applies an invalid formula, or synthesizes the final answer poorly.

This gap directly motivates the design of FinMR. FinMR is not only a new dataset; it is an attempt to make hidden capability limitations observable. Its use of 3,200 multimodal questions, 15 topics, mathematical and expertise reasoning categories, and single-image and multi-image settings is designed to evaluate the specific capabilities that the literature leaves under-tested. Its error analysis also connects benchmark results to the reliability pipeline used throughout the thesis.

Addressing the evaluation gap is foundational because improvement claims depend on what is measured. If a benchmark does not test knowledge-intensive visual evidence use, then a method can appear reliable while still failing in the setting that matters for this thesis. If evaluation does not separate reasoning types, then improvements cannot be attributed to better evidence use, better calculation, or better financial knowledge. FinMR therefore provides the diagnostic basis for the later SEA and CLER studies.

2.8.2 Evidence Acquisition Gap

Existing systems often rely on fixed contexts or predetermined workflows and lack adaptive mechanisms for identifying evidence gaps and acquiring modality-specific evidence step by step. This gap matters because multimodal financial reasoning depends on heterogeneous evidence that may be incomplete, distributed, or modality-specific. Chapter 4 addresses this gap through SEA, a step-wise multi-agent framework that uses specialized agents and query-based communication to acquire visual, textual, and financial knowledge evidence dynamically.

The evidence acquisition gap arises from a common assumption in benchmark and system design: the relevant context is either given upfront or retrieved before reasoning begins. This assumption is convenient, but it does not match many financial reasoning problems. A model may not know that a chart value is missing until it tries to compute a ratio. It may not know that a financial definition is needed until it identifies the topic. It may not know that the textual context is insufficient until the visual evidence introduces another condition. Fixed pipelines are poorly suited to this type of dependency because they decide evidence flow before the reasoning state is known.

The gap is also modality-specific. Visual evidence, textual context, and financial knowledge require different extraction strategies. An image agent may need to inspect labels, axes, legends, or layout. A context agent may need to retrieve a relevant passage or summarize a report section. A knowledge agent may need to provide a formula, definition, or domain rule. A single undifferentiated model prompt may blur

these functions, making it difficult to determine whether an error came from evidence acquisition or reasoning synthesis.

Existing agent-based systems begin to address this problem, but the literature shows several limitations. Role decomposition does not guarantee that agents ask the right questions. Debate does not guarantee that missing evidence is acquired. Controller-based systems may still follow fixed action sequences. Many systems lack explicit evidence sufficiency criteria, so they may stop too early or continue generating unnecessary calls. Agent communication protocols are also rarely compared systematically, leaving open whether improvements come from specialization, more compute, or better evidence routing.

SEA is designed as a response to this gap. It treats evidence acquisition as a step-wise reasoning activity rather than a preprocessing stage. The planner identifies evidence gaps; specialized agents return modality-specific evidence; the reasoning agent integrates the evidence; and termination depends on evidence sufficiency. This design operationalizes the literature’s movement from static contexts to dynamic multimodal interaction.

The evidence acquisition gap also creates a bridge between Chapter 3 and Chapter 4. FinMR diagnoses that MLLMs fail not only because they lack general intelligence, but also because they use evidence incompletely or inappropriately. SEA then asks whether a structured multi-agent system can reduce those failures by acquiring the right evidence at the right step.

2.8.3 Error Reflection Gap

Existing reflection methods are dominated by self-reflection and answer-level correction, with limited use of structured step-level errors from diverse MLLMs. This gap matters because even correct evidence does not guarantee correct reasoning, and final-answer feedback is too coarse to diagnose financial reasoning failures. Chapter 5 addresses this gap through FinErrorSet and CLER, which retrieve cross-model step-level errors and use them to guide reflection and revision.

The error reflection gap is visible in how many systems treat errors. If evaluation only records the final answer, then the error signal is binary and coarse. The model knows that the answer is wrong but not whether the failure occurred during visual perception, evidence selection, formula or operation selection, computation, logical inference, or answer synthesis. This is particularly problematic in finance because different errors require different corrections. A computation error may be fixed with

recalculation; a formula error requires conceptual correction; an evidence selection error requires revisiting the source; a visual error requires reinspection of the image.

Self-reflection is limited because the model often reflects from the same representation that produced the error. If the model has already misread the evidence or selected the wrong formula, it may not generate feedback that challenges that assumption. In addition, self-reflection is often answer-level: it asks the model to reconsider the whole response rather than diagnose a specific step. This can lead to unnecessary changes when the reasoning is mostly correct, or insufficient changes when the error is localized but subtle.

Structured step-level error resources offer a different path. If errors are collected from multiple MLLMs and categorized by reasoning step, they can serve as external memory. A new reasoning trace can be compared with prior errors to identify similar failure patterns. Cross-model errors are especially valuable because different MLLMs may have different strengths and weaknesses. One model’s mistake can warn another model about a likely trap, provided that the retrieved error is relevant to the current reasoning step.

CLER addresses this gap by constructing FinErrorSet and using cross-MLLMs error retrieval for reflection and revision. This connects the literature on process supervision, reflection, and verification to the financial setting. It also complements SEA: even after evidence acquisition improves, models may still make reasoning errors. CLER targets these remaining errors by making step-level failures explicit and reusable.

The error reflection gap completes the thesis progression. Chapter 3 asks how to evaluate capability limitations; Chapter 4 asks how to improve evidence acquisition; Chapter 5 asks how to improve reasoning after evidence has been acquired. Together, the three gaps show that reliability requires more than a better base model. It requires evaluation, evidence, reasoning validity, and reflection.

2.9 Chapter Summary

This chapter reviewed financial reasoning with LLMs and established the conceptual foundation for the thesis. It defined financial reasoning as a domain-specific process that integrates financial knowledge, heterogeneous evidence, numerical or logical operations, and contextual interpretation to derive justified and decision-relevant conclusions. It also explained why financial reasoning is distinct from general language understanding: it depends on domain knowledge, heterogeneous evidence, numerical

and structural precision, temporal and contextual sensitivity, decision orientation, and high reliability requirements.

The chapter organized financial reasoning tasks into five categories: financial numerical reasoning, financial tabular reasoning, financial document reasoning, financial multimodal reasoning, and financial conversational reasoning. It then reviewed four methodological paradigms: prompting and program-based methods, retrieval-augmented and evidence-grounded methods, agent-based and interactive methods, and training-, reflection-, and verification-based methods. The evaluation discussion showed that answer-level Accuracy must be complemented by program-level, evidence-level, process-level, and reliability-level metrics.

The synthesis identified three broad movements in the literature: from answer prediction to evidence-grounded financial intelligence, from static inputs to dynamic multimodal interaction, and from local verification to end-to-end reliability. These movements lead to three research gaps. Chapters 3–5 address these gaps through benchmarking, step-wise multi-agent evidence acquisition, and cross-model error reflection.

The chapter therefore positions the thesis as a coherent progression. FinMR responds to the need for comprehensive and fine-grained evaluation. SEA responds to the need for adaptive acquisition and integration of heterogeneous evidence. CLER responds to the need for step-level error reflection beyond self-correction. The remainder of the thesis develops and evaluates these three responses.

FinMR: Benchmarking Knowledge-Intensive Multimodal Financial Reasoning

Chapter 2 identified an evaluation gap in financial reasoning research: existing benchmarks only partially measure knowledge-intensive, multi-topic, and multimodal financial reasoning, and they provide limited fine-grained evidence about mathematical and financial expertise capabilities. This chapter addresses that gap by introducing FinMR, a benchmark for evaluating knowledge-intensive multimodal financial reasoning with large language models and multimodal large language models [39]. The chapter answers RQ1: *How can the knowledge-intensive multimodal financial reasoning capabilities of large language models be comprehensively evaluated?*

3.1 Motivation and Chapter Overview

Existing financial reasoning benchmarks have established important foundations for numerical question answering, table-text reasoning, document-grounded question answering, and multimodal visual understanding. However, Chapter 2 showed that the field still lacks a benchmark that jointly stresses professional financial knowledge, mathematical reasoning, visual evidence interpretation, topic diversity, and fine-grained diagnostic analysis. This limitation matters because a model that performs well on a general multimodal benchmark may still fail when it must read a financial chart, identify the relevant variable, choose a domain-specific formula, and synthesize a verifiable answer.



Rich Contextual Information

Enhanced Depth of Financial Knowledge

Diverse Visual Images

High-Quality Annotations

Advanced Reasoning

Diverse Topics

Figure 3.1: Two examples of FinMR.

The central motivation of this chapter is therefore not simply to add another dataset. The goal is to make a difficult capability measurable: knowledge-intensive multimodal financial reasoning. In realistic financial analysis, evidence is rarely presented as clean text alone, and the interpretation of financial data is shaped by measurement quality, institutional reporting practices, domain terminology, and quantitative financial concepts [70, 105, 44, 104]. Analysts encounter charts, visual tables, diagrams, time-series figures, maps, and textual contexts. They must combine visual perception with financial concepts and mathematical operations. A benchmark that provides only short text, only tabular values, or only basic visual perception questions cannot expose whether MLLMs can perform this combined reasoning reliably.

FinMR is designed to address this evaluation gap. It contains 3,200 question-answer pairs across 15 financial topics, including 1,049 mathematical reasoning questions and 2,151 financial expertise questions. It also contains 3,764 images, covering both single-image and multi-image settings. Figure 3.1 shows two representative examples. These examples illustrate the kind of reasoning targeted by this chapter: the model must interpret financial context, identify relevant visual evidence, apply financial knowledge or mathematical operations, and produce a justified answer.

This chapter proceeds as follows. Section 3.2 positions FinMR relative to directly related financial reasoning benchmarks. Section 3.3 defines the FinMR task. Section 3.4 presents the benchmark design. Section 3.5 describes data collection and construction. Section 3.6 reports dataset statistics and analysis. Section 3.7 describes the experimental setup. Section 3.8 presents the main results. Section 3.9 provides fine-grained analysis. Section 3.10 analyses model errors using the thesis-wide error taxonomy. Section 3.11 discusses the implications of FinMR and answers RQ1. Section 3.12 summarizes the chapter and motivates Chapter 4.

3.2 Positioning of FinMR

FinMR is positioned at the intersection of financial numerical reasoning, financial expertise reasoning, and multimodal financial reasoning. It is most closely related to benchmarks such as FinQA, TAT-QA, MMMU-Finance, FAMMA, and recent multimodal financial benchmarks. The comparison in this section is intentionally focused. Chapter 2 already reviewed the broader benchmark landscape; this section only discusses work that clarifies what FinMR contributes.

FinQA and TAT-QA are foundational benchmarks for financial reasoning over text and tables. FinQA evaluates numerical reasoning over financial reports and provides reasoning programs that make calculations partially inspectable [26]. TAT-QA evaluates hybrid table-text reasoning and requires models to identify evidence before applying symbolic operations. These benchmarks are important because they show that financial reasoning is not reducible to span extraction. However, they are primarily text-table benchmarks. They do not systematically evaluate visual financial evidence such as charts, diagrams, maps, and report figures, nor do they cover multi-image financial reasoning.

MMMU-Finance and FAMMA are closer to FinMR because they evaluate multimodal, knowledge-intensive financial questions [180, 168]. They show that financial knowledge can be evaluated in a multimodal format, but their scope is narrower than the setting targeted in this chapter. MMMU-Finance contains a relatively small finance subset within a broader multidisciplinary benchmark. FAMMA contains multimodal financial QA examples, but its images are dominated by tables and textbook-style contexts. These resources are useful for evaluating broad multimodal financial competence, but they do not fully test topic diversity, image-type diversity, advanced mathematical reasoning, and fine-grained diagnostic performance across financial topics.

Table 3.1: Comparison of FinMR with existing financial reasoning benchmarks.

Dataset	Modality	Main Type	Image	Core Knowledge Topics	Number	Knowledge Level	Knowledge Intensive	Perception	Numerical Math?	Advanced Math?
FinQA	Text Only	-		-	8,281	College	✓	✗	✓	✗
TAT-QA	Text Only	-		-	16,552	College	✓	✗	✓	✗
MultiHierrt	Text Only	-		-	10,440	College	✓	✗	✓	✗
FinanceMath	Text Only	-		7	1,200	College	✓	✗	✓	✗
Fin-Fact	Text & Image	News Illustrations		-	3,369	-	✗	✓	✗	✗
FinSet	Text & Image	Charts		-	100	-	✗	✓	✗	✗
Financial VQA	Text & Image	Bar Chart		-	100	-	✗	✓	✓	✗
MME-Finance	Text & Image	Charts, Screen-shots		-	2,274	-	✗	✓	✗	✗
FinMME	Text & Image	Charts, Flow, Timeseries		3	11,099	-	✗	✓	✓	✗
MMMU-Finance	Text & Image	Tables		3	490	College	✓	✓	✓	✗
FAMMA	Text & Image	Tables		6	1,945	College	✓	✓	✓	✗
FinMR	Text & Image	Charts, Tables, Diagram, Timeseries, Maps		15	3,200	College	✓	✓	✓	✓

Other multimodal financial datasets, such as FinVQA, FinSet, Fin-Fact, MME-Finance, and FinMME, further expand the visual side of the field [134, 14, 125, 52, 103]. These datasets often focus on visual perception, chart understanding, or broad financial image recognition. They are valuable because they expose visual limitations of MLLMs. Yet many of them do not require advanced financial expertise or complex mathematical reasoning at the level of professional financial analysis. This distinction is important for the thesis: a system may visually identify a chart but still fail to perform financial reasoning over it.

Table 3.1 summarizes the positioning of FinMR. Compared with the most relevant benchmarks, FinMR combines professional financial knowledge, diverse image types, mathematical reasoning, perception reasoning, and broad topic coverage. This combination is designed to make the evaluation gap identified in Chapter 2 directly observable.

3.3 Task Definition

FinMR evaluates multimodal financial reasoning in a question-answering format. Each example consists of a financial question q , a textual context C , one or more images $\mathcal{I} = \{I_1, \dots, I_m\}$, relevant financial knowledge K , an answer a , and a reasoning requirement R . The model receives the question, textual context, and visual evidence, and must produce the correct final answer:

$$\hat{a} = f(q, C, \mathcal{I}, K).$$

Although K is not always provided explicitly as a separate input, it is required by many questions. For example, the model may need to know how a financial ratio is defined, how a portfolio statistic is interpreted, or which option-pricing or risk-management formula is appropriate.

The textual context C contains the surrounding information needed to understand the question. It may include a short problem statement, a longer scenario, financial definitions, numerical conditions, or explanatory context. The image set $\mathcal{I}$ contains visual evidence such as charts, tables, time-series plots, diagrams, maps, 3-D plots, or other financial figures. In single-image questions, $m = 1$ and the main challenge is locating and interpreting evidence within one visual input. In multi-image questions, $m > 1$ and the model must decide which images are relevant and how evidence across images should be combined.

The answer a is represented as a definitive answer option or value, which enables automatic accuracy-based evaluation. This design supports controlled comparison across many LLMs and MLLMs. At the same time, the dataset includes explanations that make the reasoning process inspectable. The explanation is not merely a justification after the fact; it clarifies which evidence and financial concepts are expected to support the answer.

FinMR distinguishes two major reasoning requirements. **Mathematical questions** require the model to perform numerical or formula-based reasoning. They may involve arithmetic, percentage change, discounting, probability, risk measurement, statistical inference, or other financial calculations. Solving these questions requires variable grounding, operation selection, execution, and financial interpretation. **Financial expertise questions** require the model to apply professional financial concepts, definitions, and contextual judgment. These questions may involve accounting relationships, investment principles, risk-management concepts, market mechanisms, or interpretation of financial products. They may still contain numbers, but the main challenge is domain understanding rather than calculation alone.

This task definition makes FinMR different from generic visual question answering. The image is not simply an object to be described. It is a source of financial evidence. The model must connect visual elements with textual context, financial knowledge, and the reasoning requirement. A correct answer therefore depends on the whole pipeline introduced in Chapter 1: evidence, understanding, reasoning, answer, and reflection.

3.4 Benchmark Design

FinMR was designed around four evaluation goals. First, it should evaluate knowledge-intensive multimodal financial reasoning rather than shallow financial image recognition. Second, it should cover a broad range of financial topics so that performance is not dominated by a single area such as accounting tables or market charts. Third, it should include both mathematical reasoning and financial expertise reasoning. Fourth, it should support fine-grained analysis across topics, image types, difficulty levels, and reasoning categories.

The benchmark scope is intentionally professional and education-oriented. Questions were selected from college-level and professional financial assessment materials, where answers usually require formal domain knowledge rather than common-sense inference. This scope is appropriate for a thesis on reliable financial reasoning because professional financial tasks are high-stakes, evidence-sensitive, and often formula-sensitive. A model must not only recognize that a figure is financial; it must understand what the figure means in a financial reasoning context.

The 15-topic design supports broad coverage of finance. As shown in Table 3.2, the benchmark includes Investment, Quantitative Methods, Valuation and Risk Models, Financial Markets and Products, Financial Reporting and Analysis, Portfolio Management, Fixed Income, Credit Risk, Foundation of Risk Management, Economics, Operational Risk, Derivatives, Market Risk, Corporate Finance, and Liquidity and Treasury Risk. These topics cover both general financial analysis and specialized subfields, allowing the benchmark to reveal whether models have uniform or uneven financial reasoning capability.

Knowledge intensity is built into the dataset through source selection, topic coverage, and reasoning requirements. Many questions require specialized concepts such as risk modeling, fixed-income valuation, derivatives, portfolio management, market risk, liquidity risk, and financial reporting. Mathematical questions may require multi-step formula application, while expertise questions may require conceptual distinctions that are difficult to infer from surface text alone. This design directly responds to the benchmark limitation identified in Chapter 2: existing multimodal financial datasets often contain financial-looking images without requiring deep financial reasoning.

The multimodal design also aims to avoid overfitting evaluation to a single visual form. FinMR includes statistical charts, tables, time-series plots, diagrams, 3-D plots, maps, and other images. These image types create different reasoning demands. Tables test structural and row-column grounding. Charts test axis and legend interpretation. Time-series plots test trend and temporal reasoning. Diagrams test

Table 3.2: Financial Topic Distribution of FinMR

Topics & Abbreviation	Number	Ratio
Investment (Inv)	371	11.6%
Quantitative Methods (QM)	342	10.7%
Valuation and Risk Models (VRM)	318	9.9%
Financial Markets and Products (FMP)	297	9.3%
Financial Reporting and Analysis (FRA)	264	8.3%
Portfolio Management (PM)	258	8.1%
Fixed Income (FI)	251	7.8%
Credit Risk (CR)	170	5.3%
Foundation of Risk Management (FRM)	169	5.3%
Economics (Eco)	156	4.9%
Operational Risk (OR)	131	4.1%
Derivatives (Der)	126	3.9%
Market Risk (MR)	121	3.8%
Corporate Finance (CF)	119	3.7%
Liquidity and Treasury Risk (LTR)	107	3.3%

abstract and causal financial concepts. Maps and 3-D plots test spatial or geographical interpretation. This diversity makes it possible to diagnose visual perception and cross-modal alignment errors rather than reporting only an aggregate score.

Question format is another deliberate design choice. FinMR uses questions with definitive answers, allowing consistent automatic scoring across many models. This supports reproducibility and clear leaderboard-style comparison. At the same time, the questions are paired with explanations, enabling qualitative analysis of how evidence and reasoning should support the answer. Difficulty is represented through the number and diversity of financial knowledge points involved in a question: easy questions involve one knowledge point, medium questions involve two to three, and hard questions involve more than three.

The design also reflects the thesis-wide view that reliability is a process-level property. A benchmark that only reports final answers cannot show whether a model is unreliable because it lacks financial knowledge, misreads a figure, selects irrelevant evidence, or performs an invalid operation. FinMR therefore structures examples around attributes that expose different stages of the reasoning process. Topic labels reveal domain coverage. Reasoning-type labels separate mathematical reasoning from expertise reasoning. Image-type labels reveal visual dependency. Difficulty labels indicate the number of knowledge points involved. Single-image and multi-image indicators expose whether the model is reasoning locally or integrating evidence across sources.

The distinction between knowledge intensity and visual perception is particularly important. Some multimodal financial benchmarks include many financial images, but the questions can often be answered through surface-level chart reading or visual classification. FinMR instead emphasizes cases where the visual evidence must be interpreted through a financial concept. For example, a time-series plot may require the model to understand volatility, trend reversal, interest-rate movement, or risk exposure. A diagram may require knowledge of a financial process or risk-management structure. A table may require the model to distinguish accounting categories, periods, units, or ratios. In these cases, the visual signal becomes useful only after it is connected to domain knowledge.

The benchmark also avoids treating multimodality as a single capability. Different visual forms impose different reasoning burdens. Tables are compact and structured, but they require row-column localization and period alignment. Charts require axis reading, legend matching, and trend interpretation. Diagrams often require abstract reasoning rather than numerical extraction. Maps and 3-D plots require spatial interpretation. Multi-image cases add evidence selection and synthesis. By representing these differences explicitly, FinMR enables analysis that is more informative than a single multimodal Accuracy score.

Finally, the benchmark is designed to support subsequent method development. FinMR is not only an endpoint evaluation. It produces diagnostic evidence about where model failures occur. If a model fails mainly on diagrams, evidence acquisition may need better visual decomposition. If it fails mainly on mathematical questions, formula or operation selection and numerical computation may need stronger verification. If it fails on multi-image cases, the system may need adaptive evidence routing. These design choices explain why FinMR naturally leads to the SEA and CLER studies in later chapters.

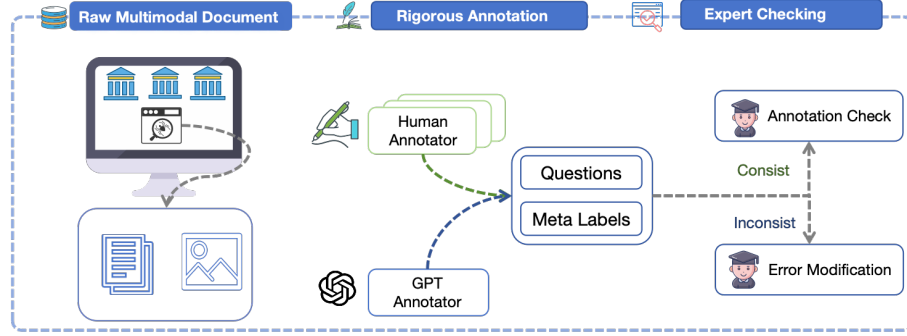


Figure 3.2: The annotation pipeline of FinMR and two examples in FinMR.

3.5 Data Collection and Construction

The construction of FinMR follows a multi-stage pipeline, shown in Figure 3.2. The pipeline was designed to preserve the authenticity of professional financial assessment materials while producing a clean, structured, and evaluable benchmark for MLLMs.

3.5.1 Data Sources and Question Collection

This study compiled financial exam papers from college-level courses and professional certification programs. The collection targeted materials from business schools that collaborate with Chartered Financial Analyst (CFA) and Financial Risk Management (FRM) programs, whose curricula are frequently used in university finance education. Questions, images, answers, and detailed explanations were extracted from past final exam papers available on the public academic resource website¹. This source choice makes FinMR different from datasets based mainly on synthetic examples or generic web images: the questions are grounded in authentic assessment contexts where financial reasoning is expected.

The initial collection step aimed for topical and modality diversity. Candidate questions were selected only when they required both financial reasoning and interpretable evidence. Questions that could be answered without financial knowledge, or that only tested generic image description, were excluded. The retained examples therefore emphasize financial concepts, calculations, charts, visual tables, or other finance-specific evidence.

¹<https://www.studocu.com/en-nz>

3.5.2 Image Extraction and Cleaning

For exam questions available in PDF format, the Mathpix API [148] was used to extract textual content, mathematical formulas, and images. Question texts and images were extracted separately so that each example could be represented in a structured format. This separation is important because multimodal financial reasoning depends on correctly aligning the question, textual context, image, answer, and explanation.

Automatic extraction required extensive cleaning. Mathpix occasionally produced misaligned or out-of-order visual materials, especially when an exam page contained multiple figures or when a table was visually embedded in a complex layout. Four annotators manually validated the alignment among questions, images, options, answers, and explanations. Images with insufficient clarity were enhanced using Image Generator Pro. Questions lacking proper explanations or containing non-English content were excluded. Questions whose images appeared only in answer options or explanations, rather than in the question context, were also excluded to keep the task format consistent.

3.5.3 Answer Verification, Duplicate Removal, and Splitting

The cleaning process produced a refined collection of 4,470 candidate QA pairs. Duplicate and near-duplicate examples were removed during manual review. This step considered repeated question stems, repeated image-question pairs, and examples that tested the same answer logic with only superficial wording changes. Removing duplicates reduces the risk that benchmark performance is inflated by memorization or repeated patterns.

Metadata labels were assigned using GPT-4o [116]. Each question was classified as either expertise-based or math-reasoning, and each was assigned a reference topic. Difficulty calibration was also performed using GPT-4o by assessing the number of distinct reasoning points or domain-specific knowledge points involved in each question. Easy questions involve a single knowledge point; medium questions involve two to three knowledge points; hard questions involve more than three knowledge points. Each question was further annotated with structured metadata, including question ID, source topic, question type, and other attributes.

Two PhD students then reviewed each explanation for validity, completeness, and clarity. Questions with insufficient or overly brief explanations were revised and expanded to include detailed reasoning steps. Grammatical and stylistic issues were also corrected. Following this expert review stage, the dataset was refined to a final collec-

tion of 3,200 high-quality QA pairs. The dataset was partitioned by topic into an 80% training/development portion and a 20% held-out test portion, following established practice in financial mathematics benchmarking [193]. The final split contains 2,560 training/development samples and 640 test samples.

The construction process also standardizes the relation among fields in each example. Each retained item contains a question, a textual context when available, one or more image references, answer options, a correct answer, an explanation, topic metadata, reasoning-type metadata, and difficulty metadata. This structured representation makes the benchmark usable by both direct MLLM evaluation and caption-based LLM evaluation. It also makes later error analysis possible because failures can be mapped back to the relevant evidence source, topic, image type, and reasoning requirement.

Quality control was intentionally conservative. Questions were removed if the source material was ambiguous, if the image could not be clearly associated with the question, if the answer could not be verified from the provided evidence and explanation, or if the reasoning depended on information outside the available context in a way that could not be fairly evaluated. This conservative filtering reduces dataset size, but it improves benchmark reliability. For a high-stakes domain such as finance, retaining ambiguous examples would make model comparison less meaningful because an incorrect response might reflect annotation noise rather than model failure.

The split design also supports fair evaluation. Because financial topics differ in difficulty and evidence structure, the held-out test set was created with topic-level coverage rather than by simply selecting a narrow subset of examples. This reduces the risk that the test set overrepresents a single domain such as investment or financial reporting. It also makes the reported topic-level analysis meaningful, because each topic remains visible in evaluation. The training/development portion can be used for prompt checking and future model development, but the results reported in this chapter are based on the held-out test set.

3.6 Dataset Statistics and Analysis

Table 3.3 summarizes the main statistical characteristics of FinMR. The benchmark contains 3,200 questions and 3,764 images. It includes 1,049 mathematical reasoning questions and 2,151 expertise reasoning questions. The split contains 2,560 training/development examples and 640 test examples. These statistics directly address the evaluation gap identified in Chapter 2: FinMR provides a larger and more sys-

Table 3.3: FinMR Benchmark Statistics

Total Questions	3200	Total Images	3764
* Test	640 (20%)	* QA/Image	2643/2643 (70%)
* Train	2560 (80%)	* QA/Images	557/1118 (30%)
Image Type		Reasoning Type	
* Chart	1393 (37%)	* Expertise	2151 (67%)
* Table	828 (22%)	* Math	1049 (33%)
* Timeseries	634 (17%)		
* Diagram	452 (12%)	Level of Difficulty	
* 3-D	188 (5%)	* Easy	1510 (47%)
* Maps	151 (4%)	* Medium	1105 (35%)
* Others	113 (3%)	* Hard	586 (18%)
Average Length			
* Context	327.74		
* Question	33.97	* Explanation	63.24

tematically categorized setting for knowledge-intensive multimodal financial reasoning than prior finance subsets or narrowly scoped visual benchmarks.

The distribution of reasoning types shows that FinMR is not a math-only benchmark. Expertise reasoning accounts for 67% of the dataset, while mathematical reasoning accounts for 33%. This balance is important because financial reasoning involves both conceptual and quantitative competence. A model may be able to compute a value from a clearly stated formula but still fail when asked to interpret a financial relationship. Conversely, a model may know a concept but fail to execute the required calculation. Separating these categories makes such differences visible.

FinMR also includes both single-image and multi-image examples. A total of 2,643 questions involve a single image, while 557 questions involve multiple images, corresponding to 1,118 images in the multi-image subset. This design addresses a key benchmark limitation from Chapter 2: many existing multimodal financial datasets center on single-image reasoning. Multi-image reasoning is important because financial analysis often requires comparing charts, connecting a visual table with a diagram, or integrating evidence across several figures.

Image-type diversity is another central property of the dataset. Figure 3.3 shows examples of the visual forms included in FinMR. Charts account for 37% of images, tables for 22%, time-series figures for 17%, diagrams for 12%, 3-D plots for 5%, maps for 4%, and other images for 3%. This distribution supports evaluation across both structured and less structured financial visuals. It also allows later analysis to ask whether model performance depends on the type of visual evidence.

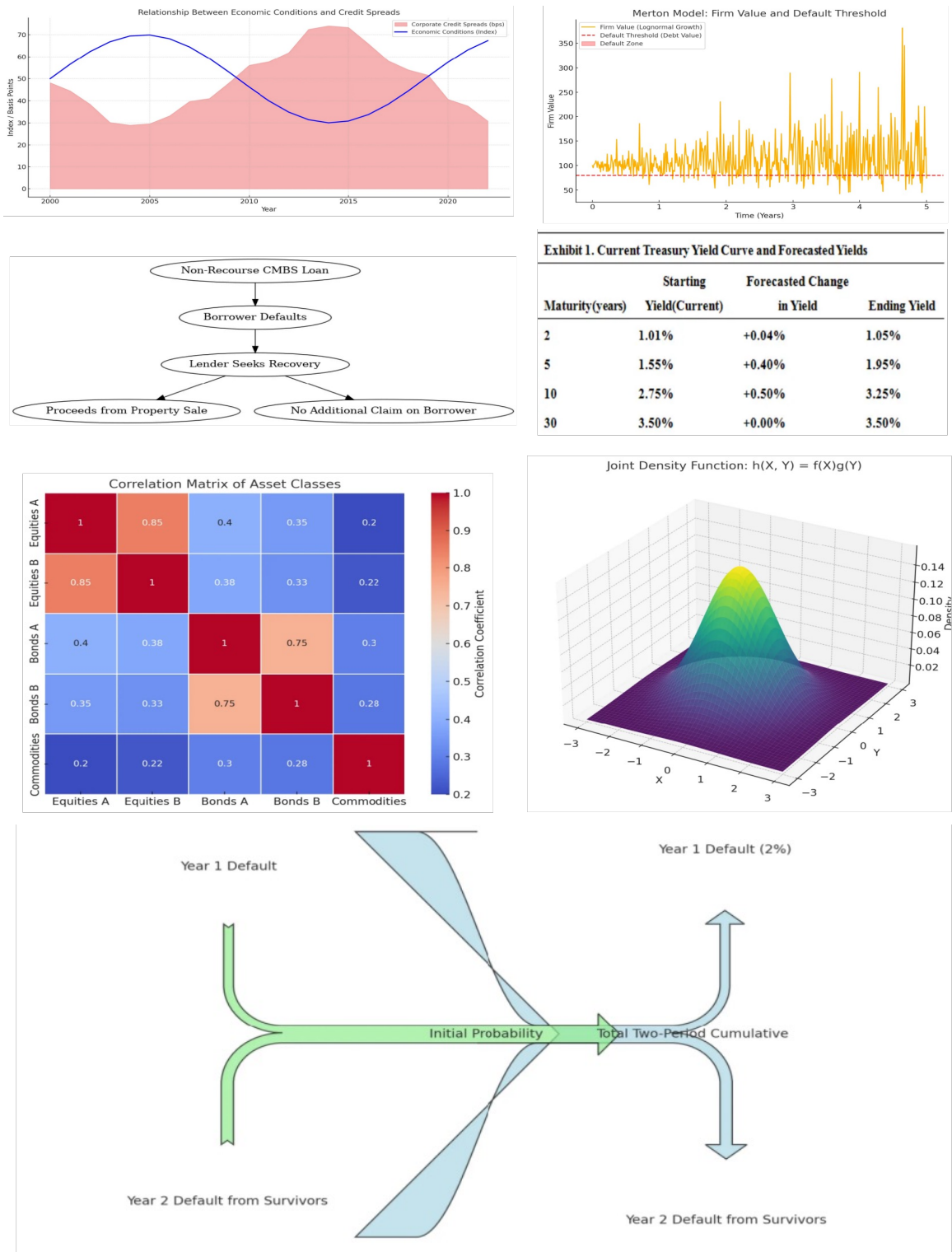


Figure 3.3: Different types of figure examples in FinMR.

The difficulty distribution further supports fine-grained evaluation. Easy questions account for 47% of the dataset, medium questions for 35%, and hard questions for 18%. This mix prevents the benchmark from being dominated by either trivial recognition or very hard expert tasks. It also creates a basis for analyzing whether models fail gradually as reasoning demands increase.

Human performance provides another indicator of benchmark difficulty. Two financial experts were invited to answer 200 randomly sampled questions from FinMR. The experts were senior instructors from institutions specializing in CFA and FRM certification training, each with over a decade of experience in financial education, examination design, and candidate assessment. The CFA expert achieved 91% accuracy, the FRM expert achieved 86% accuracy, and their average accuracy was 88.5%. Their solution process also provides useful qualitative context: experts typically identify and circle key information in the question, write concise annotations or calculations in the margins, and check whether the final answer is consistent with the financial concept being tested.

Together, these statistics show how FinMR responds to the limitations of existing benchmarks. It increases topic coverage, separates mathematical and expertise reasoning, includes diverse visual evidence, supports both single-image and multi-image settings, and provides human-validated explanations. These properties make FinMR useful not only as a leaderboard but also as a diagnostic instrument for reliable multimodal financial reasoning.

3.7 Experimental Setup

The evaluation process consists of four stages: test data input, reasoning, output, and evaluation, as shown in Figure 3.4. The input stage provides the model with the question, textual context, and either direct images or generated image captions, depending on the model type. The reasoning stage uses Chain-of-Thought prompting [155, 188], with instructions that encourage the model to reason step by step. The output stage requires the model to provide a final answer. The evaluation stage compares the parsed answer with the gold answer.

The evaluated text-only reasoning models are GPT-o1 [117], Grok-3 [162], and DeepSeek-R1 [37]. The evaluated MLLMs include GPT-4o [116], Gemini-2.5-Pro [55], Claude-3.7-Sonnet [8], Grok-2 [161], DeepSeek-VL-2 [160], Llama-4-Maverick [110], InternVL-3-78B [208], Qwen-VL-2.5-72B [12], and LLaVA-NeXT-72B [89]. These models are selected to cover frontier closed-source systems, text-only reasoning mod-

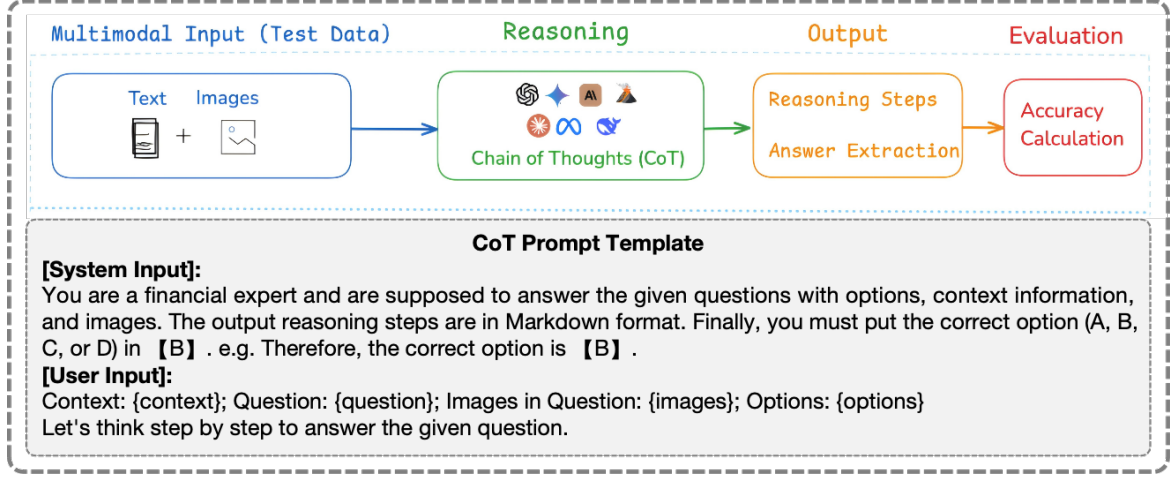


Figure 3.4: Four stages of the evaluation process.

els, and representative open-source MLLMs with direct visual input. Where public release names are available but stable dated API snapshots are not provided in the source study, this chapter reports the named model versions and treats missing access-date metadata as a reproducibility limitation rather than as evidence for cross-chapter comparison.

For LLMs without native visual input, image captions were generated using GPT-4o with the prompt: *Please describe this image in detail.* The generated descriptions were then used consistently across text-only models, following captioning strategies used in prior multimodal reasoning work such as MathVista [101]. This setup allows text-only reasoning models to be compared with direct-image MLLMs while keeping the caption source consistent.

Accuracy is the main evaluation metric because FinMR provides questions with definitive correct answers. The output parser extracts the final selected answer from each model response. When a model produces no parseable final answer, gives multiple incompatible answers, or fails to commit to a valid option, the response is counted as incorrect. This parsing rule is important for reliability because a long reasoning trace is not sufficient if the system cannot synthesize the required final answer.

The held-out test set contains 640 examples and is used for the reported model comparison. The training/development portion contains 2,560 examples and supports prompt checking, dataset inspection, and future model development. All open-source model experiments were conducted using four A100 GPUs, while closed-source model experiments were conducted through official API calls. The temperature was set to

0.7 for evaluated models. For reproducibility, the benchmark reports the model family, modality setting, prompt strategy, answer parsing rule, data split, and metric.

The evaluation protocol treats text-only LLMs and MLLMs differently because their input capabilities differ. For MLLMs, the model receives the image directly together with the question and textual context. For text-only LLMs, the model receives the question, textual context, and an image caption generated by GPT-4o. This design does not make text-only and multimodal models identical; rather, it creates a controlled comparison between caption-mediated visual access and direct visual access. The difference between these settings helps measure how much information is lost when visual evidence is converted into text before reasoning.

Output parsing is handled consistently across models. The parser searches for a final answer in the response and maps it to the expected answer format. If a model provides reasoning but no final answer, the response is counted as incorrect. If a model provides multiple contradictory final answers, the response is also counted as incorrect. This strict rule reflects the reliability requirement of the thesis. In financial applications, an answer that cannot be unambiguously interpreted is not reliable, even if parts of the reasoning appear plausible.

The use of Accuracy is appropriate for FinMR because the questions have definitive answers. However, Accuracy is not interpreted as the only reliability metric. The later analyses consider reasoning type, topic, image type, and error category. This combination of answer-level scoring and process-oriented diagnosis is important: a model can achieve the same overall Accuracy through different patterns of success and failure. One model may be stronger on expertise reasoning but weaker on mathematical questions; another may be strong on tables but weak on diagrams. The experimental setup is therefore designed to support both ranking and diagnosis.

Reproducibility is a particular concern when evaluating commercial MLLMs, since model behavior can change over time. For this reason, model names, modality settings, prompt strategy, and evaluation rules are stated explicitly. The chapter does not treat results across different model versions as permanently fixed facts, and its numerical results should not be interpreted as direct comparisons with later SEA or CLER experiments when the model versions differ. Instead, it treats FinMR as a reusable evaluation resource: as models change, the same benchmark structure can be used to remeasure knowledge-intensive multimodal financial reasoning under a consistent protocol.

Table 3.4: Reasoning performance comparison of LLMs and MLLMs on *FinMR*. The best model performance is highlighted in green (LLMs) and blue (MLLMs). The meaning of the abbreviations can be viewed from Table 3.2.

Models	Open Source?	Overall	Expertise	Math	Inv	QM	VRM	FMP	FRA	PM	FI	FRM	CR	Eco	OR	Der	MR	CF	LTR
Textual Modality: Text + Image Caption																			
DeepSeek-R1	✓	44.84	48.51	42.20	47.50	50.00	46.15	52.17	44.44	41.77	40.17	50.00	46.23	36.36	23.08	11.11	49.17	42.86	43.83
GPT-o1	✗	45.78	54.84	33.21	55.00	28.57	53.85	39.13	33.33	34.97	50.60	18.75	45.28	16.18	33.46	60.67	51.67	40.48	30.43
Grok-3	✗	21.88	22.58	20.90	30.00	21.57	19.23	13.04	33.33	11.39	31.33	25.00	18.87	18.18	7.69	44.44	20.83	22.57	15.39
Multimodality: Text + Image																			
Claude-3.7-Sonnet	✗	53.91	61.83	42.91	52.50	42.86	65.38	52.17	20.22	48.10	60.24	37.50	52.83	18.18	30.77	20.02	65.00	47.62	52.52
DeepSeek-VL-2	✓	37.81	42.47	31.34	45.00	35.71	46.15	17.39	25.22	34.18	43.37	18.75	38.68	18.18	15.38	22.62	38.33	30.95	30.43
Gemini-2.5-Pro	✗	54.76	55.11	54.28	47.50	42.86	51.92	52.17	33.33	46.84	51.19	59.60	59.43	45.45	38.46	57.00	55.00	64.29	47.83
GPT-4o	✗	50.00	50.68	47.25	60.00	42.86	44.23	52.17	33.33	53.16	46.99	60.75	50.94	45.45	23.08	55.56	49.17	50.00	62.22
Grok-2	✗	27.81	33.33	20.15	30.00	21.43	40.38	21.74	10.11	21.05	30.12	6.25	27.36	27.27	15.38	24.44	33.50	11.90	4.35
InternVL-3-78B	✓	31.87	33.06	30.22	32.50	14.29	28.85	20.43	16.22	27.85	33.73	18.75	34.91	18.18	30.77	22.22	36.33	40.38	20.74
Llama-4-Maverick	✓	27.34	30.11	23.51	27.50	21.43	23.08	39.13	0.00	25.32	36.14	14.75	23.58	10.09	15.69	11.11	32.50	35.71	21.74
LLaVA-NeXT-72B	✓	16.72	18.55	14.18	20.00	7.14	19.23	13.09	0.00	18.99	21.69	10.50	16.04	9.09	7.69	10.01	18.13	4.76	4.35
Qwen-VL-2.5-72B	✓	34.84	38.71	28.73	42.50	28.57	33.46	43.83	22.22	30.65	30.96	18.75	33.02	27.27	7.69	22.22	39.17	28.57	26.09

3.8 Main Results

Table 3.4 reports reasoning performance on FinMR. Overall, the best-performing MLLM is Gemini-2.5-Pro, with 54.76% accuracy, followed by Claude-3.7-Sonnet at 53.91% and GPT-4o at 50.00%. Among text-only models using image captions, GPT-o1 reaches 45.78%, DeepSeek-R1 reaches 44.84%, and Grok-3 reaches 21.88%. These results show that current frontier MLLMs can benefit from direct visual input, but they remain far from the human expert average of 88.5%.

The overall ranking also shows a clear gap between closed-source and open-source MLLMs. The strongest closed-source MLLMs are clustered around 50–55% accuracy, whereas most open-source MLLMs remain below 40%. DeepSeek-VL-2 reaches 37.81%, Qwen-VL-2.5-72B reaches 34.84%, InternVL-3-78B reaches 31.87%, Llama-4-Maverick reaches 27.34%, and LLaVA-NeXT-72B reaches 16.72%. This gap suggests that current open-source models struggle with the combination of financial domain knowledge, multimodal perception, and multi-step reasoning required by FinMR.

The distinction between mathematical and expertise reasoning is also informative. Gemini-2.5-Pro achieves the highest mathematical reasoning score among MLLMs at 54.28%, while Claude-3.7-Sonnet achieves the highest expertise reasoning score at 61.83%. This divergence shows that the strongest model overall is not necessarily the strongest model for every reasoning type. It also supports the benchmark design choice of separating mathematical and expertise questions. If only overall Accuracy were reported, this capability difference would be hidden.

Because the held-out test set contains 640 examples, small differences in overall Accuracy should be interpreted cautiously. Using a binomial approximation for overall Accuracy, Gemini-2.5-Pro’s 54.76% score corresponds to an approximate 95%

confidence interval of about ± 3.9 percentage points; Claude-3.7-Sonnet’s 53.91% and GPT-4o’s 50.00% have similar uncertainty widths. The difference between Gemini-2.5-Pro and Claude-3.7-Sonnet is therefore much smaller than the approximate interval width. FinMR is useful for ranking models, but its main value for this thesis lies in fine-grained diagnostic analysis rather than only in marginal leaderboard differences.

3.9 Fine-Grained Analysis

Fine-grained analysis shows that FinMR reveals several dimensions of model capability that are obscured by overall Accuracy. These dimensions include reasoning type, topic, modality, image type, difficulty, and model complementarity.

Mathematical versus expertise reasoning. Mathematical reasoning is difficult because it requires precise variable grounding, formula selection, and computation. Expertise reasoning is difficult because it requires domain concepts and contextual interpretation. The results show that these abilities are not identical. Claude-3.7-Sonnet is strongest on expertise questions, while Gemini-2.5-Pro is strongest on math questions. GPT-4o performs more evenly across the two categories, with 50.68% expertise Accuracy and 47.25% math Accuracy. This pattern supports the definition in Section 3.3: mathematical and expertise questions should be treated as related but separable reasoning requirements.

Topic-level performance. Performance varies substantially across the 15 financial topics. Closed-source MLLMs often perform strongly in topics such as Investment, Fixed Income, Corporate Finance, and Liquidity and Treasury Risk, while performance is weaker or more uneven in topics such as Financial Reporting and Analysis, Economics, Operational Risk, and some derivatives-related settings. These differences show that financial reasoning is not a single homogeneous capability. A model may perform well on investment-style questions but fail on operational-risk or reporting questions because the evidence structure and required concepts differ.

Modality dependency. Direct visual input improves performance compared with caption-based input for many models, but captioned text remains competitive in some cases. GPT-o1 with GPT-4o-generated captions reaches 45.78%, which is below the strongest MLLMs but above several open-source MLLMs using direct images. This suggests that visual access alone is not sufficient. The model must also know how to transform visual evidence into financial variables and reasoning steps. Conversely, a high-quality caption can help text-only reasoning, but it may omit visual details that become decisive for some questions.

Single-image and multi-image reasoning. FinMR includes both single-image and multi-image questions, allowing the benchmark to test whether models can move beyond localized visual inspection. Single-image questions mainly test direct evidence extraction and local cross-modal alignment. Multi-image questions require comparison, evidence selection, and synthesis across visual sources. Even when aggregate results are reported at model level, this dataset property is important because it exposes failure modes that cannot be observed in single-image-only benchmarks: a model may inspect the wrong image, ignore an image, or combine incompatible evidence from multiple figures.

Difficulty. The easy, medium, and hard labels encode the number of financial knowledge points involved in a question. This helps distinguish shallow perception failures from multi-step reasoning failures. Easy questions are useful for testing whether a model can identify direct evidence. Medium and hard questions test whether the model can chain concepts, formulas, and visual evidence. The presence of hard examples is particularly important for evaluating professional-level financial reasoning because many real analytical tasks require more than one knowledge point.

3.9.1 Performance Correlation and Model Complementarity

Figure 3.5 illustrates performance correlations among financial reasoning dimensions. The heatmap shows that model strengths are not uniform across the benchmark. Some dimensions, such as Expertise, Math, Investment, Credit Risk, Market Risk, and Liquidity and Treasury Risk, show stronger correlation, suggesting that models performing well on one of these dimensions often perform well on related dimensions. Other dimensions, such as Financial Reporting and Analysis, Economics, and Valuation and Risk Models, show lower or even negative correlation with several categories. This implies that success in one financial subdomain does not automatically transfer to another.

This complementarity is important for reliable financial reasoning systems. If different models have different strengths across financial topics or reasoning types, then a single model leaderboard may be insufficient. Model selection, routing, or ensemble design may benefit from knowing which model is stronger for which topic or modality. This finding also anticipates Chapter 5, where errors from different MLLMs are treated as potentially useful resources for cross-model reflection.

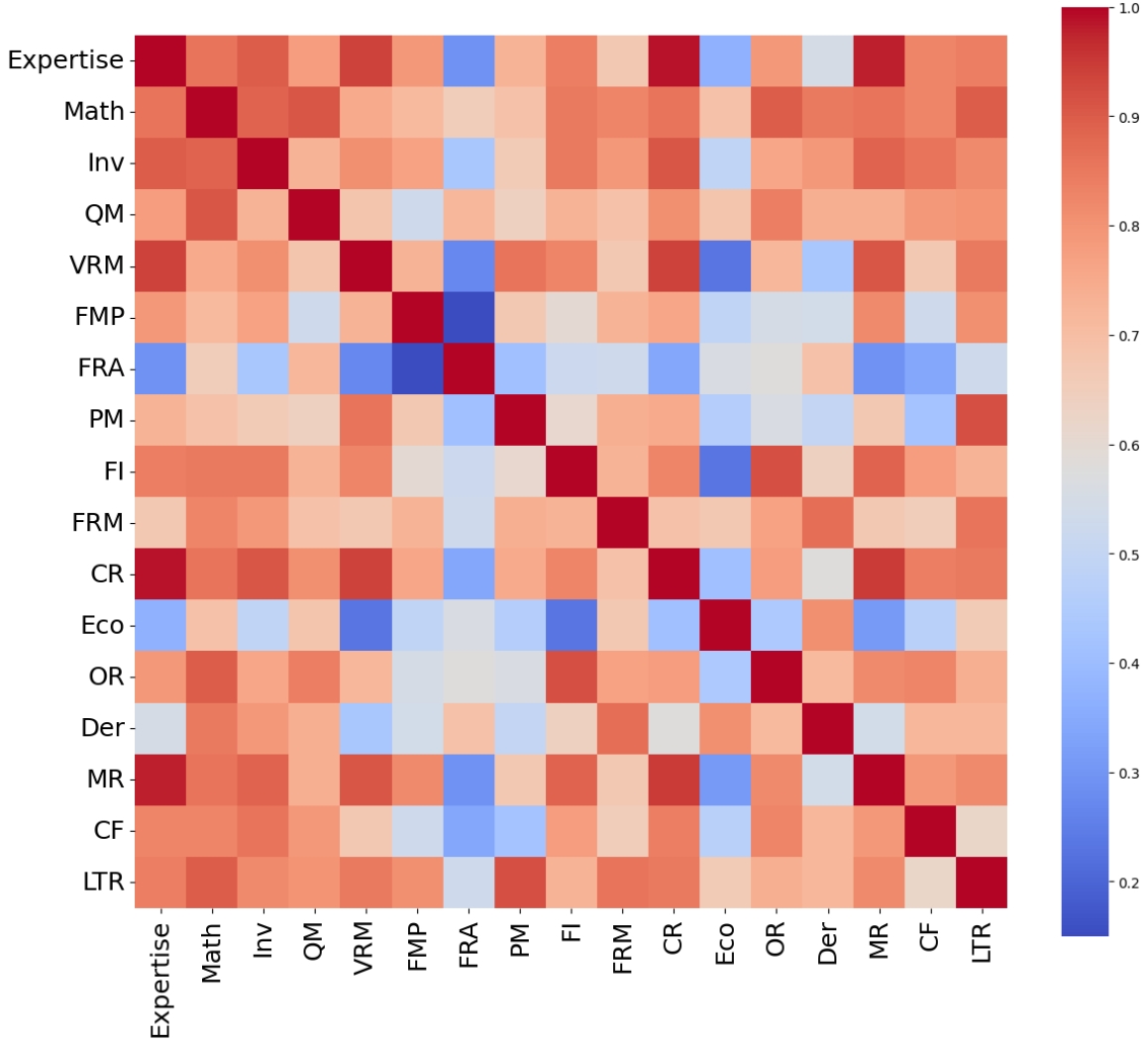


Figure 3.5: MLLMs’ reasoning performance correlation across all financial subsets in FinMR

3.9.2 Image-Type Dependency

Figure 3.6 analyzes GPT-4o’s performance across image types. GPT-4o performs best on tables, achieving 67.90%, which suggests relatively strong structured-data interpretation. Performance drops on charts at 54.50% and on other visual types at 51.77%. It declines further on maps at 49.77% and 3-D plots at 47.60%. The lowest accuracy is on diagrams, at 41.00%, indicating difficulty with abstract, causal, or process-oriented financial visualizations.

This analysis shows that visual evidence type matters. Financial tables are closer to structured text and may be easier for current MLLMs to parse. Diagrams, maps, and

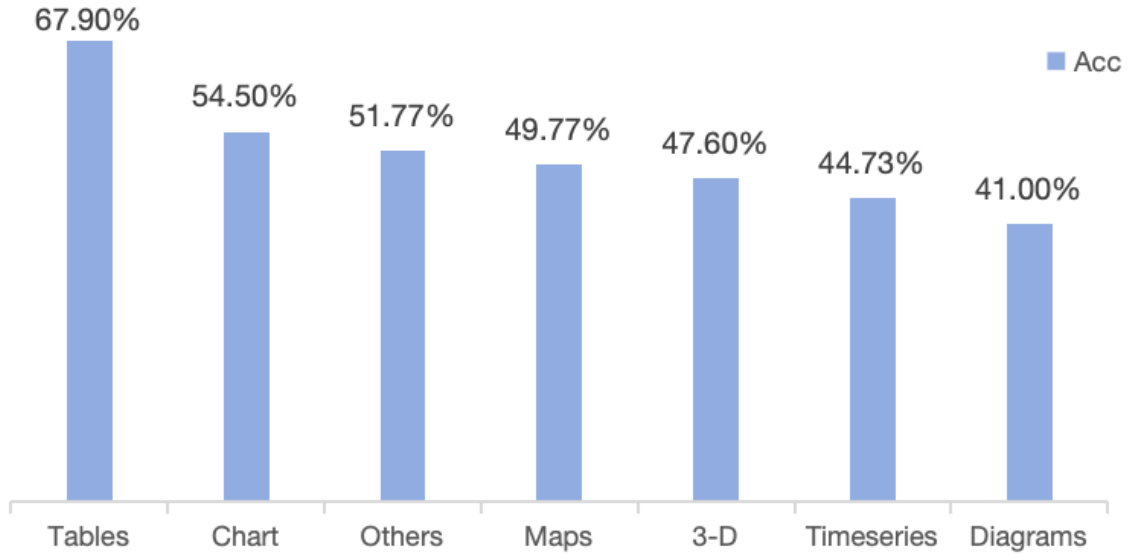


Figure 3.6: Performance of GPT-4o on financial multimodal questions categorized by image types.

3-D plots require more abstraction, spatial interpretation, or domain-specific mapping between visual elements and financial concepts. This provides one reason why overall Accuracy alone is insufficient: two models with similar aggregate scores may have very different visual strengths and weaknesses.

The image-type analysis also helps interpret the difference between perception and reasoning. A low score on diagrams does not necessarily mean that the model cannot see the diagram. It may mean that the model can describe the diagram but cannot connect its components to the financial concept being tested. Similarly, a chart error may arise because the model misreads an axis, because it selects the wrong series, or because it fails to apply the relevant financial operation after reading the chart. FinMR makes these possibilities visible by pairing image categories with reasoning categories and explanations.

The topic-level and image-type analyses together suggest that future systems should not treat all financial images in the same way. A visual table may benefit from table-structure extraction; a time-series chart may require trend and volatility analysis; a diagram may require concept recognition; a map may require spatial grounding; and a multi-image question may require evidence routing. This observation is directly relevant to SEA, where different evidence agents are responsible for different forms of

Main Error Type	Model	Reasoning Steps	Human Check
Image Recognition Problem (73%)	GPT-4o	... based on the graph alone, the spot rate should be understood as 4.0%. ❌	$r(5) = 5\sqrt{\frac{1.0437}{0.8394}} - 1 = 4.453\%$ The model needs to extract data from the image for calculation instead of relying solely on textual information.
Question Misunderstanding (9%)	LLaVa-NEXT-70B	This appears to be a task related to logic puzzles specifically an example of a "river crossing" problem where you need to ... ❌	No "river crossing" problem in this case.
Wrong Financial Math Formula (4%)	LLaVa-NEXT-70B; Qwen2.5-VL-70B	The distance to default: $DD = \frac{\ln\left(\frac{V}{D}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$ ❌	The distance to default: $DD = \frac{\text{Asset Value} - DP}{\text{Asset Volatility}}$
Repetition Problem (3%)	LLaMa-4-Maverick	Now, let's calculate the present value of the face : Now, let's calculate the present value of the face: ❌	The reasoning step repetition results in no final answer.

Figure 3.7: Model reasoning error analysis.

evidence. The benchmark results therefore do more than describe model performance: they identify where adaptive evidence acquisition may help.

Fine-grained analysis also clarifies why model complementarity matters. If one model performs better on mathematical reasoning while another performs better on expertise reasoning, then errors from one model may be informative for another. If one model fails on visual evidence but another produces a different failure pattern, then cross-model comparison can expose hidden assumptions. This insight connects FinMR to CLER, where step-level errors from different MLLMs are used as reflection evidence. The fine-grained structure of FinMR therefore supports both evaluation and later error-learning methods.

3.10 Error Analysis

To understand why models fail on FinMR, this chapter analyzes model errors at both the empirical category level and the thesis-wide taxonomy level. The original benchmark study categorized errors into five primary types: image recognition failure, question misunderstanding, incorrect formula application, answer not found, and others. Figure 3.7 provides representative error examples and the percentage of each error type.

Image recognition failure is the dominant category, accounting for 73% of representative errors. These failures occur when a model cannot correctly read a chart, identify the visual signal, recognize a table structure, or extract the value needed for reasoning. Many FinMR images contain implicit financial information rather than direct answer text. A model may therefore need to infer a trend, map visual elements to a financial variable, or recognize how a diagram encodes a concept. Current models often struggle with this first step.

Question misunderstanding is another common failure mode. Many FinMR questions are written in professional financial language and require the model to understand the intent of the task before extracting evidence. A model may solve a nearby but different problem, overlook a condition in the prompt, or misunderstand what is being asked. This error type is especially important because it can make later reasoning appear coherent while being directed toward the wrong target.

Incorrect formula application appears when the model recognizes the question and evidence but chooses an invalid operation or formula. This failure is common in harder questions that require cross-topic knowledge or multi-step mathematical reasoning. It shows that financial reasoning is not only a perception problem. Even when the visual input is correctly read, the model may lack the domain knowledge needed to select the correct financial operation.

Answer-not-found errors occur when a model produces a long or repetitive reasoning trace without giving a parseable final answer. This problem is particularly visible in weaker open-source models. From a reliability perspective, this is an answer synthesis error: partial reasoning is not useful if the system cannot commit to a valid final output. It also illustrates why output parsing rules and answer-format control are part of benchmark design.

For consistency across the thesis, these empirical categories are aligned with the eight-part error taxonomy introduced in Chapters 1 and 2. First, *visual perception errors* occur when the model fails to identify or read visual elements. Second, *evidence selection errors* occur when the model attends to irrelevant evidence or ignores relevant evidence. Third, *cross-modal alignment errors* occur when the model fails to connect visual evidence with textual context or the question. Fourth, *financial knowledge errors* occur when the model lacks or misuses a financial concept. Fifth, *formula or operation selection errors* occur when the model chooses the wrong calculation or analytical operation. Sixth, *numerical computation errors* occur when the selected operation is appropriate but executed incorrectly. Seventh, *logical inference errors* occur when intermediate conclusions do not follow from the evidence. Eighth, *answer*

synthesis errors occur when the model cannot express the correct final answer in the required format.

This taxonomy reveals that many FinMR failures are compound. A visual perception error can cause evidence selection to fail. An evidence selection error can lead to a correct formula being applied to the wrong values. A financial knowledge error can lead to both formula error and answer synthesis error. This compounding structure is crucial for the rest of the thesis. Chapter 4 addresses evidence-related failures through adaptive evidence acquisition, while Chapter 5 addresses remaining step-level reasoning failures through cross-model error reflection.

3.11 Discussion

FinMR answers RQ1 by operationalizing knowledge-intensive multimodal financial reasoning as a benchmark that combines professional financial topics, mathematical reasoning, financial expertise reasoning, and diverse visual evidence. Its 3,200 questions, 3,764 images, 15 financial topics, single-image and multi-image cases, and distinction between math and expertise questions provide a more comprehensive evaluation setting than benchmarks focused only on text, tables, or isolated visual perception.

The main strength of current MLLMs is that they can use visual input and financial context to solve a subset of professional financial questions. The best MLLMs substantially outperform weaker open-source models and text-only models without direct visual access. They also show particular strengths on structured image types such as tables and on some financial topics. These results indicate that multimodal models have begun to acquire useful financial reasoning capabilities.

The limitations are equally important. Current MLLMs remain far from human expert performance. Their errors are not caused only by missing financial knowledge. They also arise from incomplete or inappropriate use of heterogeneous evidence. Models may misread visual inputs, select the wrong evidence, fail to align a chart with the question, apply the wrong formula, make a calculation error, or generate an invalid final answer. These failures support the thesis argument that reliability requires evaluation at multiple levels: answer correctness, evidence use, reasoning process, and error type.

FinMR therefore contributes a new evaluation capability. It does not claim to solve reliable financial reasoning by itself. Instead, it provides a diagnostic benchmark that exposes where current models succeed and fail. The benchmark makes it possible

to analyze performance by topic, reasoning type, image modality, and error category. This diagnostic role is the reason FinMR serves as the first technical study in the thesis progression.

The answer to RQ1 can be stated more concretely through four evaluation requirements. First, comprehensive evaluation requires *domain breadth*: the benchmark must cover enough financial topics to reveal whether model performance generalizes beyond one familiar subdomain. FinMR addresses this through 15 topics that span investment, valuation, reporting, markets, risk, derivatives, fixed income, and treasury-related reasoning. Second, comprehensive evaluation requires *reasoning-type separation*: the benchmark must distinguish mathematical reasoning from financial expertise reasoning, because a model can be strong in one and weak in the other. FinMR addresses this through its explicit math/expertise labels and corresponding result breakdowns. Third, comprehensive evaluation requires *modality-sensitive design*: the benchmark must represent different visual forms and both single-image and multi-image cases. FinMR addresses this through charts, tables, time-series plots, diagrams, maps, 3-D plots, and multi-image examples. Fourth, comprehensive evaluation requires *diagnostic analysis*: the benchmark must show why errors occur rather than only whether the final answer is wrong. FinMR addresses this through topic-level, image-type, correlation, and error-taxonomy analyses.

This formulation also clarifies the boundary of the contribution. FinMR is not intended to replace document-grounded benchmarks such as FinanceBench, program-supervised benchmarks such as FinQA, or conversational benchmarks such as ConvFinQA. Instead, it complements them by focusing on a capability that they only partially measure: knowledge-intensive multimodal financial reasoning. A reliable financial reasoning system should ultimately be evaluated across all of these settings, but FinMR provides a focused test of the multimodal component that is central to this thesis.

The benchmark further shows that evaluation should preserve evidence diversity. If the benchmark contains mainly tables, then systems optimized for table parsing may appear more reliable than they are. If it contains mainly simple charts, then systems may appear visually competent without demonstrating financial expertise. If it contains mainly textbook-style questions without rich context, then evidence selection and contextual interpretation remain under-tested. FinMR’s mixture of contexts, image types, topics, and reasoning requirements reduces these risks and makes the evaluation closer to real financial analysis, where the decisive evidence is often heterogeneous and unevenly structured.

3.11.1 Implications for Benchmark Design

The design of FinMR suggests several principles for future benchmark construction. First, benchmarks should separate reasoning types that impose different cognitive demands. Mathematical reasoning and financial expertise reasoning overlap, but they are not interchangeable. Second, benchmarks should cover multiple financial topics because financial knowledge is not uniform across subdomains. Third, multimodal benchmarks should include diverse image types and both single-image and multi-image settings. Fourth, benchmark reports should include error analysis rather than only leaderboards. A leaderboard can motivate model improvement, but error analysis explains where improvement is needed.

Fifth, benchmark design should preserve answer verifiability. FinMR uses questions with definitive answers, which enables accuracy-based comparison across a wide set of models. This choice does not capture every form of financial reasoning, especially open-ended report generation or professional advisory dialogue, but it supports controlled evaluation. Future benchmarks can extend this design to longer answers and interactive tasks if evidence annotation and scoring remain rigorous.

3.11.2 Limitations and Future Directions

This study has several limitations. First, FinMR’s scope is strictly within the financial domain and does not include broader business topics such as marketing or management strategy. This keeps the benchmark focused but limits its generalization to adjacent business reasoning tasks. Second, multi-answer questions were excluded to simplify evaluation, although such questions appear in real exams and professional assessments. Third, the benchmark is primarily static. It does not evaluate real-time data retrieval, changing market conditions, or interactive clarification. Fourth, the reported experiments evaluate model outputs at a fixed point in time, and results for commercial systems may shift with model updates.

These limitations suggest future research directions. FinMR can be extended with multi-answer questions, real-time financial evidence, multilingual settings, and interaction-based tasks. It can also be connected with process-level annotations that identify the exact reasoning step where an error occurs. Such extensions would support stronger evaluation of evidence acquisition, verification, and reflection.

The findings indicate that model failures frequently arise from incomplete or inappropriate use of heterogeneous evidence. Chapter 4 therefore investigates adaptive

multi-agent evidence acquisition. SEA builds on FinMR’s diagnostic finding that evidence use is not merely preprocessing: it is part of financial reasoning itself.

More broadly, the chapter shows why evaluation should be treated as the first layer of reliability. Before a system can improve evidence acquisition or error correction, it must be possible to observe which capability is failing. FinMR provides that observation layer. It reveals whether a model is weak on a particular topic, reasoning type, image type, or error category. This does not make the system reliable by itself, but it makes reliability research more targeted. Without such diagnostic evaluation, a method may improve average performance while leaving the most important financial reasoning failures unresolved.

The chapter also shows that evidence and reasoning cannot be cleanly separated in multimodal finance. Evidence is not simply given to the model; it must be recognized, selected, aligned, and interpreted. A chart value is not evidence until it is linked to the question and financial concept. A table cell is not evidence until the relevant row, column, period, and unit are identified. A diagram is not evidence until the model understands what financial process or relationship it encodes. This is why the next chapter treats evidence acquisition as an active reasoning process rather than as a retrieval step performed before reasoning begins.

3.12 Chapter Summary

This chapter introduced FinMR, a benchmark for knowledge-intensive multimodal financial reasoning. It positioned FinMR relative to FinQA, TAT-QA, MMMU-Finance, FAMMA, and related multimodal financial benchmarks; defined the task; described benchmark design and construction; reported dataset statistics; evaluated leading LLMs and MLLMs; and analyzed performance and errors.

The chapter answered RQ1 by showing that comprehensive evaluation requires more than overall Accuracy. It requires a benchmark that covers financial topics, reasoning types, image types, single-image and multi-image settings, and error categories. FinMR reveals that current MLLMs have meaningful but uneven financial reasoning capabilities and remain vulnerable to visual perception errors, evidence selection errors, cross-modal alignment errors, financial knowledge errors, formula or operation selection errors, numerical computation errors, logical inference errors, and answer synthesis errors. The next chapter builds on this diagnosis by treating evidence acquisition as an adaptive, step-wise process.

Multi-Agent SEA: Step-Wise Evidence Acquisition for Multimodal Financial Reasoning

Chapter 3 showed that MLLM failures on FinMR often arise from incomplete visual interpretation, insufficient use of textual context, and weak application of financial knowledge. These findings suggest that evidence acquisition is part of reasoning rather than merely preprocessing. This chapter addresses RQ2 through SEA [?]: *How can specialized agents dynamically acquire and integrate heterogeneous evidence to improve multimodal financial reasoning?*

4.1 Motivation and Chapter Overview

Recent advancements in multimodal large language models (MLLMs) have significantly improved AI systems' capacity to integrate visual and textual information [115, 36, 7]. These models have demonstrated strong performance across a wide range of tasks [45], including visual question answering [95], and multimodal reasoning [178]. However, their performance on multimodal financial reasoning and question-answering tasks remains far from satisfactory [169, 103, 39]. Financial problems often require precise interpretation of numerical information, domain-specific knowledge, and multi-step reasoning over heterogeneous inputs such as text, tables, and financial charts [25, 137]. These characteristics pose significant challenges for current multimodal models.

The latest financial benchmarks, e.g., FinMMR [141], FAMMA [169], and FinMME [103], find that state-of-the-art MLLMs consistently fail in reliable financial

Context (c): Financial analysts often assess the value of fixed-income securities when comparing different investment opportunities. Estimating the present value of future payments helps determine whether a bond offers a reasonable return.

Question (q): A government bond has a face value of \$1,000 which matures in 5 years. Given an interest rate of 4%, what is the present value of this bond?

Image (I):



Options (O): A: \$1133.56, B: \$1148.65, C: \$1159.27, D: \$1142.15

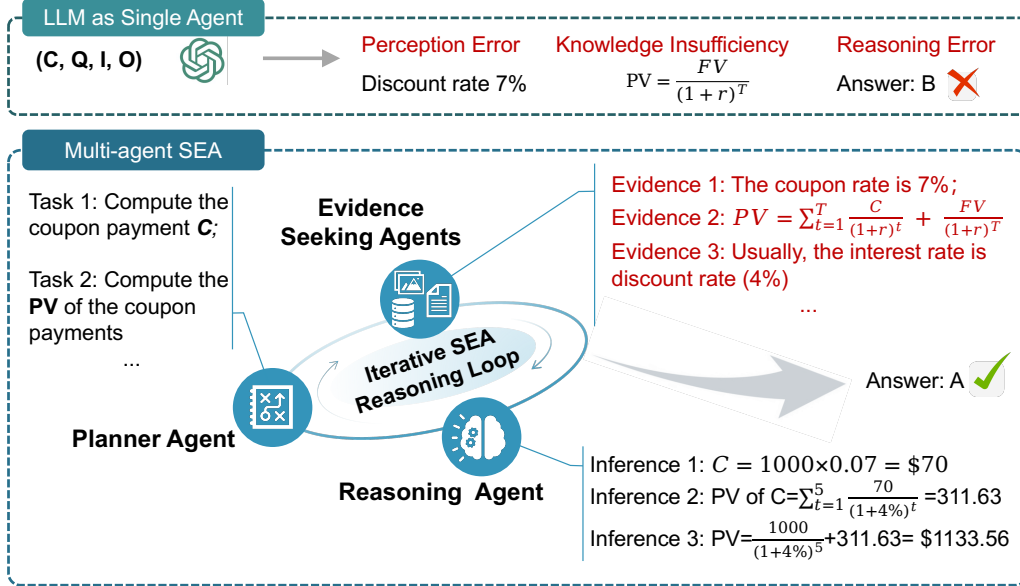


Figure 4.1: An illustrative example of financial multimodal reasoning via single MLLM and SEA.

reasoning. These failures are primarily attributed to perception errors (misreading charts), context misunderstanding (misinterpreting constraints), and knowledge insufficiency (lacking domain formulas), as illustrated in Figure 4.1. While diverse in origin, these issues converge on a single critical failure: the inability to acquire the correct intermediate evidence. Consequently, models proceed with reasoning based on incomplete or irrelevant information, leading to cascading errors.

In contrast, human expert reasoning follows a distinct, decomposed pattern [18]. Aligned with information processing theory [10] and bounded rationality [109], experts adopt a step-wise strategy to manage limited cognitive resources. Rather than simultaneous processing, they employ goal-directed attention [32] to identify and extract task-relevant evidence from heterogeneous sources dynamically. In this framework,

each reasoning step determines the specific evidence required for the next, making accurate evidence acquisition the bottleneck for successful inference. This suggests that financial reasoning performance is limited less by raw logic than by the ability to retrieve precise, task-conditioned information during intermediate steps.

This observation raises a fundamental question: *Is multimodal financial reasoning fundamentally limited by reasoning capability, or by the ability to acquire accurate and task-relevant evidence during intermediate steps?* The hypothesis of this chapter is that the primary source of reasoning errors lies in incorrect or incomplete intermediate evidence, rather than deficiencies in reasoning itself.

This chapter introduces SEA (Step-wise Evidence Acquisition) for multi-agent systems to address the challenge of cascading reasoning failures caused by poor intermediate evidence quality. SEA explicitly models multimodal financial reasoning as a step-wise process of evidence acquisition and reasoning. Instead of forcing a single model to process all multimodal inputs in a single pass, SEA decomposes the reasoning process into a sequence of sub-tasks, where each step identifies and retrieves the specific evidence required for subsequent inference. Specifically, a planning agent first decomposes the task into sub-tasks and determines the information needed at each step. It then generates targeted evidence-seeking queries, which are routed to specialized agents responsible for extracting information from different sources, including financial images, textual context, and external financial knowledge. The retrieved evidence is accumulated in a shared evidence memory and used to guide step-wise reasoning. To support collaboration among agents, SEA adopts a query-based communication mechanism that enables targeted evidence exchange among agents. Rather than relying on complex structured protocols, the planner issues concise natural-language queries, and specialized agents return focused responses. This design mirrors how human experts solve financial problems by asking targeted questions to obtain necessary information, reducing redundant information, and improving the quality of intermediate evidence.

This chapter evaluates SEA on three benchmarks: FinMR, FAMMA, and MMMU-Finance [39, 169, 182]. The experimental results show that SEA improves over representative baselines, yielding average gains of 13.68 percentage points in Accuracy. Furthermore, dynamic task decomposition yields 6.14 percentage-point Accuracy gains on average compared with the static strategy. Finally, the query-based communication mechanism exceeds pipeline-based [202], controller-based [157], and debate-based [92] mechanisms by at least 4.17 percentage points, indicating that query-based communication can better align collaboration among agents and deliver evidence with higher utility for downstream reasoning.

This chapter’s contributions can be summarized as follows:

- This chapter introduces SEA, a multi-agent framework that explicitly models task decomposition, evidence acquisition, and evidence-guided reasoning, enabling dynamic integration of visual, textual, and knowledge-based information.
- This chapter introduces a query-based communication mechanism in which agents interact via concise, step-conditioned natural-language queries, enabling effective and interpretable evidence exchange.
- Experiments on three multimodal financial reasoning benchmarks show that SEA improves over strong baselines and reduces several perception- and knowledge-related failures by improving intermediate evidence quality.

The remainder of this chapter follows the evaluation-to-evidence-to-reflection progression established in Chapter 3. Section 4.2 formulates the problem. Sections 4.3–4.6 present the SEA framework, step-wise evidence acquisition, specialized agent roles, and query-based communication. Sections 4.7–4.13 evaluate SEA through main results, communication comparison, ablations, evidence perturbation, cost and statistical analysis, and error analysis. Section 4.14 answers RQ2 and discusses implications, limitations, and the transition to Chapter 5. Section 4.15 summarizes the chapter.

4.1.1 Positioning Relative to Prior Work

SEA is positioned at the intersection of financial multimodal reasoning, active evidence acquisition, and multi-agent reasoning. Existing financial benchmarks such as FinMR [39], FAMMA [169], FinMME [103], and FinMMR [141] show that current MLLMs still struggle with financial charts, tables, domain formulas, and contextual interpretation. These failures are not only answer-level errors; they often occur because the model fails to acquire the right intermediate evidence before reasoning.

Related work on active computation, active feature acquisition, active perception, and visual information selection shows that selective information use can improve efficiency and reasoning quality [57, 69, 19, 112, 11, 50, 198]. However, these methods usually optimize feature cost, token selection, or demonstration selection rather than step-wise evidence utility for multimodal financial reasoning. SEA instead asks what visual evidence, textual context, or financial knowledge is required at each reasoning step.

Tool-augmented and multi-agent reasoning systems, including ReAct [176], ToolFormer [130], AutoGen [157], role-aware agents [202], and debate-based systems [92],

demonstrate the value of external actions and role decomposition. Yet role assignment alone does not guarantee useful evidence. In contrast, SEA uses specialized agents under a query-based protocol so that communication is driven by explicit evidence gaps. Table 4.1 compares these representative techniques and clarifies the position of SEA.

4.2 Problem Formulation

This chapter studies multimodal financial reasoning as a sequential evidence-grounded decision problem over heterogeneous inputs. Each instance is defined as

$$X = (C, I, Q, O), \quad (4.1)$$

where C denotes the textual context (e.g., the context of a question), $I = \{i_1, \dots, i_S\}$ denotes the set of visual inputs (e.g., charts, tables, or figures), Q is the question, and $O = \{o_1, \dots, o_M\}$ is the set of candidate answers (options). The ground-truth answer is $Y^* \in O$.

Unlike conventional single-pass prediction, SEA models reasoning as an iterative process that alternates between *planning*, *evidence acquisition*, and *evidence-guided reasoning*. At reasoning step n , the system maintains: an evidence state E_n , which stores all retrieved evidence up to step n ; and a reasoning trajectory R_n , which stores all intermediate reasoning outputs up to step n . Formally,

$$E_n = \{e_1, \dots, e_{|E_n|}\}, \quad R_n = \{r_1, \dots, r_n\}. \quad (4.2)$$

The process starts from empty memory: $E_0 = \emptyset, R_0 = \emptyset$.

The goal is to predict the final answer $\hat{Y} \in O$ by progressively collecting task-relevant evidence and conditioning each reasoning step on the accumulated evidence. This formulation explicitly separates *what information is needed* from *how the answer is inferred*, thereby reducing hallucination and improving interpretability. For clarity, the main notations used in the subsequent formulation and analysis are summarized in A.1.

4.3 Framework Overview

SEA (Step-wise Evidence Acquisition) is a multi-agent framework that solves multimodal financial reasoning through a closed-loop interaction among five specialized agents:

Technique	Representative Works	Main Idea	Advantages	Limitations
Multimodal CoT	FinMR [39]; FAMMA [169]; FinMME [103]; FinMMR [141]	Evaluates models on financial questions involving textual context, charts, tables, and domain knowledge.	Provides realistic benchmarks for assessing financial expertise, numerical reasoning, and visual understanding.	Existing MLLMs often suffer from perception errors, weak domain knowledge, and insufficient cross-modal alignment.
LLM-based financial agents	TradingAgents [163]; LLM trading agent survey [41]	Uses multiple LLM agents to support financial decision-oriented workflows, including market monitoring and risk assessment.	Demonstrates the potential of role-specialized agents in financial decision-support systems.	Mainly focuses on trading or portfolio actions rather than evidence-grounded multimodal financial reasoning.
Active evidence acquisition	Adaptive computation time [57]; active feature acquisition [69, 19]; active perception [112, 11]; adaptive token selection [50]	Dynamically selects features, tokens, or visual regions to reduce unnecessary computation or acquisition cost.	Improves computational efficiency and encourages selective information processing.	Usually optimizes efficiency or feature cost, rather than task-conditioned evidence utility for multi-step reasoning.
Visual information selection	Visual In-Context Learning [198]	Selects and summarizes task-relevant visual demonstrations or visual information for MLLM reasoning.	Shows that selective visual evidence can improve multimodal reasoning.	Focuses on demonstration selection or visual prompting, without explicit step-wise evidence acquisition.
Tool-augmented reasoning	ReAct [176]; ToolFormer [130]	Enables LLMs to interact with external tools such as search engines, calculators, or code executors.	Extends LLMs beyond passive generation and supports external knowledge or computation.	Tool use is often implicit or loosely controlled, and may not retrieve the precise evidence needed at each reasoning step.
Retrieval-augmented reasoning and KG-based methods	RARE [144]; RSA [5]; AKR [4]	Incorporates external knowledge, retrieval actions, or structured relational semantics into reasoning.	Improves factual grounding and highlights the value of structured knowledge and contextual enrichment.	Usually focuses on knowledge retrieval, factuality, or entity alignment, rather than multimodal financial evidence acquisition.
Multi-agent reasoning	AutoGen [157]; MAM [199]; role-aware agents [202]; debate-based agents [92]	Decomposes complex tasks into multiple specialized agents that collaborate through predefined roles or communication protocols.	Supports modular problem solving and role-specialized reasoning.	Existing collaboration is often prompt-driven, static, or debate-oriented, which may introduce redundant or misaligned evidence.
SEA	This work	Performs step-wise task decomposition, targeted evidence acquisition, and evidence-guided reasoning through query-based multi-agent communication.	Dynamically retrieves task-relevant evidence from visual inputs, textual context, and financial knowledge, improving reasoning reliability and interpretability.	Still depends on the quality of planner-generated queries and evidence returned by specialized agents.

Table 4.1: Comparison of representative techniques, including their main ideas, advantages, and limitations.

1. a Planner $\mathcal{A}_P$ that decomposes the problem into step-wise sub-tasks and issues evidence-seeking queries;

2. an Image Evidence Agent $\mathcal{A}_I$ that extracts visual evidence from charts, tables, diagrams, and figures;
3. a Context Evidence Agent $\mathcal{A}_C$ that identifies relevant textual evidence, variables, constraints, and assumptions;
4. a Knowledge Evidence Agent $\mathcal{A}_K$ that retrieves financial concepts, formulas, and domain principles from an external knowledge base;
5. a Reasoning Agent $\mathcal{A}_R$ that performs step-wise reasoning conditioned on the accumulated evidence and reports whether evidence issues remain.

As illustrated in Figure 4.2, SEA proceeds iteratively. At each step, the planner determines the current sub-task, formulates a set of targeted natural-language queries, and decides whether the current evidence is sufficient to stop. The queries are routed to the appropriate evidence-seeking agents, whose outputs are integrated into the shared evidence state. The reasoning agent then updates the reasoning trajectory based on the current sub-task and accumulated evidence. The process repeats until the planner determines that the available evidence is sufficient to answer the original question.

This design explicitly treats multimodal financial reasoning as a process of *demand-driven evidence acquisition*: the system does not attempt to process all information uniformly at once, but instead retrieves the specific evidence required by the current reasoning step.

4.4 Step-Wise Evidence Acquisition

Step-wise evidence acquisition is the core mechanism of SEA. The process begins with initial planning, where the planner identifies the current reasoning objective and the evidence gaps that prevent a reliable answer. The planner then generates targeted queries, routes them to the appropriate evidence agents, integrates the returned evidence into the shared evidence state, and updates the reasoning trajectory. If the accumulated evidence remains insufficient, the system continues with a new planning step; if the planner judges that the evidence is sufficient, SEA terminates and produces the final answer.

4.4.1 Initial Planning and Evidence-Gap Identification

The planner agent $\mathcal{A}_P$ controls the overall reasoning process. At step n , given the input instance X , the previous evidence state E_{n-1} , and the previous reasoning trajectory

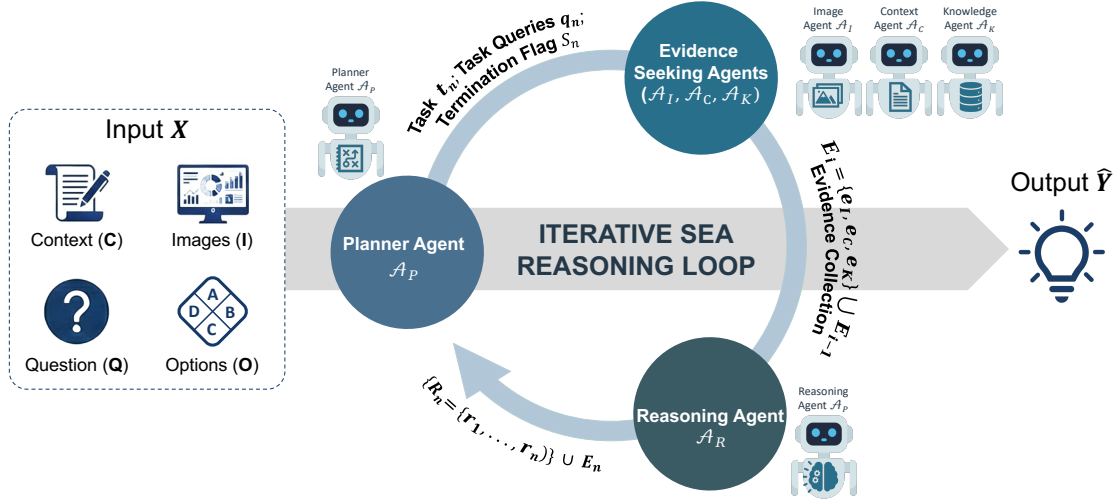


Figure 4.2: Overview of SEA, a multi-agent framework for step-wise evidence acquisition. The planner generates sub-task-conditioned queries under a unified communication protocol, which are routed to multi-source evidence agents (image, text, and knowledge). Retrieved evidence is accumulated into a shared evidence state and used by the reasoning agent for step-wise inference. The entire process forms a closed-loop interaction that tightly couples planning, evidence acquisition, and reasoning.

R_{n-1} , the planner outputs a sub-task t_n describing the current reasoning objective; a set of natural-language queries $q_n = \{q_{n,1}, \dots, q_{n,L_n}\}$ for evidence acquisition; and a termination flag $s_n \in \{\text{continue}, \text{stop}\}$. Formally,

$$(t_n, q_n, s_n) = \mathcal{A}_P(X, E_{n-1}, R_{n-1}). \quad (4.3)$$

The sub-task t_n specifies the information needed to advance the reasoning process, such as extracting a numerical quantity from a chart, identifying a constraint from the context, or retrieving a domain formula. The query set q_n operationalizes this sub-task into concrete information requests that can be answered by specialized agents. The termination flag s_n indicates whether the current evidence state is already sufficient to answer the original question Q :

$$s_n = \begin{cases} \text{stop}, & \text{if } \mathcal{A}_P \text{ judges that } (E_{n-1}, R_{n-1}) \text{ is sufficient for } Q, \\ \text{continue}, & \text{otherwise.} \end{cases} \quad (4.4)$$

This formulation makes the planner adaptive rather than static. Instead of committing to a fixed chain of sub-tasks in advance, the planner revises subsequent reasoning goals according to the evidence accumulated so far. Such adaptive decompo-

sition is particularly important in financial reasoning, where relevant information is often distributed across text, tables, charts, and external knowledge. The complete planner-agent prompt design is provided in A.2.1.

4.4.2 Query Generation, Routing, and Evidence Integration

Given the query set $q_n = \{q_{n,1}, \dots, q_{n,L_n}\}$ generated at step n , each query is routed to the most appropriate evidence-seeking agent:

$$a_{n,\ell} = \rho(q_{n,\ell}), \quad a_{n,\ell} \in \{\mathcal{A}_I, \mathcal{A}_C, \mathcal{A}_K\}, \quad (4.5)$$

where $\rho(\cdot)$ denotes the routing function. The corresponding evidence is then retrieved as

$$e_{n,\ell} = \begin{cases} \mathcal{A}_I(I, q_{n,\ell}), & \text{if } a_{n,\ell} = \mathcal{A}_I, \\ \mathcal{A}_C(C, q_{n,\ell}), & \text{if } a_{n,\ell} = \mathcal{A}_C, \\ \mathcal{A}_K(\mathcal{D}, q_{n,\ell}), & \text{if } a_{n,\ell} = \mathcal{A}_K, \end{cases} \quad (4.6)$$

where $\mathcal{D}$ denotes the external financial knowledge base. The evidence returned at step n is collected into

$$e_n = \{e_{n,1}, \dots, e_{n,L_n}\}, \quad (4.7)$$

and the evidence state is updated by

$$E_n = E_{n-1} \cup e_n. \quad (4.8)$$

4.5 Specialized Agent Roles

SEA depends on specialization, but specialization is useful only when the roles are linked to explicit inputs, outputs, and limitations. The Planner receives the question, accumulated evidence, and previous reasoning trajectory; its function is to identify the next evidence gap; its output is a sub-task, a query set, and a stopping decision; and its main limitation is that a weak or incomplete plan can send the system toward irrelevant evidence. The Image Evidence Agent receives visual inputs and image-directed queries; its function is to extract values, labels, trends, relations, and visual structures; its output is concise visual evidence; and its limitation is fine-grained perception, especially when charts are dense or multiple images must be combined. The Context Evidence Agent receives the textual context and context-directed queries; its function is to locate assumptions, variables, constraints, and definitions; its output is textual

evidence; and its limitation is long-context alignment. The Knowledge Evidence Agent receives the financial knowledge base and knowledge-directed queries; its function is to retrieve formulas, concepts, and principles; its output is domain evidence; and its limitation is adapting general knowledge to scenario-specific assumptions. The Reasoning Agent receives the current sub-task, evidence state, and reasoning trajectory; its function is to update reasoning and report evidence issues; its output is an intermediate reasoning step; and its limitation is that relevant evidence can still be misused during synthesis.

The three evidence-seeking agents serve complementary roles. **Image agent** $\mathcal{A}_I$ extracts structured visual evidence such as values, trends, and relations from charts, tables, and figures. **Context agent** $\mathcal{A}_C$ identifies task-relevant information from the textual context, including variables, assumptions, and problem constraints. **Knowledge agent** $\mathcal{A}_K$ retrieves financial concepts, formulas, and domain-specific principles from the external knowledge base $\mathcal{D}$.

A key property of SEA is that evidence acquisition is *step-conditioned*. The system does not passively summarize all available inputs at once. Rather, it retrieves evidence only when demanded by the current sub-task. This improves both the relevance and the precision of intermediate evidence, which is essential for reliable multi-step financial reasoning. The complete prompts used by the three evidence-seeking agents are provided in A.2.2. These prompts specify how the Image Agent, Context Agent, and Knowledge Agent respond to planner-generated queries by extracting visual evidence from I , textual evidence from C , and financial knowledge from $\mathcal{D}$, respectively.

Financial knowledge base To alleviate the domain-knowledge limitations of general-purpose MLLMs, this chapter constructs an external financial knowledge base $\mathcal{D}$ from the training sets of FinMR and FAMMA. Specifically, answer explanations are extracted because they often contain financial concepts, formulas, and reasoning procedures, and an LLM is used to identify key terms together with their definitions and mathematical expressions across 15 financial topics. The resulting database provides structured domain knowledge that supports downstream evidence retrieval. The statistics of $\mathcal{D}$ are reported in Table 4.2.

4.5.1 Reasoning Agent and Evidence Sufficiency

After updating the evidence state, the reasoning agent $\mathcal{A}_R$ performs the current inference step. At step n , it takes as input the sub-task t_n , the updated evidence state E_n , and the previous reasoning trajectory R_{n-1} , and produces an intermediate reasoning

	Number	Ratio
Database Size	8440	–
Knowledge Type		
Financial Concept	5698	67.51%
Math Formula	2742	32.49%
Topic Diversity		
Investment	1063	12.60%
Quantitative Methods	819	9.70%
Valuation and Risk Models	785	9.30%
Financial Markets and Products	776	9.20%
Financial Reporting and Analysis	701	8.30%
Portfolio Management	684	8.10%
Fixed Income	658	7.80%
Credit Risk	456	5.40%
Foundation of Risk Management	447	5.30%
Economics	414	4.90%
Operational Risk	346	4.10%
Derivatives	338	4.00%
Market Risk	329	3.90%
Corporate Finance	329	3.90%
Liquidity and Treasury Risk	295	3.50%

Table 4.2: Statistics of the constructed financial knowledge database $\mathcal{D}$, including knowledge types and topic diversity.

output r_n :

$$r_n = \mathcal{A}_R(t_n, E_n, R_{n-1}). \quad (4.9)$$

The reasoning trajectory is then updated as

$$R_n = R_{n-1} \cup \{r_n\}. \quad (4.10)$$

The interaction between evidence acquisition and reasoning is tightly coupled. If the currently accumulated evidence is insufficient (i.e., the reasoning agent reports issues as a reasoning step), the planner generates a new sub-task and additional queries at the next step. If the planner determines that the evidence is sufficient, the iterative process terminates and the final answer is produced:

$$\hat{Y} = \mathcal{A}_P^{\text{final}}(X, E_N, R_N), \quad (4.11)$$

where N is the final reasoning step satisfying $s_N = \text{stop}$.

This design ensures that the final decision is grounded in an explicit evidence trail rather than implicit latent reasoning alone. Each reasoning step is therefore traceable to retrieved evidence, improving robustness and interpretability. The implementation details of the reasoning agent are provided in A.2.3.

4.6 Query-Based Communication

SEA uses an effective query-based communication protocol to coordinate agents. Instead of relying on complex symbolic message passing or open-ended multi-agent debate, the planner communicates with evidence-seeking agents through concise natural-language queries. This design has two advantages. First, it aligns naturally with the capabilities of modern LLM/MLLM agents. Second, it constrains information exchange to the evidence required by the current sub-task, reducing redundancy and improving alignment between planning and retrieval.

Query-based communication is particularly suited to heterogeneous financial evidence because it makes missing information explicit. A pipeline can process image, context, and knowledge in a fixed order, but it cannot easily revise the order when the first evidence is insufficient. A controller can assign work to agents, but it may delegate broad tasks rather than ask for the exact figure value, textual constraint, or formula required by the current step. Debate can expose disagreement, but open-ended argument often increases communication cost and may not retrieve the missing evidence. SEA instead asks targeted questions such as which chart value is relevant, which sentence defines the constraint, or which financial formula applies. This makes communication both adaptive and evidence-driven.

The protocol operates as follows. At each step, the planner first decides whether reasoning should continue. If $s_n = \text{stop}$, the system terminates evidence acquisition and produces the final answer using the accumulated evidence and reasoning history. Otherwise, the planner generates a query set q_n , each query is routed to a specialized evidence-seeking agent, and the returned evidence is added to the shared memory for downstream reasoning. Algorithm 1 summarizes the complete SEA procedure.

4.7 Experimental Setup

4.7.1 Model and Inference Settings

SEA instantiates agents based on three frontier models: GPT-4o [115], Gemini-2.5-Flash [36], and Claude-4.5-Sonnet [7]. These models are selected because they represent widely used MLLMs with native visual input and strong instruction-following ability. For each experiment, all agents are instantiated using the same backbone model to ensure a fair comparison across different model families. To improve reproducibility and reduce randomness in multi-step reasoning, deterministic decoding is used for all agents, with the temperature set to 0.0. Each agent is prompted to

Algorithm 1 SEA: step-wise evidence acquisition for multimodal financial reasoning.

Require: Input instance $X = (C, I, Q, O)$, knowledge base $\mathcal{D}$, maximum number of steps $N_{\max}$
Ensure: Predicted answer $\hat{Y}$

```

1: Initialize evidence state  $E_0 \leftarrow \emptyset$ 
2: Initialize reasoning trajectory  $R_0 \leftarrow \emptyset$ 
3: for  $n = 1$  to  $N_{\max}$  do
4:    $(t_n, q_n, s_n) \leftarrow \mathcal{A}_P(X, E_{n-1}, R_{n-1})$ 
5:   if  $s_n = \text{stop}$  then
6:      $\hat{Y} \leftarrow \mathcal{A}_P^{\text{final}}(X, E_{n-1}, R_{n-1})$ 
7:     return  $\hat{Y}$ 
8:   end if
9:    $e_n \leftarrow \emptyset$ 
10:  for each query  $q_{n,\ell} \in q_n$  do
11:     $a_{n,\ell} \leftarrow \rho(q_{n,\ell})$ 
12:    if  $a_{n,\ell} = \mathcal{A}_I$  then
13:       $e_{n,\ell} \leftarrow \mathcal{A}_I(I, q_{n,\ell})$ 
14:    else if  $a_{n,\ell} = \mathcal{A}_C$  then
15:       $e_{n,\ell} \leftarrow \mathcal{A}_C(C, q_{n,\ell})$ 
16:    else if  $a_{n,\ell} = \mathcal{A}_K$  then
17:       $e_{n,\ell} \leftarrow \mathcal{A}_K(\mathcal{D}, q_{n,\ell})$ 
18:    end if
19:     $e_n \leftarrow e_n \cup \{e_{n,\ell}\}$ 
20:  end for
21:   $E_n \leftarrow E_{n-1} \cup e_n$ 
22:   $r_n \leftarrow \mathcal{A}_R(t_n, E_n, R_{n-1})$ 
23:   $R_n \leftarrow R_{n-1} \cup \{r_n\}$ 
24: end for
25:  $\hat{Y} \leftarrow \mathcal{A}_P^{\text{final}}(X, E_{N_{\max}}, R_{N_{\max}})$ 
26: return  $\hat{Y}$ 

```

return structured JSON outputs, including the planner’s stopping decision, subtask, target agent, and evidence-seeking queries. The full prompt templates for all agents are provided in A.2. Because these commercial model versions differ from some models evaluated in Chapters 3 and 5, SEA results are interpreted within the Chapter 4 experimental setting rather than as direct cross-chapter model-version comparisons. Exact API access dates are not available in the source study and are therefore treated as part of the API-version-drift limitation discussed in Chapter 6.

To address the domain knowledge limitations of general-purpose MLLMs, an external financial knowledge base $\mathcal{D}$ is further constructed using expert-annotated explanatory data. See more details of the financial knowledge base in Section 4.5. These human-annotated resources provide detailed explanations of financial concepts, formulas, and application scenarios, and therefore serve as a high-quality repository of external financial expertise. The knowledge agent retrieves relevant financial con-

Dataset	Test Size	Math	Expertise	Main Image Type	Knowledge Level	Knowledge Intensive?	Numerical Math?	Advanced Math?
FinMR	640	242 (37.8%)	398 (62.2%)	Charts, tables, diagrams, time series, maps	College	Yes	Yes	Yes
FAMMA	200	100 (50.0%)	100 (50.0%)	Charts, tables, diagrams	College	Yes	Yes	Yes
MMMU-Finance	1100	1100 (100.0%)	–	Tables	College	Yes	Yes	No

Table 4.3: Statistics and characteristics of the evaluation datasets, including test set size, task-type distribution, main image types, knowledge level, and reasoning requirements.

cepts, formulas, and domain-specific rules from this database according to the query generated by the planner.

To keep the overall reasoning process computationally tractable and to avoid unnecessarily long planning chains, the maximum number of planning steps is set to $N_{\max} = 6$. At each step, the planner generates at most three evidence-seeking queries, corresponding to the image agent, context agent, and knowledge agent. Therefore, the maximum number of agent invocations for one sample is bounded by $1 + N_{\max}(1 + 3 + 1)$, where the three terms inside the parentheses correspond to one planner call, up to three evidence-agent calls, and one reasoning-agent call at each step, and the additional call denotes the final answer generation. In practice, the reasoning process may terminate earlier once the planner determines that the accumulated evidence is sufficient.

4.7.2 Benchmarks

This chapter evaluates SEA on three financial multimodal reasoning benchmarks: FinMR, FAMMA, and MMMU-Finance. These benchmarks are selected because they differ in domain scope, question type, evidence dependency, and difficulty level, thereby supporting evaluation of whether SEA generalizes across different financial reasoning scenarios. To prevent knowledge leakage, all experiments are conducted on the test sets only, since the training sets of FinMR and FAMMA are used to construct the external financial knowledge base. Table 4.3 summarizes the statistics and characteristics of the three testing datasets.

FinMR [39] is an expert-level financial multimodal reasoning benchmark consisting of 3,200 financial reasoning questions. Each instance includes detailed textual context, one or more financial images, candidate options, and expert-annotated explanations. FinMR is designed to evaluate professional-level financial reasoning, covering diverse financial topics and requiring the integration of mathematical reasoning, domain knowledge, and visual interpretation. In the evaluation subset used in this chapter, FinMR contains both mathematical reasoning questions and financial expertise questions. The mathematical subset mainly requires numerical calculation, formula application, and quantitative comparison, while the expertise subset focuses more on conceptual understanding, financial interpretation, and domain-specific judgment. This makes FinMR the most knowledge-intensive and expert-oriented benchmark in this evaluation.

FAMMA [169] is a multilingual financial multimodal reasoning benchmark designed to assess college-level and advanced financial expertise. The original benchmark contains financial question-answer pairs collected from textbooks and exams, spanning multiple finance subfields such as corporate finance, asset management, and financial engineering. Each question is paired with one or more images, such as tables, charts, and diagrams, requiring models to combine textual understanding, visual evidence extraction, and financial knowledge. Compared with FinMR, FAMMA emphasizes multilinguality and academic financial knowledge. In the evaluation subset used in this chapter, FAMMA is evenly divided into mathematical reasoning and expertise-oriented questions, allowing the experiments to test whether SEA is effective across both quantitative and conceptual financial reasoning.

MMMU-Finance [182] is a finance-focused subset of the Multi-discipline Multimodal Understanding benchmark. MMMU is designed to evaluate college-level multimodal understanding and deliberate reasoning across a wide range of academic disciplines. This chapter uses the finance-related subset, which contains samples drawn from Finance, Economics, and Accounting¹. Unlike FinMR and FAMMA, which are specifically constructed for financial multimodal reasoning, MMMU-Finance provides a broader academic setting with more standardized exam-style questions. Notably, approximately 95% of the visual inputs in this subset are financial tables, making it especially useful for evaluating table-based numerical reasoning and structured financial information extraction.

¹Economics and Accounting are closely related subfields within the broader financial domain and frequently appear in financial reasoning tasks.

Overall, the three benchmarks provide complementary evaluation settings. FinMR tests expert-level financial reasoning with rich textual and visual evidence; FAMMA evaluates multilingual and college-level financial reasoning across heterogeneous visual inputs; and MMMU-Finance examines broader academic finance reasoning, with a strong emphasis on table-based visual evidence. Therefore, evaluating SEA on these datasets allows us to assess its generalization across different levels of difficulty, evidence sources, and financial reasoning types.

4.7.3 Baselines

The following baseline methods are compared in the experiments:

CoT [156] is a method that instructs LLMs to generate intermediate reasoning steps (i.e., *Let’s think step by step*) before producing the final answer. It serves as a strong single-agent baseline by improving reasoning transparency and decomposing complex financial problems into sequential inference steps.

MAD [74] studies the impact of decision-making protocols in multi-agent debate systems. Multiple agents first generate candidate solutions and exchange reasoning through several discussion rounds. A predefined decision protocol, such as voting or consensus, is then applied to determine the final answer.

RAA [202] is a role-aware multi-agent framework for financial question answering, comprising an Evidence Retriever, a CoT-based Generator, and an Expert Reviewer. The retriever gathers financial knowledge, the generator produces an initial answer, and the reviewer refines it through critique and correction.

4.8 Main Results

This section evaluates SEA’s overall effectiveness by comparing it with representative baselines on three financial multimodal reasoning benchmarks: FinMR, FAMMA, and MMMU-Finance. Later sections analyze communication paradigms, ablations, evidence perturbations, cost, statistical significance, and remaining error patterns.

Table 4.4 reports the performance comparison of different reasoning strategies across three multimodal financial reasoning benchmarks: FinMR, FAMMA, and MMMU-Finance. Overall, SEA achieves the best performance across all models and datasets in this evaluation, supporting the effectiveness of demand-driven evidence acquisition for multimodal financial reasoning.

Model	Method	FinMR (Accuracy %)			FAMMA (Accuracy %)			MMMU-Finance (Accuracy %)
		ALL	Math(37.8%)	Expertise(62.2%)	ALL	Math(50%)	Expertise(50%)	Math
GPT-4o	CoT	54.17	49.40	57.07	54.50	53.00	56.00	62.77
	MAD	58.31 (↑4.14)	54.39 (↑4.99)	60.70 (↑3.63)	58.5 (↑4.00)	58.00 (↑5.00)	59.00 (↑3.00)	67.00 (↑4.23)
	RAA	59.17 (↑5.00)	53.21 (↑3.81)	62.79 (↑5.72)	60.00 (↑5.50)	59.00 (↑6.00)	61.00 (↑5.00)	69.00 (↑6.23)
	SEA(this chapter)	62.37 (↑8.23)	56.62 (↑7.22)	65.85 (↑8.78)	63.00 (↑8.50)	61.00 (↑8.00)	65.00 (↑9.00)	74.00 (↑11.23)
Gemini-2.5-Flash	CoT	61.20	56.79	63.88	61	58.00	64.00	63.77
	MAD	66.19 (↑4.99)	59.79 (↑3.11)	70.01 (↑6.13)	63.00 (↑2.00)	59.00 (↑1.00)	67.00 (↑3.00)	69.44 (↑5.67)
	RAA	69.42 (↑7.22)	62.44 (↑5.65)	72.06 (↑8.18)	64.50 (↑3.50)	60.00 (↑2.00)	62.00 (↑5.00)	71.23 (↑7.46)
	SEA(this chapter)	74.80 (↑13.60)	66.20 (↑9.41)	80.00 (↑16.12)	69.10 (↑8.10)	65.00 (↑7.00)	73.20 (↑9.20)	79.00 (↑15.23)
Claude-4.5-Sonnet	CoT	60.89	58.59	62.28	60.00	59.00	61.00	63.78
	MAD	74.46 (↑13.58)	72.31 (↑13.72)	75.77 (↑13.49)	61.00 (↑1.00)	60.00 (↑1.00)	62.00 (↑1.00)	73.55 (↑9.77)
	RAA	71.49 (↑10.61)	70.11 (↑11.52)	72.33 (↑10.05)	61.00 (↑1.00)	59.00 (-)	63.00 (↑2.00)	78.00 (↑14.22)
	SEA(this chapter)	78.49 (↑17.61)	76.50 (↑17.91)	79.70 (↑17.42)	64.50 (↑4.5)	63.00 (↑4.00)	66.00 (↑5.00)	82.70 (↑18.92)

Table 4.4: Performance comparison across three financial multimodal reasoning benchmarks: FinMR, FAMMA, and MMMU-Finance. Results are reported as Accuracy (%). Math and Expertise denote subsets requiring quantitative reasoning and domain-specific financial knowledge, respectively. Numbers in parentheses indicate the absolute improvement over the CoT baseline. Bold indicates the best result in each column.

Across all benchmarks, SEA significantly outperforms the standard Chain-of-Thought (CoT) baseline. For example, on the FinMR benchmark, SEA improves the overall Accuracy from 54.17% to 62.37% for GPT-4o, from 61.20% to 74.80% for Gemini-2.5-Flash, and from 60.89% to 78.49% for Claude-4.5-Sonnet. Notably, the improvement for Claude-4.5-Sonnet exceeds 17 percentage points, indicating that SEA can improve the observed reasoning performance of strong base models in this setting.

Compared with existing multi-agent reasoning methods such as MAD and RAA, SEA achieves consistently higher performance across all benchmarks. While MAD relies on multi-agent debate and RAA introduces role-aware reasoning agents, both approaches still process information from a static input context. In contrast, SEA explicitly models *evidence acquisition as a decision process*, enabling the system to dynamically retrieve task-relevant information from different sources before performing reasoning. This design leads to more reliable intermediate reasoning steps and reduces error propagation during multi-step reasoning.

The improvement of SEA is particularly pronounced on the mathematical problems of the three benchmarks, which require precise numerical reasoning and formula application. For example, on the FinMR Math subset, SEA achieves 66.20% Accuracy with Gemini-2.5-Flash and 76.50% with Claude-4.5-Sonnet, substantially outperforming other reasoning strategies. This suggests that dynamically retrieving relevant formulas and financial concepts helps mitigate calculation errors and improves reasoning reliability.

Another important observation is that SEA improves performance across all evaluated base models, including GPT-4o, Gemini-2.5-Flash, and Claude-4.5-Sonnet. Despite the large differences in model architectures and capabilities, SEA yields stable

Communication Method	FinMR	FAMMA	MMMU-Finance
Pipeline-based: Role-aware Agent [202]	70.31	60.33	67.66
Controller-based: AutoGen-style [157]	67.44	56.78	65.59
Debate-based: Multi-agent Debate [92]	72.33	57.32	70.69
Query-based: SEA	78.49	64.50	82.70

Table 4.5: Comparison with representative multi-agent communication paradigms. All methods are implemented using the same backbone model and input setting for fair comparison. The results show that SEA’s query-based communication paradigm outperforms pipeline-based, controller-based, and debate-based alternatives across all three benchmarks in this evaluation.

gains across FinMR, FAMMA, and MMMU-Finance. This result indicates that SEA is not tied to a single model family in the evaluated setting.

4.9 Comparison of Communication Paradigms

To evaluate the effectiveness of different collaboration mechanisms, this chapter compares four representative paradigms: 1) pipeline-based collaboration, where agents operate sequentially without interaction; 2) controller-based coordination, where a central planner assigns subtasks to different modules without explicit query–response communication; 3) debate-based collaboration, where agents engage in multi-round adversarial discussions; and 4) SEA’s query-driven interactive collaboration, which enables structured query–response communication between the planner and evidence-seeking agents.

For a fair comparison, all communication baselines are implemented using the same backbone model and the same input context. The pipeline-based baseline follows a sequential evidence-retrieval-and-reasoning process, similar to role-aware financial agents [202]. The controller-based baseline follows the centralized coordination paradigm of AutoGen-style systems [157], where a controller assigns subtasks to specialized agents but does not use explicit query–response evidence acquisition. The debate-based baseline follows multi-agent debate protocols [92], where multiple agents independently generate answers and exchange critiques before producing a final answer. In contrast, SEA uses planner-generated, step-conditioned queries to explicitly acquire missing evidence from the most relevant agent at each reasoning step.

As shown in Table 4.5, query-driven interaction outperforms all other collaboration paradigms across all benchmarks in this comparison. On FinMR, SEA achieves 78.49% Accuracy, significantly surpassing pipeline-based (+8.18%), controller-based (+11.05%), and debate-based (+6.16%) collaboration. Similar improvements are ob-

served on FAMMA and MMMU-Finance, where query-driven interaction yields the best performance, with particularly notable gains on MMMU-Finance (+17.11% over controller-based and +15.04% over pipeline-based).

The results reveal several important insights. First, purely sequential collaboration is insufficient for complex multimodal financial reasoning. Pipeline-based methods lack the ability to dynamically acquire missing evidence, leading to incomplete or incorrect intermediate evidence. This limitation is especially pronounced in tasks requiring cross-modal integration and domain knowledge. Second, controller-based coordination, despite introducing task routing, remains limited in its ability to support evidence-grounded reasoning. Although the planner can decide which module to invoke, it delegates entire subtasks rather than explicitly identifying and retrieving missing evidence. As a result, the reasoning process remains implicit, and the system may fail to obtain the precise evidence required for accurate answer synthesis. Third, while debate-based collaboration introduces stronger interaction, unstructured or adversarial communication does not necessarily improve reasoning performance. The relatively lower performance on FAMMA suggests that multi-round discussions may introduce redundancy or distract from the core task, without effectively addressing missing evidence. Moreover, debate-based methods incur higher communication overhead without guaranteeing more accurate evidence acquisition.

In contrast, SEA’s query-driven framework achieves the best performance by enabling targeted and efficient evidence acquisition. At each step, the planner explicitly identifies missing evidence and formulates task-specific queries to retrieve evidence from the most relevant agent. This structured communication paradigm avoids both the rigidity of pipeline-based execution and the inefficiency of open-ended debate, while overcoming the implicit reasoning limitations of controller-based coordination. As a result, it leads to more accurate intermediate evidence and, consequently, more reliable final answers. These results demonstrate that not all forms of interaction are equally effective. Instead of no interaction (pipeline), implicit task delegation (controller-based), or excessive interaction (debate), minimal yet targeted query-based communication provides the most suitable balance between efficiency and reasoning Accuracy for multimodal financial tasks.

4.10 Ablation Studies

To quantify the contribution of each component in SEA, this chapter conducts ablation studies by systematically removing key modules from the full framework. Since the

Method	FinMR Accuracy	FAMMA Accuracy	MMMU-Finance Accuracy
Full SEA	78.49	64.50	82.70
w/o Planner	63.79	60.00	65.30
w/o Agent Communication	64.69	61.00	73.21
w/o Evidence Seeking Agents	72.19	62.30	78.22

Table 4.6: Ablation study of SEA on FinMR, FAMMA, MMMU-Finance, using base model Claude-4.5-Sonnet.

full SEA system built on Claude-4.5-Sonnet achieves the strongest overall performance in Table 4.4, this model is used as the base for all ablation experiments. These analyses are intended to examine how visual grounding, multi-agent communication, and specialized evidence acquisition jointly affect performance on financial multimodal reasoning.

SEA w/o Planner. In the full framework, the planning agent decomposes the task into step-wise sub-tasks and generates targeted queries to guide evidence acquisition at each reasoning step. In this variant, the planning agent is removed entirely. Instead, each evidence-seeking agent operates based on a fixed, generic instruction (i.e., *understanding the image, context, and question to extract relevant evidence, or retrieving knowledge from a financial knowledge base*), without step-specific guidance. The collected evidence is then directly used by the reasoning agent. As shown in Table 4.6, this variant achieves performance slightly better than standard CoT reasoning in Table 4.4, but remains significantly inferior to the full SEA framework. This observation highlights that while evidence-seeking agents are beneficial, their effectiveness critically depends on fine-grained, step-conditioned guidance. Without an explicit planning mechanism, evidence retrieval becomes coarse and less aligned with the evolving reasoning needs, leading to suboptimal downstream reasoning performance.

SEA w/o Agent Communication. In this variant, the iterative communication mechanism among the planning agent, the evidence-seeking agents, and the reasoning agent is removed. The planner generates the reasoning plan and queries only once, the evidence-seeking agents retrieve information in a single pass, and the reasoning agent produces the final answer from the collected evidence. Compared with the full SEA system, this variant yields the largest overall degradation, dropping to 64.69% on FinMR, 61.00% on FAMMA, and 73.21% on MMMU-Finance. These results indicate that iterative multi-agent communication substantially improves the quality of intermediate evidence and enables more effective coordination between planning, evidence acquisition, and reasoning.

SEA w/o Evidence-Seeking Agents. To evaluate the importance of targeted evidence acquisition, the evidence-seeking agents are removed and only the planner and reasoning agent are retained. Under this setting, the system can no longer extract evidence from different sources through specialized agents. Performance again declines across all datasets, reaching 72.19% on FinMR, 62.30% on FAMMA, and 78.22% on MMMU-Finance. Although this variant remains stronger than the settings without planner vision input or agent communication, it is still consistently worse than the full SEA system, showing that dedicated evidence-seeking agents contribute meaningfully to accurate financial multimodal reasoning.

Overall, the ablation results confirm that all three components are beneficial, while their contributions are not equally strong. Iterative agent communication produces the largest performance gain, suggesting that coordination among agents is central to SEA. Direct planner access to visual input is also important, particularly for tasks requiring precise visual grounding. Specialized evidence-seeking agents further improve performance by supporting targeted evidence acquisition from heterogeneous sources. The full SEA system, therefore, achieves the best results on FinMR, FAMMA, and MMMU-Finance in this ablation setting, demonstrating the effectiveness of the SEA evidence acquisition framework.

4.10.1 Utility Analysis of Dynamic Sub-task Generation

This chapter compares dynamic sub-task generation with a static decomposition strategy on the FinMR dataset, where all sub-tasks are predefined at the beginning and remain fixed throughout the reasoning process. As illustrated in Figure 4.3, dynamic sub-task generation consistently outperforms static decomposition across all base models. In particular, the performance gains are substantial for Gemini-2.5-Flash (+9.43%) and noticeable for GPT-4o (+3.93%), indicating that the benefit of dynamic planning is robust across models of varying capabilities.

The performance improvement stems from the adaptive nature of dynamic planning. In static decomposition, the reasoning trajectory is determined without access to intermediate evidence, making it vulnerable to early-stage planning errors that cannot be corrected later. This limitation is especially pronounced in FinMR, where relevant information is often fragmented across textual context, visual inputs, and domain-specific knowledge, requiring flexible and iterative reasoning. In contrast, dynamic sub-task generation enables the planner to iteratively update subsequent sub-tasks conditioned on the evolving evidence state. This allows the system to refine reasoning objectives based on newly acquired evidence, correct earlier misinterpretations, and

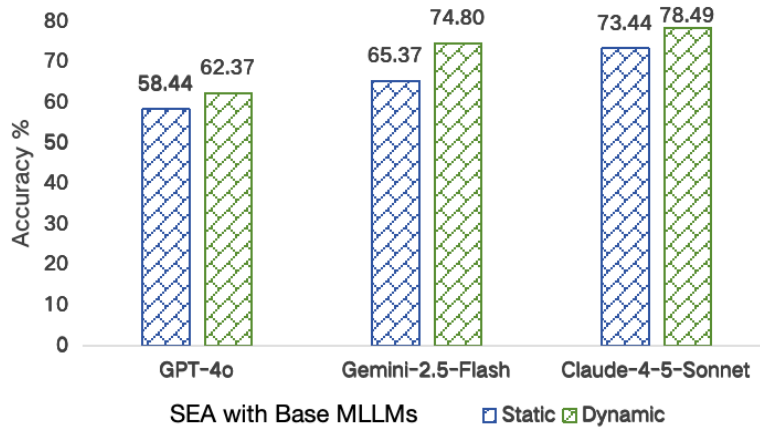


Figure 4.3: Static vs. dynamic task decomposition on the FinMR benchmark. Dynamic sub-task generation consistently outperforms static planning across all models, highlighting the effectiveness of iterative, evidence-conditioned task refinement.

focus on the most relevant information at each step. As reflected in Figure 4.3, this adaptive process leads to consistent and sometimes large performance improvements across all evaluated models.

4.11 Evidence Perturbation Analysis

To evaluate the quality of evidence acquired by different agents, this chapter conducts a fine-grained analysis on the FinMR, FAMMA, and MMMU-Finance benchmarks.

Case-based Analysis For each dataset, 100 intermediate queries generated during the reasoning process are randomly sampled, and the correctness of the corresponding agent responses is assessed. Different evaluation strategies are adopted for each modality. For the image agent, correctness is assessed via human evaluation due to the complexity of visual interpretation. For the context agent, GPT-5 is used to judge whether the generated answers are correct given the query and context, following prior work on LLM-based evaluation [97, 179]. For the knowledge agent, its responses are compared with ground-truth explanations provided in the datasets, and an LLM is employed to determine correctness.

Figure 4.4 indicates that all three agents produce high-quality evidence overall, while exhibiting distinct characteristics across modalities. The context agent achieves consistently strong performance across datasets (e.g., 95% on FinMR), indicating its reliability in extracting and reasoning over textual context. The knowledge agent also

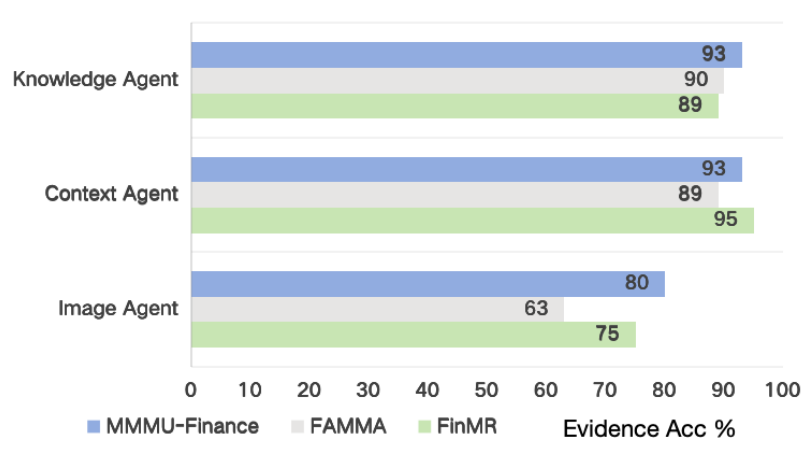


Figure 4.4: Evidence quality of different agents on FinMR, FAMMA, and MMMU-Finance. Context and knowledge agents achieve consistently high Accuracy, while the image agent shows relatively lower performance due to the challenges of fine-grained visual understanding.

performs robustly (up to 93% on MMMU-Finance), demonstrating its effectiveness in retrieving and applying domain-specific financial knowledge. In contrast, the image agent shows comparatively lower Accuracy (e.g., 63% on FAMMA), reflecting the inherent difficulty of fine-grained visual perception in financial charts and tables. Nevertheless, its performance remains reasonably strong (75% on FinMR), suggesting that it still provides useful complementary evidence, particularly when visual information is essential.

The differences across agents highlight an important insight: evidence quality is modality-dependent, and errors in multimodal reasoning often originate from weaker perception modules. This observation further justifies the design of SEA, where multiple specialized agents collaboratively contribute evidence, allowing stronger modalities (e.g., the context and knowledge agents) to compensate for weaker ones (e.g., the image agent). Overall, these results demonstrate that SEA can reliably acquire high-quality intermediate evidence across modalities, which serves as a critical foundation for improving downstream reasoning performance.

Evidence Perturbation Analysis To further examine how intermediate evidence quality affects final reasoning outcomes, this chapter conducts an evidence perturbation analysis using Claude-4.5-Sonnet as the backbone model. For each benchmark, knowledge evidence, image evidence, and textual-context evidence are independently dropped at ratios of 0%, 25%, 50%, 75%, and 100%, and the resulting final reasoning

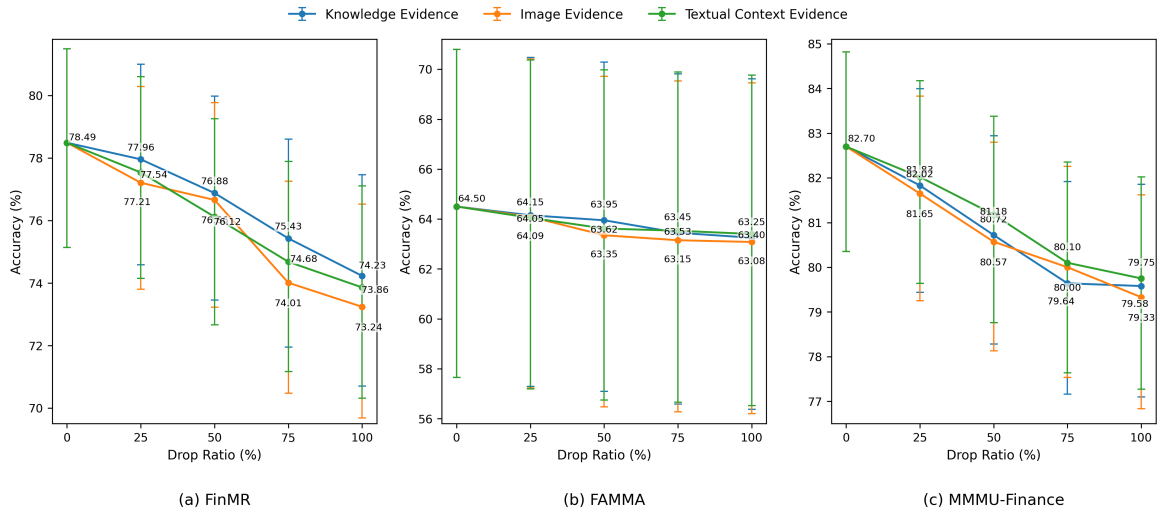


Figure 4.5: Evidence perturbation analysis under different evidence drop ratios on three benchmarks. (a) FinMR, (b) FAMMA, and (c) MMMU-Finance. For each benchmark, knowledge, image, and context evidence are progressively dropped from 0% to 100%, and the corresponding reasoning Accuracy is reported. Error bars indicate 95% Wilson confidence intervals.

Accuracy is evaluated. This controlled setting quantifies how the completeness of different evidence types affects the final answer. To provide a more statistically reliable assessment, 95% Wilson confidence intervals are reported for each Accuracy value and visualized as error bars in Figure 4.5.

As shown in Figure 4.5, Accuracy generally decreases as the evidence drop ratio increases across all three benchmarks, indicating that intermediate evidence directly supports the final reasoning process. On FinMR, Accuracy drops from 78.49% to 74.23%, 73.86%, and 73.24% when knowledge, textual-context, and image evidence are fully removed, respectively. The largest degradation comes from removing image evidence, suggesting that visual grounding is particularly important for FinMR. On FAMMA, the performance decreases from 64.50% to 63.25%, 63.08%, and 63.40% under full removal of knowledge, image, and textual-context evidence, respectively. Although the absolute drops are smaller, the consistent downward trend shows that all three evidence types contribute to the final answer. On MMMU-Finance, Accuracy decreases from 82.70% to 79.58%, 79.33%, and 79.75% when knowledge, image, and textual-context evidence are fully removed, respectively, again showing that both multimodal evidence and financial knowledge are important for reliable reasoning.

Overall, these perturbation results, together with the reported confidence intervals, provide direct empirical evidence that SEA’s effectiveness is closely tied to the

completeness of intermediate evidence. The performance degradation under increasing drop ratios confirms the utility and quality of the intermediate evidence: if the acquired evidence were mostly noisy, redundant, or weakly related to the reasoning task, removing a larger portion of it would not lead to a consistent decline in performance. Instead, the observed degradation suggests that SEA acquires evidence that contributes positively to reasoning by constraining candidate answers and supporting intermediate decisions. Since only one evidence type is removed at a time, the remaining evidence sources can still partially compensate for the missing modality, highlighting the complementary nature of SEA’s multi-agent evidence acquisition mechanism.

4.12 Cost, Latency, and Statistical Analysis

Because SEA uses multiple agents, its accuracy gains must be interpreted together with computational cost and latency. The framework controls cost through two design choices. First, the planner is allowed to terminate early when the evidence state is sufficient, so not every question uses the maximum number of reasoning steps. Second, each planner step can issue at most three evidence-seeking queries, one for each evidence source. With $N_{\max} = 6$, the theoretical upper bound on model calls for one sample is

$$1 + N_{\max}(1 + 3 + 1),$$

where the terms denote the final answer call, the planner call, up to three evidence-agent calls, and one reasoning-agent call at each step. In practice, the number of calls is lower whenever the planner stops early or does not require all evidence agents.

This bound makes the accuracy–cost trade-off explicit. Compared with single-pass CoT, SEA introduces additional model calls and therefore higher token cost and latency. Compared with debate-based systems, however, SEA’s communication is targeted rather than open-ended: each query requests a specific item of visual evidence, textual context, or financial knowledge. This makes the additional cost easier to control and audit. The framework is therefore most appropriate for settings where improved evidence grounding and reasoning reliability justify extra inference cost, such as financial education, analyst support, audit preparation, and high-value document analysis.

To further assess the reliability of the observed improvements, this chapter conducts a non-parametric statistical significance analysis based on the Accuracy scores reported in Table 4.4. Since the compared methods are evaluated under the same

Statistical Test	Comparison	Result
Friedman test	CoT vs. MAD vs. RAA vs. SEA	$\chi_F^2(3) = 55.23, p = 6.14 \times 10^{-12}$
Friedman average rank	SEA / RAA / MAD / CoT	1.00 / 2.38 / 2.69 / 3.93
Nemenyi post-hoc test	SEA vs. all baselines	Significant at $\alpha = 0.05$
Wilcoxon signed-rank test	SEA vs. CoT	$p = 9.54 \times 10^{-7}$
Wilcoxon signed-rank test	SEA vs. MAD	$p = 5.94 \times 10^{-5}$
Wilcoxon signed-rank test	SEA vs. RAA	$p = 5.93 \times 10^{-5}$

Table 4.7: Statistical significance analysis across 21 model–dataset–subset evaluation settings. The significance level is set to $\alpha = 0.05$.

model–dataset–subset setting and the normality assumption may not hold for Accuracy scores, non-parametric tests are used. Specifically, each model–dataset–subset combination is treated as one evaluation setting, resulting in 21 paired observations for each method. The Friedman test is first applied to compare whether there exist overall performance differences among CoT, MAD, RAA, and SEA. The Friedman test indicates a significant difference among the compared methods, $\chi_F^2(3) = 55.23$, $p = 6.14 \times 10^{-12}$, with the significance level $\alpha = 0.05$.

The average ranks of the methods are further compared. SEA obtains the best average rank of 1.00, followed by RAA (2.38), MAD (2.69), and CoT (3.93). The Nemenyi post-hoc test shows that the rank differences between SEA and the baseline methods are larger than the critical difference at $\alpha = 0.05$, indicating that SEA performs significantly better than the compared baselines across evaluation settings. In addition, two-sided Wilcoxon signed-rank tests further confirm the improvements of SEA over CoT ($p = 9.54 \times 10^{-7}$); MAD ($p = 5.94 \times 10^{-5}$); and RAA ($p = 5.93 \times 10^{-5}$). These results suggest that the performance gains of SEA are statistically significant under model–dataset–subset setting rather than being driven by isolated cases.

The statistical analysis in Table 4.7 evaluates aggregate paired Accuracy scores across model–dataset–subset settings. When per-sample prediction logs are available, the evaluation can be strengthened with bootstrap confidence intervals and McNemar’s test. Bootstrap confidence intervals quantify uncertainty around Accuracy by repeatedly resampling test instances with replacement. McNemar’s test is appropriate for paired binary outcomes because it compares whether two methods make different errors on the same instances. These tests complement Friedman and Wilcoxon analyses: Friedman and Wilcoxon summarize method-level differences across settings, while bootstrap and McNemar analyses inspect uncertainty and paired disagreement at the instance level.

Aspect	Treatment in SEA
Maximum calls	Bounded by $1 + N_{\max}(1 + 3 + 1)$, with $N_{\max} = 6$.
Early stopping	The planner terminates when the accumulated evidence is sufficient.
Token cost	Increases with planner, evidence-agent, and reasoning-agent calls, but remains query-bounded.
Latency	Higher than single-pass CoT, but lower than unconstrained open-ended debate in comparable settings.
Bootstrap confidence interval	Used as a resampling-based robustness check for accuracy estimates when sample-level predictions are available.
McNemar’s test	Used for paired method comparisons on the same test instances when per-sample predictions are available.
Friedman and Wilcoxon tests	Used for aggregate paired comparisons across model–dataset–subset settings, as reported in Table 4.7.

Table 4.8: Cost, latency, and statistical analysis considerations for SEA.

Overall, this cost and statistical analysis clarifies the practical boundary of SEA. The framework improves reliability by acquiring better evidence, but this improvement is not free. Its deployment should therefore be guided by task criticality, acceptable latency, token budget, and the availability of per-instance logs for statistical verification.

4.13 Error Analysis

Although SEA achieves strong performance on FinMR and MMMU-Finance, relatively lower accuracy is observed on the FAMMA benchmark. To examine the limitations of SEA on FAMMA, this chapter conducts a detailed error analysis. Following prior works [196, 21, 187], 100 error cases were randomly sampled from the FAMMA dataset. Figure 4.6 demonstrates the error distribution intuitively. The errors were categorized as follows:

Cross-Image Reasoning Errors (46%): The largest portion of errors occurs in one question with multiple charts or tables. These questions require the model to extract numerical information from several visual sources and aggregate them correctly. In many cases, the system fails to accurately combine information across different charts, leading to incorrect intermediate values and final answers. In the thesis-wide taxonomy, these cases mainly correspond to visual perception errors, evidence selection errors, cross-modal alignment errors, numerical computation errors, and logical inference errors.

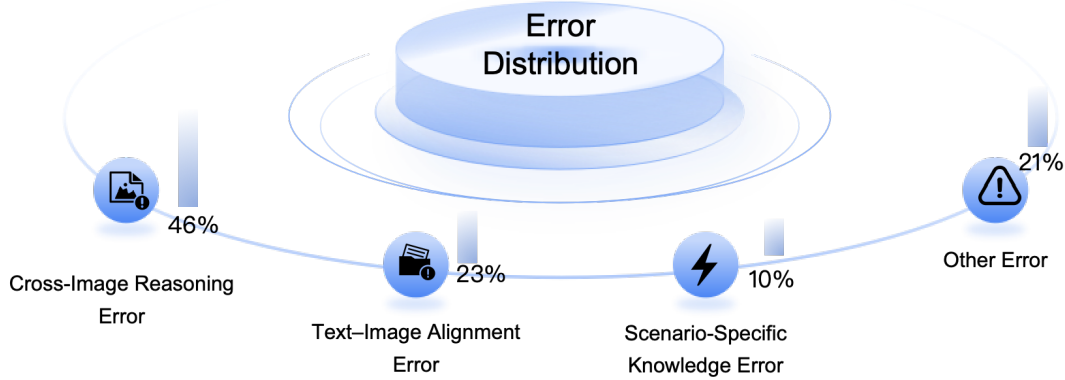


Figure 4.6: Error distribution in FAMMA.

Text-Image Alignment Errors (23%): Another common error arises when questions contain long textual contexts accompanied by images. In such cases, the model sometimes struggles to properly align textual descriptions with relevant visual elements. As a result, important information from either the text or the image may be overlooked during reasoning. These failures correspond most directly to evidence selection errors and cross-modal alignment errors.

Scenario-Specific Knowledge Errors (10%): The analysis also observes errors related to financial knowledge application. Although the system can retrieve general financial formulas or concepts, some questions require adapting these formulas to specific financial scenarios. In these cases, the model occasionally applies standard formulas without correctly accounting for contextual assumptions, leading to reasoning mistakes. These cases align with financial knowledge errors and formula or operation selection errors.

Others (21%): The remaining errors include numerical calculation mistakes, occasional reasoning inconsistencies, premature answer synthesis, and failures to express the final answer in the required format. These cases correspond to numerical computation errors, logical inference errors, and answer synthesis errors.

Beyond the above empirical categories, these errors also reveal several component-level limitations of SEA. First, the framework depends on the quality of planner-generated subtasks. When a question requires integrating multiple charts, tables, or textual constraints, a wrong query or incomplete subtask decomposition may cause the system to overlook necessary evidence. Second, although evidence-seeking agents improve modularity, they may still return incomplete or partially misaligned evidence, especially when long textual contexts must be matched with specific visual elements.

Third, the current framework does not explicitly resolve conflicts among evidence returned by different agents. As a result, contradictory or insufficient evidence may propagate to the reasoning agent and lead to incorrect intermediate reasoning. Fourth, the planner may terminate prematurely if the evidence state appears sufficient even though a hidden assumption remains unresolved. Finally, although the Knowledge Agent can retrieve general financial concepts and formulas, some questions require adapting these principles to scenario-specific assumptions, which remains challenging. These limitations suggest that future work should incorporate planner verification, cross-source evidence consistency checking, and more robust mechanisms for aggregating evidence across multiple visual and textual sources.

4.14 Discussion

This chapter presents SEA, a multi-agent framework for multimodal financial reasoning that explicitly models the interaction between task decomposition, evidence acquisition, and reasoning. By introducing a planner-guided, query-based communication protocol, SEA enables dynamic retrieval of step-conditioned evidence from heterogeneous sources, including visual, textual, and knowledge-based inputs. The key insight of SEA is that reasoning errors in multimodal financial tasks often stem from insufficient or incorrect intermediate evidence. By maintaining an explicit evidence state and grounding each reasoning step in retrieved evidence, SEA improves evidence grounding and inspectability. Empirical results show gains across multiple benchmarks, particularly in scenarios requiring cross-modal integration and domain-specific knowledge. Overall, SEA integrates perception, knowledge, and reasoning in complex multimodal settings and offers a reliability-oriented approach to evidence acquisition.

The results support the effectiveness of dynamic evidence acquisition. SEA improves over CoT, MAD, and RAA because it does not assume that the initial input already contains all information in a form the model can use. Instead, the planner repeatedly asks whether the current evidence state is sufficient and, if not, generates new queries that target the missing visual, textual, or knowledge-based information. This is especially important in financial reasoning because the relevant evidence may be a chart value, a textual assumption, a reporting-period constraint, or a formula that is not directly stated in the question. Dynamic evidence acquisition therefore changes the role of context from a passive input into an actively managed reasoning resource.

The experiments also clarify when agent specialization is useful. Specialization alone is not enough; assigning an agent the role of image reader, retriever, or critic does not guarantee reliable reasoning. The ablation and communication analyses show that specialization becomes effective when agents exchange evidence through an adaptive protocol. In SEA, the Image Evidence Agent, Context Evidence Agent, and Knowledge Evidence Agent are not asked to produce broad summaries. They respond to step-conditioned queries produced by the planner. This makes their outputs easier for the Reasoning Agent to use and reduces the chance that irrelevant or redundant evidence distracts from the current sub-task.

The comparison of communication paradigms further shows why query-based communication is central to SEA. Pipeline-based systems are efficient but rigid; controller-based systems provide routing but may still delegate overly broad tasks; debate-based systems increase interaction but can introduce unnecessary argumentation. Query-based communication occupies a middle position. It is more adaptive than a fixed pipeline, more evidence-specific than controller delegation, and more efficient than open-ended debate. This balance explains why SEA achieves strong gains without relying on training or complex external tools.

The main trade-off is cost. SEA introduces additional planner, evidence-agent, and reasoning-agent calls, so it is not intended as the cheapest possible inference method. Its value is strongest when correctness, traceability, and evidence grounding matter more than minimal latency. This boundary is important for financial settings. SEA should be viewed as a reliability-oriented reasoning framework for tasks such as financial education, analyst assistance, document interpretation, and audit preparation, rather than as a low-cost replacement for every single-pass financial QA setting.

4.14.1 Chapter-Level Limitations

While SEA demonstrates strong performance improvements, several limitations remain. First, the framework relies on multiple large model-based agents, which introduces additional computational overhead compared to single-model approaches. Although the query-based communication protocol is simple, the iterative interaction among agents may increase latency in real-world applications. Second, the effectiveness of SEA depends on the quality of the evidence-seeking agents. Errors in visual perception, information extraction, or knowledge retrieval can still propagate through the system, potentially affecting downstream reasoning. Finally, the experiments focus on financial multimodal benchmarks, and the generalization of SEA to other domains, such as scientific or medical reasoning, remains to be further explored.

4.14.2 Answer to RQ2

SEA answers RQ2 by decomposing multimodal financial reasoning into planner-guided, query-based interactions among specialized evidence agents and a reasoning agent. Instead of requiring a single model to passively consume all inputs at once, SEA enables the system to ask step-conditioned evidence queries, route them to image, context, or knowledge agents, accumulate the returned evidence, and terminate when the evidence state is sufficient for reasoning. The experimental results show that this adaptive process improves performance across FinMR, FAMMA, and MMMU-Finance, and that query-based communication outperforms pipeline-based, controller-based, and debate-based alternatives in the evaluated setting.

The chapter also clarifies the boundary of agent specialization. Specialization is useful when agents exchange targeted evidence under an adaptive protocol; role assignment alone is insufficient. SEA’s remaining errors show that relevant evidence does not guarantee correct reasoning, because models may still propagate wrong intermediate steps, misapply financial formulas, or synthesize inconsistent answers. Chapter 5 addresses this limitation through cross-MLLM error reflection.

4.15 Chapter Summary

This chapter introduced SEA, a multi-agent framework for step-wise evidence acquisition in multimodal financial reasoning. It formulated evidence acquisition as an iterative reasoning process, described specialized agents and query-based communication, and evaluated the framework through main results, ablations, evidence perturbation, communication comparisons, statistical analysis, and error analysis. The central conclusion is that adaptive evidence acquisition improves reliability, but it must be complemented by mechanisms that identify and correct reasoning-step errors.

CLER: Cross-MLLM Error Reflection for Multimodal Financial Reasoning

Chapter 4 improved evidence acquisition, but its remaining errors show that correct or relevant evidence does not automatically produce correct reasoning. Models may still misread a reasoning step, select an inappropriate operation, perform an incorrect calculation, or fail to notice that their own reflection repeats the original bias. This chapter addresses RQ3 through CLER [?]: *How can step-level errors produced by different multimodal large language models be exploited to improve financial reasoning?*

5.1 Motivation and Chapter Overview

*A smart man learns from his own mistakes,
a wise man learns from the mistakes of others.*

— Otto von Bismarck

Financial reasoning improves the Accuracy and interpretability of financial analysis by providing a structured and transparent thinking process. Large Language Models (LLMs) have shown strong performance on financial reasoning tasks [204, 192, 44], yet the knowledge-intensive and multimodal nature of financial data makes the domain especially suitable for Multimodal LLMs (MLLMs). Financial report analysis, risk assessment support, and financial document understanding particularly benefit from models capable of jointly processing text and visual evidence. However, recent benchmarks [103, 126, 51, 169, 38] reveal that mainstream MLLMs (e.g., GPT-4o, Claude-3.5) still perform poorly on financial reasoning, with Accuracy often below

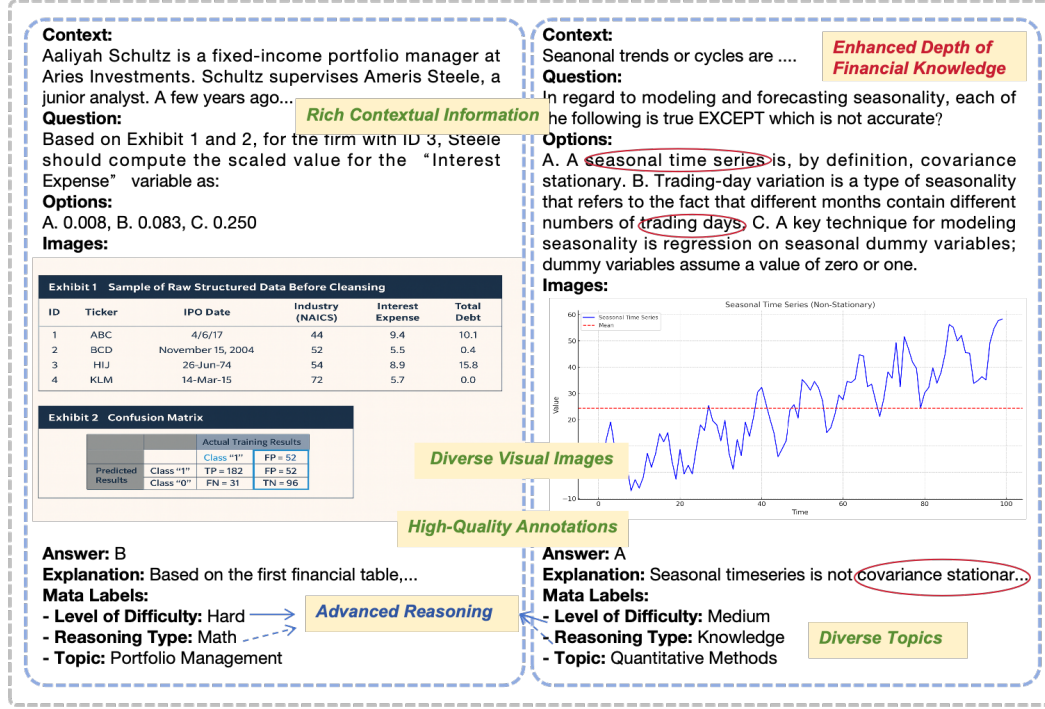


Figure 5.1: An illustrative example of financial multimodal reasoning via CoT and CLER.

60%. These evaluations highlight persistent challenges in contextual integration, logical reasoning, and numerical computation. Moreover, closed-source models, despite generally stronger performance, incur high inference costs due to the token explosion from image processing. Specifically, as the image resolution increases, the number of vision tokens rises quadratically, significantly escalating computational complexity and inference latency [190, 91, 59].

Learning from mistakes, particularly those made by others, is a cognitively grounded and empirically supported strategy for improving reasoning [109]. *Social Cognitive Theory* [13] suggests that individuals effectively acquire knowledge and behavioral strategies by observing others, a process known as *vicarious learning*. Learning from others' mistakes is also a useful strategy for mitigating the risk of costly failures. Studies on erroneous example-based learning [146, 48] further support this view, showing that exposure to incorrect examples promotes deeper understanding by prompting learners to reflect, diagnose, and revise errors. This aligns with the notion of *negative knowledge* [145], which emphasizes learning what not to do in complex tasks. Recent studies show that negative examples can guide LLMs to reflect and avoid previous errors [6, 143, 200, 184, 124].

Inspired by these cognitive foundations, this chapter explores whether MLLMs can improve financial reasoning by learning from each other’s errors. Specifically, it examines how lower-performing open-source MLLMs can provide informative negative examples that help closed-source models revise step-level reasoning in multimodal financial tasks. This raises the central question: *To what extent can MLLMs benefit from cross-model error reflection compared to naive reasoning?*

However, applying the principle of learning from mistakes across different MLLMs presents several non-trivial challenges. First, identifying and extracting meaningful error instances from model-generated reasoning processes is difficult due to the knowledge-intensive and multimodal nature of financial reasoning tasks, which involve advanced financial concepts, intricate logical and numerical reasoning, and high context-dependency. Second, transferring negative knowledge between heterogeneous MLLMs is non-trivial because their reasoning styles often differ substantially, making direct reuse of error patterns ineffective. Third, in multi-step reasoning tasks, general errors (e.g., applying incorrect formulas or misinterpreting financial logic) are often overly broad and nonspecific, offering little actionable guidance for targeted model improvement.

To address these challenges, this chapter introduces Contrastive Learning for Error Retrieval (CLER), a cross-MLLM error reflection framework that uses errors from cost-effective open-source MLLMs as external reflection signals for financial multimodal reasoning (Figure 5.1). CLER addresses these challenges by: (1) training a contrastive retrieval model that captures both semantic and structural similarities between reasoning steps to identify meaningful, task-specific error instances; (2) enabling individualized reflection by allowing each MLLM to adapt retrieved errors to its own reasoning style rather than directly transferring generalized error rules; and (3) retrieving fine-grained, step-level mistake-correction pairs, enabling targeted recognition of context-dependent errors (e.g., misapplying a formula at step 3 or omitting a negative sign at step 4). This fine-grained, context-aware error reflection supports more precise revision in multimodal financial reasoning.

The contributions are summarized as follows:

- This chapter introduces **CLER**, a cross-MLLM error reflection framework that leverages reasoning errors made by cost-effective open-source MLLMs to improve step-level reflection for closed-source MLLMs.
- This chapter introduces *FinErrorSet*, a large-scale financial reasoning error dataset consisting of over 8,000 mistake–correction pairs at the reasoning-step level.

- This chapter trains a contrastive retrieval model on *FinErrorSet* to retrieve semantically and structurally similar wrong steps from an external mistake database and use them as reflection signals.

5.1.1 Positioning Relative to Prior Work

Financial Multimodal Reasoning

In the financial sector, characterized by domain-specific expertise and sophisticated analysis, LLMs have demonstrated substantial potential in financial reasoning tasks such as financial question answering, report interpretation, and numerical reasoning [25, 158, 186, 172, 165, 195, 194, 42]. Notably, recent evaluations indicate that advanced LLMs now surpass human performance on established financial reasoning datasets, including TAT-QA [204], TabMWP [204], FinQA [25], and DocMath [195].

Beyond text-based analysis, financial multimodal reasoning introduces additional complexity and potential for innovation. Generally, financial multimodal reasoning tasks can be categorized into two types: *knowledge-driven reasoning* and *perception-driven reasoning*. Knowledge-driven reasoning [38, 51, 169, 46] requires MLLMs to leverage deep financial expertise, handle advanced financial terminologies, interpret nuanced contextual information, and analyze complex financial visual data such as charts, graphs, and tables. In contrast, perception-driven reasoning [103, 125, 52, 15] mainly assesses basic visual comprehension capabilities, requiring interpretation of simpler visual elements without extensive integration of textual context.

Despite the expectation that MLLMs approach human-level reasoning performance on multimodal tasks, empirical evaluations on financial multimodal benchmarks such as FinMR [38], FAMMA [170], and FinMME [103] demonstrate performance consistently below 60%, indicating significant limitations. These findings highlight persistent challenges faced by MLLMs in effectively integrating complex textual and visual financial data, underscoring the critical need for further advancement in multimodal financial reasoning methodologies.

Learning from Mistakes

Researchers have recently proposed various methods for integrating negative or erroneous examples to enhance the performance of LLMs [6, 143, 90, 150, 184, 138, 71]. One prominent approach involves fine-tuning LLMs with negative data to guide them away from common errors [109]. However, this strategy is impractical for closed-source models (e.g., GPT series and major MLLMs).

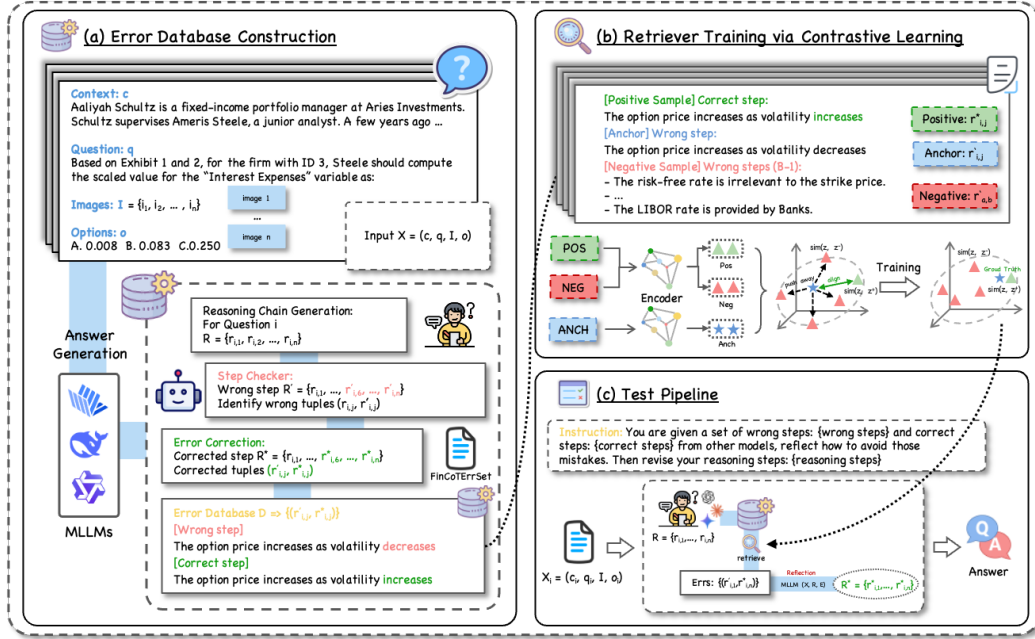


Figure 5.2: The framework of CLER: (a) error database construction, where three open-source models (Qwen-VL-2.5, DeepSeek-VL-2, and InternVL-3) solve multimodal financial problems step by step; (b) retriever training with contrastive learning; and (c) reflection and correction, where target models generate reasoning chains, retrieve similar wrong-step and corrected-step pairs from other models, reflect on whether similar errors appear in their own reasoning, and revise the answer.

An alternative to fine-tuning is to store historical mistakes in external databases and retrieve these examples dynamically to inform model reasoning. Methods such as Contrastive Chain-of-Thought [28], which explicitly provide both positive and negative reasoning examples within prompts, have been proposed. Additionally, recent techniques like LEAP [184] and RICP [138] utilize generalized insights and principles derived from past errors to guide LLM reasoning.

However, directly applying these methods to multimodal financial reasoning tasks incurs prohibitively high token-processing costs due to the complexity and size of multimodal data. Consequently, a more cost-effective strategy involves learning from errors generated by open-source models, avoiding the resource-intensive demands of closed-source multimodal models. Moreover, rather than generalizing principles broadly, adopting a step-level individualized error learning mechanism could yield more precise and effective corrections tailored to each reasoning scenario.

5.2 Problem Definition

This chapter treats multimodal financial reasoning as a process that produces both an answer and a reasoning trace. Given an input $X = (c, q, I, O)$, where c is textual context, q is a financial question, I is a set of visual inputs, and O is a set of candidate answers, a reasoner generates a trace

$$\hat{\mathcal{R}} = \{\hat{r}_1, \hat{r}_2, \dots, \hat{r}_T\}$$

and a predicted answer $\hat{A}$. A trace is useful only if its steps are grounded in the multimodal evidence and if the final answer follows from those steps. In practice, a trace may contain one or more incorrect steps even when some evidence is correct. These incorrect steps can cause error propagation, where a local mistake becomes the basis for later reasoning.

An incorrect reasoning step is defined as an intermediate step that violates at least one of the following conditions: it misreads visual evidence, selects irrelevant or incomplete evidence, aligns textual and visual evidence incorrectly, applies invalid financial knowledge, selects an inappropriate formula or operation, performs an incorrect numerical computation, draws an invalid inference, or synthesizes an answer that does not follow from the preceding steps. This definition follows the thesis-wide error taxonomy and makes the unit of correction smaller than the whole answer.

The goal of CLER is to use a database of prior step-level errors to improve the current reasoning trace. Let $\mathcal{D}$ denote an error database containing pairs (r'_k, r_k^*) , where r'_k is a wrong step generated by a source MLLM and r_k^* is a corrected version. For each current step $\hat{r}_t$, CLER retrieves a relevant pair from $\mathcal{D}$ and asks the target MLLM to reflect on whether the retrieved mistake resembles its own current step. The model then revises its reasoning only when the retrieved error provides an applicable warning or correction. The task is therefore not to copy another model’s correction, but to use another model’s failure as an external reflection signal.

This formulation differs from answer-level correction in two ways. First, the retrieved evidence concerns the reasoning process rather than only the final answer. Second, the source of reflection is cross-model rather than self-generated. This is important because self-reflection may preserve the same bias that caused the original error, while cross-model errors can expose alternative failure modes. The central hypothesis of this chapter is that structurally and semantically relevant step-level errors from diverse MLLMs can improve the reasoning reliability of a target MLLM without requiring fine-tuning of that target model.

5.3 FinErrorSet Construction

5.3.1 Question Sources and Source Models

FinErrorSet is constructed to provide step-level negative knowledge for multimodal financial reasoning. The question sources are the training splits of FinMR and FAMMA, which provide multimodal financial questions, textual context, visual inputs, options, and expert explanations. These sources are used because they cover both mathematical reasoning and financial expertise reasoning, and because their explanations make it possible to judge whether an intermediate step is consistent with the expected reasoning process.

5.3.2 Construction Pipeline Overview

The construction pipeline uses multiple open-source MLLMs as error sources. DeepSeek-VL-2, InternVL-3, and Qwen-VL-2.5 are selected because they are capable of generating multimodal reasoning traces but still exhibit diverse reasoning failures. This diversity is important: FinErrorSet is intended to capture errors that a stronger target model may not produce in exactly the same way, but that may still be structurally relevant as a warning signal.

As illustrated in Figure 5.2, CLER consists of three key stages: 1) constructing an error database (*FinErrorSet*) that records erroneous reasoning steps along with their corresponding corrections, 2) training a contrastive retriever that identifies semantically and structurally similar errors, and 3) facilitating model-specific reflection to promote effective self-correction. In the construction stage, source models first generate reasoning traces. These traces are then checked at the step level, incorrect steps are extracted, corrected versions are generated using reference explanations, and low-quality or ambiguous pairs are filtered before the final resource is used for retrieval.

5.3.3 Reasoning Trace Generation and Wrong-Step Extraction

To facilitate fine-grained learning from diverse errors, this chapter constructs an error database $\mathcal{D}$, referred to as *FinErrorSet* (see Figure 5.2(a)). To be specific, multiple open-source MLLMs, including DeepSeek-VL-2 [160], Qwen-VL-2.5 [123], and InternVL-3 [24], are employed to generate reasoning chains $\mathcal{R}$ based on the FinMR

[38] and FAMMA [169] training datasets. The corresponding reasoning chains are generated using the following prompt:

[System Input]: You are a financial expert and are supposed to answer the given questions with options, context information, and images. Put the correct option (A, B, C, or D) in []. e.g. Therefore, the correct option is [B].

[User Input]: Context: {context}; Question: {question}; Images: {images}; Options: {options}

Let's think step by step to answer the given question.

Each generated reasoning chain $\mathcal{R}$ consists of multiple intermediate steps $(r_{i,j})$, where i denotes the step index and j represents the question index. Each step is then independently evaluated by the same models acting as step checkers, which compare each generated step against ground-truth annotations, in Figure 5.2(a). Once an erroneous step $r'_{i,j}$ is identified, it is revised to obtain the corrected version $r^*_{i,j}$, forming an error-correction pair $(r'_{i,j}, r^*_{i,j})$.

[System Input]: You are a step checker, you compare your wrong reasoning steps and correct explanations. Accordingly, revise your wrong reasoning steps.

[User Input]: Context: {context}; Question: {question}; Images: {images}; Options: {options}; a set of wrong steps: {wrong steps} and correct explanations: {explanations}.

5.3.4 Filtering, Quality Control, and Human Validation

The raw mistake–correction pairs are filtered before being added to *FinErrorSet*. Pairs are removed when the wrong step is too vague to diagnose, when the correction merely paraphrases the original step, when the error cannot be grounded in the question or reference explanation, or when the pair depends on information unavailable to the target reasoner. This filtering process is important because CLER relies on retrieved errors as reflection signals; noisy or ambiguous errors can cause negative transfer.

Quality control is performed at two levels. At the local level, each wrong step must identify a concrete process-level mistake, such as an incorrect formula, a mismatched unit, or an invalid visual interpretation. At the pair level, the corrected step must directly address the wrong step and remain consistent with the reference explanation. A separate human validation study is reported in Section 5.11, where sampled mistake–correction pairs are evaluated for step validity and usefulness as reflection examples.

5.4 Dataset Statistics and Error Taxonomy

	Number	Ratio
Total Questions	3258	–
Total Wrong Steps	8852	–
Reasoning Type		
* Expertise Reasoning	5138	58.0%
* Math Reasoning	3714	42.0%
Source Models		
* DeepSeek-VL-2	2668	30.1%
* InternVL-3	2689	30.4%
* Qwen-VL-2.5	3495	39.5%

Table 5.1: Statistics of the Error Database: *FinErrorSet*.

Formally, the final database can be represented as: $\mathcal{D} = \{(r'_{1,1}, r^*_{1,1}), \dots, (r'_{i,j}, r^*_{i,j})\}$. Table 5.1 shows the statistics of *FinErrorSet*. It captures a diverse set of error types produced by different MLLMs, including numerical miscalculations, financial logic misunderstandings, and incorrect multimodal integration. This structured error database serves as the basis for the retrieval model, supporting fine-grained reflection and correction for other models.

FinErrorSet contains 3,258 questions and 8,852 wrong reasoning steps. Among the wrong steps, 5,138 are associated with expertise reasoning and 3,714 with mathematical reasoning. This distribution shows that the resource is not limited to arithmetic mistakes; it also covers conceptual and interpretive failures that are common in financial expertise tasks. The source-model distribution is intentionally diverse: DeepSeek-VL-2 contributes 2,668 wrong steps, InternVL-3 contributes 2,689 wrong steps, and Qwen-VL-2.5 contributes 3,495 wrong steps. This diversity supports the core assumption of CLER: different MLLMs expose different failure modes, and these failure modes can serve as useful cross-model reflection signals.

The error taxonomy follows the thesis-wide taxonomy introduced in Chapters 3 and 4. Each wrong step can be interpreted through one or more of the following categories: visual perception error, evidence selection error, cross-modal alignment error, financial knowledge error, formula or operation selection error, numerical computation error, logical inference error, and answer synthesis error. This alignment is important for thesis coherence. FinMR uses the taxonomy to diagnose base-model failures, SEA

uses it to analyze remaining evidence-related failures, and CLER uses it to define the step-level errors stored in *FinErrorSet*.

The taxonomy also clarifies why step-level annotation is necessary. A final wrong answer may be caused by a visually misread chart, an omitted textual constraint, a correct formula applied to the wrong variable, or a correct calculation interpreted in the wrong direction. These failures require different corrections. By storing wrong steps and corrected steps, *FinErrorSet* makes the correction target explicit and allows CLER to retrieve errors that match the current reasoning step rather than only the current question.

5.5 CLER Framework

CLER improves multimodal financial reasoning through a five-stage pipeline:

Current Reasoning $\rightarrow$ Error Representation $\rightarrow$ Cross-Model Error Retrieval
 $\rightarrow$ Reflection $\rightarrow$ Revision.

The target MLLM first generates a current reasoning trace for a financial multimodal question. Each step in that trace is represented by a learned encoder. The retriever searches *FinErrorSet* for a structurally and semantically relevant mistake–correction pair produced by another MLLM. The retrieved pair is not treated as a new ground truth answer; instead, it is used as a warning that helps the target model diagnose whether its own reasoning step may contain a similar error. Finally, the model revises its reasoning and generates a final answer from the revised trace.

This framework deliberately separates retrieval from reflection. Retrieval identifies a potentially relevant past error, while reflection decides whether and how that error applies to the current reasoning step. This separation prevents CLER from simply copying corrections from unrelated examples. It also makes the framework compatible with closed-source MLLMs: the target model does not need to be fine-tuned, and the error memory can be updated externally as new source-model errors are observed.

The design of CLER is complementary to SEA. SEA improves the acquisition of evidence before and during reasoning, whereas CLER checks whether the reasoning trace produced from available evidence resembles known step-level mistakes. The framework therefore targets the residual error mode identified at the end of Chapter 4: relevant evidence can still lead to an incorrect answer when the model applies a wrong

operation, performs an invalid inference, or fails to recognize its own flawed reasoning step.

5.6 Error Representation and Contrastive Learning

Different models exhibit distinct reasoning styles due to variations in their architectures, training datasets, internal mechanisms, and development methodologies. To robustly identify and retrieve meaningful reasoning errors from the meticulously constructed error database $\mathcal{D}$ (*FinErrorSet*), this chapter employs a contrastive learning approach. Specifically, it trains a contrastive retrieval model designed to accurately capture and retrieve reasoning errors that are semantically and structurally similar in nature. Figure 5.2(b) outlines the training approach, with detailed parameter settings provided in Algorithm 2. Figure 5.1 further illustrates an example where the retriever successfully identifies a relevant erroneous reasoning step along with its corresponding correction, demonstrating the retriever’s effectiveness in capturing fine-grained reasoning similarities across diverse financial scenarios.

5.7 Reflection and Reasoning Revision

When presented with the same error cases, individuals often exhibit diverse interpretations and reasoning paths. Similarly, different LLMs demonstrate varying reflective behaviors when addressing the same erroneous reasoning. To avoid ineffective guidance resulting from the homogenization of principles, this chapter introduces a model-specific reflection strategy. In particular, models are encouraged to engage in individual reflection, guided by referenced error cases, enabling them to autonomously refine their reasoning chains and improve their answers. In the inference stage, the MLLM generates a multi-step reasoning chain $\mathcal{R} = \{r_1, r_2, \dots, r_T\}$. For each reasoning step $r_t \in \mathcal{R}$, the retrieval model queries the constructed error database $\mathcal{D}$ to identify the top-1 semantically and structurally similar erroneous step $r'_{i,j}$ and its corresponding correction $r^*_{i,j}$.

The MLLM then performs error reflection by analyzing and summarizing these retrieved pairs and generating a corrective insight $\mathcal{I}(r_t, r'_{i,j}, r^*_{i,j})$. This insight is applied to revise the current reasoning step via a reflection function:

$$\hat{r}_t = \text{Self-correction}\left(r_t, \mathcal{I}(r_t, r'_{i,j}, r^*_{i,j})\right),$$

Algorithm 2 Retriever Training via Contrastive Learning

Input: Error database $\mathcal{D} = \{(r'_{i,j}, r^*_{i,j})\}$, batch size $B = 128$, epochs $E = 5$, temperature $\tau = 0.07$, encoder $En(\cdot)$: Sup-SimCSE

Output: Trained encoder $En(\cdot)$

```

1: Initialize encoder  $En(\cdot)$ 
2: for epoch  $e = 1, 2, \dots, E$  do
3:   Shuffle error database  $\mathcal{D}$ 
4:   for each batch  $\{(r'_i, r^*_i)\}_{i=1}^B \subseteq \mathcal{D}$  do
5:     Encode anchor and positive  $\mathbf{e}_i = En(r'_i)$ ,  $\mathbf{e}_i^+ = En(r^*_i)$ 
6:      $\mathcal{L}_{\text{batch}} \leftarrow 0$ 
7:     for  $i = 1, \dots, B$  do
8:        $\text{sim}_i^+ = \frac{\mathbf{e}_i \cdot \mathbf{e}_i^+}{\|\mathbf{e}_i\| \|\mathbf{e}_i^+\|}$ 
9:        $\mathcal{N}_i \leftarrow \{En(r'_{a,b}) \mid (a,b) \neq (i,j)\}$ 
10:      Compute  $\text{sim}_i^- = \{\cos(\mathbf{e}_i, \mathbf{e}^-) \mid \mathbf{e}^- \in \mathcal{N}_i\}$ 
11:       $\text{denom}_i = \exp(\text{sim}_i^+/\tau) + \sum_{\text{sim} \in \text{sim}_i^-} \exp(\text{sim}/\tau)$ 
12:       $\mathcal{L}_i = -\log \frac{\exp(\text{sim}_i^+/\tau)}{\text{denom}_i}$ 
13:       $\mathcal{L}_{\text{batch}} \leftarrow \mathcal{L}_{\text{batch}} + \mathcal{L}_i$ 
14:    end for
15:     $\mathcal{L}_{\text{batch}} \leftarrow \mathcal{L}_{\text{batch}}/B$ 
16:    Update  $En(\cdot)$  via gradient descent on  $\mathcal{L}_{\text{batch}}$ 
17:  end for
18: end for
19: return trained encoder  $En(\cdot)$ 

```

yielding the refined reasoning chain $\hat{\mathcal{R}} = \{\hat{r}_1, \hat{r}_2, \dots, \hat{r}_T\}$. This reflection mechanism enables step-wise self-correction, allowing the MLLM to iteratively revise flawed reasoning steps and improve accuracy. Below is the reflection prompt template.

[**System Input**]: You are given a set of wrong steps: wrong steps and correct steps: correct steps from other models, reflect how to avoid those mistakes. Then revise your reasoning steps: reasoning steps

[**User Input**]: Context: {context}; Question: {question}; Images: {images}; Options: {options}.

5.8 Experimental Setup

5.8.1 Benchmarks

This chapter evaluates the proposed approach on three financial multimodal reasoning benchmarks, each requiring integration of domain-specific knowledge, textual compre-

hension, and visual perception, such as tables and charts: 1) **FinMR** [38]: An expert-level benchmark with 3,200 questions that require complex financial reasoning. Each instance includes detailed textual context, one or more financial images, and expert-annotated explanations. 2) **FAMMA** [169]: A multilingual financial multimodal reasoning benchmark designed to reflect college-level expertise. Each financial question is paired with one or more relevant images, requiring advanced domain knowledge to support sophisticated reasoning. 3) **MMMU-Finance** [181]: A specialized subset of the open-domain Multi-discipline Multimodal Understanding benchmark, comprising 1,110 samples selected from Finance, Economics, and Accounting domains¹. Notably, 95% of the visual inputs in this subset are financial tables.

5.8.2 Baselines

This chapter considers the following baseline methods: 1) **CoT** [156]: A method that instructs LLMs to generate intermediate reasoning steps (i.e., *Let’s Think Step by Step*) before producing the final answer. 2) **LEAP** [184]: This method first generates incorrect reasoning samples using zero-shot CoT responses, then derives explicit guiding principles from them alongside the correct answers. During inference, the model leverages both the learned principles and provided examples to improve answer Accuracy. 3) **RICP** [138]: This method employs a teacher-student framework, where the teacher model identifies and analyzes the student’s mistakes. During inference, relevant error cases are dynamically retrieved to construct question-specific principles for targeted guidance.

5.8.3 Implementation Details

The error database is constructed using three open-source source models: Qwen-VL-2.5 [123], DeepSeek-VL-2 [160], and InternVL-3 [24]. These source models are selected because they can generate multimodal reasoning traces while exhibiting diverse step-level financial reasoning errors. All experiments are conducted using four NVIDIA A100 GPUs. The target reasoners are GPT-4o [115], Gemini-2.0-Flash [35], and Claude-4-Sonnet [7]. They are selected as closed-source target reasoners because they provide strong multimodal instruction following and are suitable for testing whether cross-model errors can improve stronger models without fine-tuning. The temperature parameter is set to 0.7 for all closed-source target models.

¹In the broader domain of Finance, topics such as Economics and Accounting are two closely related subfields that underpin advanced financial reasoning tasks.

5.8.4 Evaluation Protocol

Accuracy is used as the primary metric because all three benchmarks are evaluated as multiple-choice financial reasoning tasks. For each question, the model is instructed to provide a reasoning trace and place the final option in square brackets. The final answer is parsed from the last valid option marker in the response. When a response contains multiple option markers, the final explicit answer is used; when no valid option marker can be parsed, the response is treated as incorrect. Improvements are reported as absolute percentage-point gains over the corresponding CoT baseline.

The evaluation separates overall performance from reasoning-type performance whenever benchmark annotations support this distinction. FinMR and FAMMA are reported with overall, mathematical reasoning, and financial expertise reasoning results. MMMU-Finance is reported as a math-heavy finance subset. This separation is important because CLER is expected to help both formula-driven calculation and expertise-driven interpretation, but the error mechanisms are not identical. Mathematical questions often fail through operation selection, numerical computation, or unit alignment, whereas expertise questions more often involve financial knowledge, cross-modal alignment, or answer synthesis.

The model order and dataset order are kept consistent with the rest of the thesis. The datasets are reported as FinMR, FAMMA, and MMMU-Finance. The target reasoners are reported as GPT-4o, Gemini-2.0-Flash, and Claude-4-Sonnet. This consistent ordering supports readability across Chapters 3–5, but the numerical results are not treated as direct cross-chapter model-version comparisons when the version names differ. For example, Chapter 4 evaluates Gemini-2.5-Flash and Claude-4.5-Sonnet, whereas this chapter evaluates Gemini-2.0-Flash and Claude-4-Sonnet. Exact API access dates are not available in the source study and are therefore treated as a reproducibility limitation. The results should be interpreted as evidence about CLER under the model versions named in this chapter, not as immutable properties of the model families.

During inference, CLER retrieves the top-1 most relevant mistake–correction pair for each reasoning step unless otherwise stated in the ablation analysis. This setting is chosen to balance usefulness and prompt cost. More retrieved examples may provide broader coverage, but they also increase the risk of irrelevant guidance and raise token cost. The reflection prompt therefore asks the target model to use the retrieved pair as a diagnostic warning rather than as an instruction that must be followed.

Model	Method	FinMR (Accuracy %)			FAMMA (Accuracy %)			MMMU-Finance (Accuracy %)
		ALL	Math(37.8%)	Expertise(62.2%)	ALL	Math(50%)	Expertise(50%)	Math
GPT-4o	CoT	54.17	49.40	57.07	54.50	53.00	56.00	62.77
	LEAP	57.66 (↑3.49)	42.97 (↓6.43)	66.59 (↑9.52)	54.50 (-)	49.00 (↓4.00)	60.00 (↑4.00)	63.56 (↑0.79)
	RICP	58.12 (↑3.95)	51.00 (↑1.60)	62.44 (↑5.37)	58.00 (↑3.50)	55.00 (↑2.00)	61.00 (↑5.00)	63.99 (↑1.22)
	CLER(this chapter)	61.18 (↑7.01)	53.41 (↑4.01)	65.90 (↑8.83)	60.00 (↑5.50)	56.00 (↑3.00)	64.00 (↑8.00)	66.78 (↑4.01)
Gemini-2.0-Flash	CoT	56.15	50.20	59.76	56.50	53.00	60.00	56.85
	LEAP	57.81 (↑1.66)	49.80 (↓0.40)	62.68 (↑2.92)	61.00 (↑4.50)	58.00 (↑5.00)	64.00 (↑4.00)	58.03 (↑1.18)
	RICP	59.64 (↑3.49)	52.61 (↑2.41)	63.90 (↑4.14)	62.50 (↑6.00)	60.00 (↑7.00)	65.00 (↑5.00)	59.77 (↑2.92)
	CLER(this chapter)	62.12 (↑5.97)	57.03 (↑6.83)	65.21 (↑5.45)	65.00 (↑8.50)	64.00 (↑11.00)	66.00 (↑6.00)	60.66 (↑3.81)
Claude-4-Sonnet	CoT	57.36	48.59	62.28	60.00	59.00	61.00	63.78
	LEAP	60.89 (↑3.53)	52.21 (↑3.62)	66.10 (↑3.82)	61.00 (↑1.00)	57.00 (↓2.00)	65.00 (↑4.00)	65.72 (↑1.94)
	RICP	61.91 (↑4.55)	53.41 (↑4.82)	67.07 (↑4.79)	64.00 (↑4.00)	62.00 (↑3.00)	66.00 (↑5.00)	66.66 (↑2.88)
	CLER(this chapter)	65.10 (↑7.74)	56.63 (↑8.04)	70.24 (↑7.96)	67.00 (↑7.00)	65.00 (↑6.00)	69.00 (↑8.00)	68.77 (↑4.99)

Table 5.2: Result comparison of different models and methods across all datasets. MMMU-Finance consists of math reasoning questions. ↑ for increase, ↓ for decrease. Bold indicates the best result in each column. Underlined values indicate a drop.

5.9 Main Results

As shown in Table 5.2, CLER outperforms the selected baselines across all datasets in this evaluation. CLER’s improvement over the CoT baseline supports the usefulness of fine-grained contrastive learning combined with error reflection. On FinMR (Expertise), CLER achieves 65.90% Accuracy with GPT-4o, representing an 8.83 percentage-point gain over CoT. Both LEAP and RICP generally outperform the CoT baseline, confirming that explicit error analysis and retrieved contextual principles can improve model performance. CLER further improves on these methods by using step-level, structurally matched errors for reflection.

When applying LEAP, several performance drops are also observed (indicated with underlines), such as GPT-4o on FinMR Math subset (-4.43%), Gemini-2.0-Flash on FinMR Math (-0.40%), and Claude-4-Sonnet on FAMMA Math (-2.00%). These drops typically occur when retrieved error cases lack structural similarity to the current reasoning step, resulting in misleading principles. Such issues are common in baseline methods, highlighting the limitations of surface-level textual similarity without incorporating deeper logical structures. In contrast, CLER captures structural-semantic similarity, which reduces misleading retrieval in the evaluated setting.

The MMMU-Finance dataset, which is dominated by math-based reasoning tasks, exhibits a slightly higher baseline Accuracy. Nevertheless, CLER achieves meaningful gains, such as a 4.01 percentage-point improvement over CoT with GPT-4o, showing that step-level error reflection remains useful even in homogeneous, math-intensive reasoning settings. On datasets requiring substantial financial expertise (e.g., FinMR and FAMMA’s expertise categories), CLER demonstrates particularly strong gains, underscoring the benefits of detailed, fine-grained error reflection in complex multi-modal reasoning tasks.

Claude-4-Sonnet outperforms the other target reasoners in this experiment, indicating strong intrinsic reasoning capabilities in financial multimodal tasks. Despite this intrinsic strength, CLER further improves performance, supporting the applicability of cross-model error reflection across the evaluated target reasoners.

5.10 Ablation Studies

To assess the contribution of each module in CLER, this chapter conducts a comprehensive ablation study across all financial reasoning datasets using GPT-4o.

- **w/o CL Retriever & Reflection:** the CoT baseline without retrieval or reflection;
- **w/o Reflection:** step-level retrieval is performed, but MLLMs do not reflect on the retrieved information;
- **w/o CL Retriever:** reflection is retained, but the retriever is replaced by a semantic similarity-based one;
- **CLER:** the full method combining contrastive retrieval with step-level reflection.

Table 5.3 reports the performance of four configurations across all datasets. Consistent performance gains are observed as more components of CLER are activated. Specifically, FinMR-All improves from 54.17% to 61.18%, while FinMR-Math increases from 49.40% to 53.41%. Similar trends are observed on FAMMA and MMMU-Finance, with gains up to +3.78% over the baseline on math-heavy subsets (e.g., 66.78% on MMMU-Math vs. 62.77%).

These results reveal three key insights: 1) Reflection is essential, as removing it results in significant drops across all subsets (e.g., FinMR-Expertise: 65.90% $\rightarrow$ 60.33%), 2) The contrastive retriever consistently outperforms semantic-based retrieval, underscoring the value of structural alignment in retrieving relevant error patterns, 3) Gains are particularly evident in math-heavy subsets, such as FinMR-Math (49.40% $\rightarrow$ 53.41%) and MMMU-Math (62.77% $\rightarrow$ 66.78%), indicating that CLER is especially effective in correcting step-level numerical reasoning errors. These tasks often involve formula misapplication, incorrect intermediate steps, or numerical condition confusion – error types that benefit most from fine-grained structural retrieval and targeted insights. Overall, this ablation confirms that both modules — contrastive retrieval and reflection — are indispensable and synergistic within CLER.

Table 5.3: Ablation study results on GPT-4o across FinMR, FAMMA, and MMMU-Finance.

Method	FinMR ALL	FinMR Math	FinMR Expertise	FAMMA ALL	FAMMA Math	FAMMA Expertise	MMMU Math
w/o CL + Ref	54.17	49.40	57.07	54.50	53.00	56.00	62.77
w/o Reflection	56.78 (+2.61)	50.93 (+1.53)	60.33 (+3.26)	55.50 (+1.00)	54.00 (+1.00)	57.00 (+1.00)	62.99 (+0.22)
w/o CL Retriever	58.52 (+4.35)	51.03 (+1.63)	63.07 (+6.00)	58.00 (+3.50)	54.00 (+1.00)	62.00 (+6.00)	63.00 (+0.23)
CLER	61.18 (+7.01)	53.41 (+4.01)	65.90 (+8.83)	60.00 (+5.50)	56.00 (+3.00)	64.00 (+8.00)	66.78 (+4.01)

Beyond the four main variants in Table 5.3, the ablation design examines several choices that determine whether error reflection is useful or misleading. The first is *same-model versus cross-model retrieval*. Same-model retrieval uses errors generated by the same source model family whenever available, whereas cross-model retrieval deliberately draws errors from different MLLMs. The cross-model setting is more consistent with the motivation of this chapter: a target model may benefit from error patterns that it did not generate itself. It also reduces the risk that reflection merely reinforces the same model-specific bias.

The second comparison is *step-level versus answer-level reflection*. Answer-level reflection retrieves or summarizes complete failed examples, but it does not identify where the reasoning failed. Step-level reflection retrieves a concrete wrong step and its correction, allowing the target model to inspect formula choice, visual interpretation, calculation, and answer synthesis separately. The experimental trend supports the step-level design: the strongest gains appear when the model receives fine-grained mistake–correction pairs rather than broad final-answer feedback.

The third comparison concerns the retriever. Semantic-only retrieval can return examples that use similar financial vocabulary while requiring different reasoning operations. Random retrieval provides a lower bound that tests whether any negative example is useful simply because it encourages caution. The contrastive retriever outperforms these alternatives because it is trained to align wrong steps with corrected steps while separating unrelated errors within the batch. This is especially important for financial reasoning, where superficially similar terms may imply different formulas, time periods, or units.

The fourth comparison examines the number of retrieved errors. Retrieving a single highly relevant pair gives the target model a focused warning and keeps the prompt compact. Retrieving more examples can increase coverage, but it may also introduce conflicting signals. In practice, CLER’s top-1 setting provides a favourable balance between correction benefit and inference cost. The database-size analysis in Figure 5.4 further shows that retrieval quality depends not only on the number of

examples placed in the prompt, but also on the breadth and diversity of the external error memory.

Together, these ablations show that CLER’s improvement is not caused by reflection prompting alone. The gains rely on the interaction of four elements: cross-model error diversity, step-level granularity, contrastive retrieval, and model-specific revision. Removing any of these elements weakens the correction signal or increases the risk of negative transfer.

5.10.1 Additional Ablations and Sensitivity Analyses

This section examines CLER in more detail to better understand *why* it achieves its improvements.

Individualized Insights of Reflection

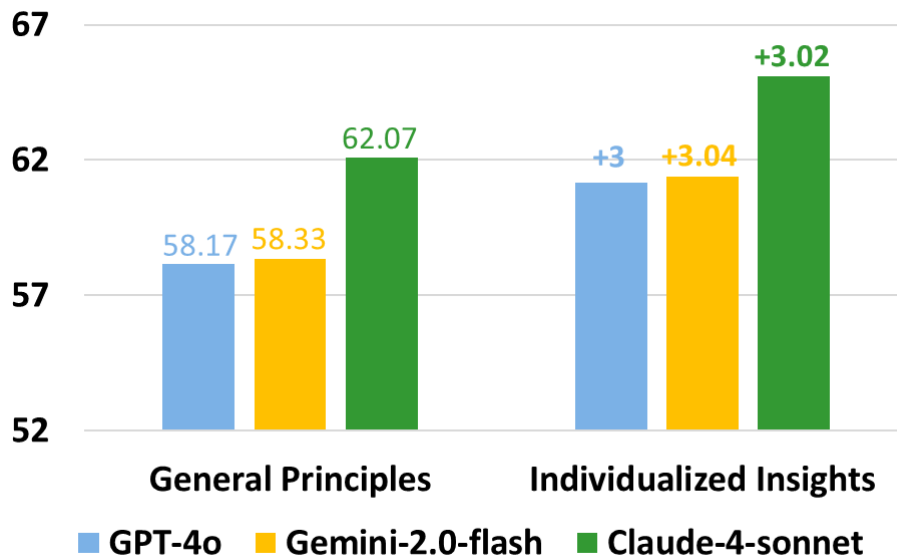


Figure 5.3: Performance Difference between General Principles and Individualized Insights on FinMR.

In the final stage of CLER, although error-correction pairs are retrieved from *FinErrorSet*, the model does not directly adopt the principles generated by others. This is by design: principles derived from one model’s reasoning may not generalize well to another, due to differences in reasoning styles and error patterns. Instead, CLER encourages models to reflect *independently* on each retrieved pair. By comparing each retrieved erroneous and corrected step to its current reasoning, the model generates

insights tailored to its own reasoning trajectory. This design enables the model to generate context-sensitive corrections aligned with its own reasoning structure.

To assess the impact of this design, this chapter compares general principles (pre-generated by open-source models before testing) with individualized insights (generated by closed-source models dynamically at test time). As shown in Figure 5.3, tailored insights consistently improve performance across all models, with gains of 3.0% for GPT-4o, 4.0% for Gemini-2.0-Flash, and 3.02% for Claude-4-Sonnet. This confirms that *individualized reflection is more effective than merely borrowing fixed principles from others*.

Impact of Using Different Versions of MLLMs

Table 5.4: Improvements from Claude-3.7-Sonnet to Claude-4.0-Sonnet across financial datasets.

Dataset	v3.7	v4.0	Gain
FinMR	62.85	65.10	+2.25
FinMR-Math	54.21	56.63	+2.42
FinMR-Expertise	68.10	70.24	+2.14
FAMMA	65.50	67.00	+1.50
FAMMA-Math	64.00	65.00	+1.00
FAMMA-Expertise	67.00	69.00	+2.00
MMMU-Finance	67.72	68.77	+1.05

To examine the effect of model version, this chapter compares Claude-3.7-Sonnet and Claude-4.0-Sonnet, two successive releases from the same model family within the CLER setting. As shown in Table 5.4, Claude-4.0-Sonnet outperforms Claude-3.7-Sonnet across all evaluated benchmarks. Specifically, on FinMR overall, Claude-4.0-Sonnet achieves 65.10% Accuracy compared to 62.85% for Claude-3.7-Sonnet, reflecting a 2.25 percentage-point improvement. Performance gains are also evident in both FinMR-Math (2.42 percentage points) and FinMR-Expertise (2.14 percentage points), indicating improved capability in numerical and domain-specific reasoning in this experiment. These results should be interpreted as a within-family version sensitivity analysis rather than as a direct comparison with Chapter 4, which uses Claude-4.5-Sonnet. They further motivate the thesis-wide caution that model-version drift and missing access-date metadata limit reproducibility for commercial MLLM evaluations.

Benefit of Error Database Size

To evaluate the impact of error database size on retrieval-augmented reasoning, this chapter varies the size of the error database used in the CLER framework. Specifically,

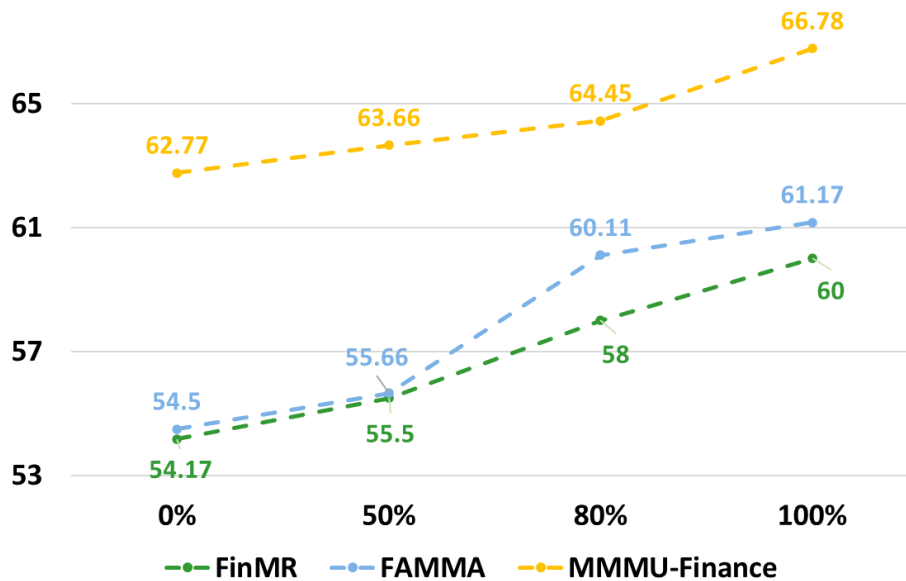


Figure 5.4: Effect of error database size on Accuracy across FinMR, FAMMA, and MMMU-Finance.

three versions of the database are constructed, containing 100%, 80%, and 50% of the original error instances. In each setting, the database is randomly subsampled while preserving the original error-type distribution to ensure fairness. As shown in Figure 5.4, reducing the size of the error database results in a clear and consistent decline in performance across all benchmarks. These results demonstrate that database scale is critical for effective retrieval and correction. A larger database not only increases the likelihood of retrieving highly relevant errors but also offers more diverse and informative counterexamples, thereby enhancing model reflection and learning. These findings highlight the importance of maintaining a comprehensive and well-curated error memory in retrieval-augmented reasoning systems.

5.11 Human Evaluation

To assess the quality of the step-level error resource, this chapter includes a human validation study on 200 sampled mistake-correction pairs from *FinErrorSet*. The purpose of the study is to verify whether the automatically identified wrong steps and revised steps are suitable for use as external reflection signals. This is important because CLER depends not only on the size of the error database, but also on the correctness and usefulness of the stored error pairs.

The annotation protocol contains three criteria. First, annotators judge whether the marked wrong step is genuinely incorrect in the context of the question, image, options, and reference explanation. Second, they judge whether the corrected step addresses the identified error rather than merely rewriting the original sentence. Third, they assess whether the mistake–correction pair is useful as a reflection example for another model facing a similar reasoning step. Each sample is therefore evaluated at the level of process validity, not only final-answer correctness. Agreement is assessed by comparing judgements on these three criteria before adjudication, and only samples with consistent or adjudicated judgements are counted in the reported validation result.

The human validation result indicates that more than 93% of the sampled mistake–correction pairs are judged valid for step-level error reflection. This supports the reliability of *FinErrorSet* as an external error memory. It also suggests that errors generated by open-source MLLMs can provide useful negative knowledge for stronger closed-source MLLMs, provided that errors are filtered, corrected, and retrieved at the reasoning-step level.

This validation has limitations. The evaluation uses a 200-sample subset rather than the full error database, so it estimates rather than exhaustively proves the quality of all entries. The annotation also focuses on local step validity and does not fully evaluate cross-step interactions, where several individually plausible steps may still form an invalid reasoning trajectory. Finally, the judgement of usefulness for reflection may depend on the target model and benchmark. These limitations motivate future work on larger human validation studies, inter-annotator agreement reporting, and continuous updating of the error database as new model failures are observed.

5.12 Error Analysis

Although CLER improves performance across FinMR, FAMMA, and MMMU-Finance, its remaining errors show that step-level reflection is not sufficient for every multimodal financial reasoning failure. This section analyzes the main success and failure patterns observed after applying CLER. The analysis follows the thesis-wide error taxonomy used in Chapters 3 and 4, while focusing on the specific behaviour of cross-model error retrieval and reflection.

Successful correction. CLER is most effective when the current reasoning step and the retrieved error share the same underlying financial operation. Typical successful cases involve formula or operation selection errors, numerical computation errors,

and answer synthesis errors. For example, when a target model uses the wrong denominator in a ratio calculation, a retrieved mistake–correction pair can warn the model that the relevant quantity should be normalized by total assets rather than revenue. Similarly, in chart-based questions, retrieved errors can help the model notice that a visual trend should be interpreted relative to a specific reporting period rather than as an overall increase or decrease. These cases illustrate the value of step-level retrieval: the retrieved pair gives a concrete warning at the exact point where the trace is vulnerable.

Irrelevant error retrieval. Some failures occur when the retriever returns a mistake that is semantically similar but structurally different. This is especially common when two reasoning steps share financial terms such as margin, return, revenue growth, or cash flow, but require different operations. In such cases, the retrieved correction may encourage the model to reconsider a step that is already valid, or it may redirect attention to an irrelevant financial relation. This failure mode shows that retrieval relevance must be judged at the level of reasoning structure, not only lexical overlap. The contrastive retriever reduces this risk compared with semantic-only retrieval, but it cannot eliminate it completely.

Negative transfer. Retrieved errors can also cause negative transfer when the target model over-applies a correction. A correction that is valid in the source example may not be valid in the current context because of a different unit, time period, accounting convention, or answer format. For instance, an error pair about using year-over-year percentage change may be misleading for a question that asks for an absolute difference. Negative transfer is therefore not simply a retrieval problem; it is also a reflection problem. The target model must decide whether the retrieved warning applies to the current trace rather than treating it as an instruction to revise.

Unseen financial knowledge. CLER depends on the coverage of *FinErrorSet*. When a question requires a financial concept that is absent from the error database, the retrieved error may be only weakly related. This affects specialized topics where the decisive step depends on domain knowledge rather than general arithmetic, such as interpreting a specific accounting ratio, understanding a risk disclosure, or distinguishing between similar valuation concepts. In such cases, CLER may still encourage more cautious reflection, but it cannot introduce missing financial knowledge unless that knowledge is present in the retrieved correction or already known by the target model.

Multiple interacting errors. Another difficult case occurs when several local mistakes interact. A reasoning trace may first select incomplete evidence, then apply a

partially correct formula, and finally synthesize an answer from inconsistent intermediate values. CLER retrieves errors step by step, so it can correct one vulnerable step, but the revised trace may still contain earlier or later inconsistencies. This limitation is important for multimodal financial reasoning because visual perception, evidence selection, formula choice, and numerical computation often depend on each other. Future systems may need joint trace-level verification in addition to step-level error reflection.

Correct answer with wrong reasoning. Some model outputs reach the correct final option despite containing flawed intermediate reasoning. These cases are challenging because answer-level evaluation would mark the output as correct, while process-level analysis reveals an unreliable explanation. CLER is useful in such cases because it inspects reasoning steps directly, but the method still depends on the quality and faithfulness of the generated trace. If the trace does not expose the model’s true decision process, reflection can only revise the visible explanation. This reinforces the broader thesis argument that reliable financial reasoning should be evaluated through evidence, process, and answer correctness rather than final-answer Accuracy alone.

Overall, the error analysis indicates that CLER is strongest as a process-level correction mechanism. It can reduce local reasoning failures by exposing the target model to relevant cross-model mistakes, especially when those mistakes involve formula choice, numerical operations, or answer synthesis. Its limitations arise when retrieval is weak, when the source error is not transferable, when required financial knowledge is absent, or when several errors interact across the trace. These findings motivate the integrated reliability discussion in Chapter 6, where error reflection is combined with evaluation and evidence acquisition rather than treated as a standalone guarantee of correctness.

5.13 Discussion

This chapter introduces CLER, a cross-MLLM error reflection framework for multimodal financial reasoning. CLER retrieves step-level mistake–correction pairs from a curated error database and prompts model-specific reflection, enabling context-aware reasoning revision. It also introduces *FinErrorSet*, which is constructed from open-source MLLM errors and therefore reduces the cost of preparing an external error resource with commercial closed-source models. Experiments on financial benchmarks show that CLER improves over strong baselines in both math-heavy and expertise-driven tasks, with ablation studies confirming the complementary roles of contrastive

retrieval and structured reflection. The contribution is bounded: CLER improves step-level reflection, but it does not guarantee correctness and does not replace evidence acquisition, verification, or human oversight. CLER’s performance depends on the quality and diversity of the error database, which may limit generalization to novel error types. It also focuses on step-level reflection, potentially overlooking cross-step or higher-order reasoning patterns.

From the perspective of the thesis, CLER addresses the part of reliability that remains after evaluation and evidence acquisition. FinMR shows that current MLLMs make diverse financial reasoning errors. SEA shows that targeted evidence acquisition can improve performance, but it also leaves a class of errors that occur after relevant evidence has been acquired. CLER targets this residual space. It assumes that the model has produced a trace and that the trace can be inspected at the step level. The method then asks whether similar errors have been observed before and whether those errors can guide the current model away from a likely mistake.

This design has three implications. First, error resources should be structured. A collection of wrong final answers is too coarse to support precise correction because it does not reveal which intermediate step failed. FinErrorSet instead records wrong steps and corrected steps, making the error memory process-aware. Second, retrieval should capture both semantic and structural similarity. Two financial questions may use different wording but require the same formula or the same correction pattern. Conversely, two steps may share similar words while involving different financial logic. Contrastive learning is used to make retrieval sensitive to this distinction. Third, reflection should be individualized. A retrieved correction is not automatically valid for the target model’s current trace; the model must compare the retrieved mistake with its own reasoning and decide whether a revision is justified.

CLER also clarifies the role of negative knowledge in reliable reasoning. Financial reasoning systems usually aim to accumulate positive evidence: relevant values, definitions, formulas, and facts. CLER shows that negative evidence can also be useful. Knowing that a similar reasoning step has previously failed can make a model more cautious about formula choice, sign direction, unit alignment, or answer synthesis. This does not make the system infallible, but it gives the reasoning process an additional verification signal.

5.13.1 What CLER Can and Cannot Solve

CLER is most suitable for errors that are local enough to be represented as a reasoning step and common enough to have analogues in the error database. These include

using the wrong financial ratio, selecting an inappropriate arithmetic operation, confusing the direction of a percentage change, omitting a condition from the question, or synthesizing an answer that contradicts the preceding calculation. In these cases, a retrieved mistake–correction pair can function as a compact diagnostic example.

CLER is less effective for failures that originate before the reasoning trace is formed. If the model cannot perceive a chart, cannot locate the relevant table row, or lacks access to necessary textual evidence, error reflection may only revise an incomplete trace. It is also less effective when the required financial knowledge is absent from both the target model and the error database. These limitations reinforce the relationship between CLER and SEA: evidence acquisition and error reflection address different reliability bottlenecks.

The method also cannot guarantee that a revised explanation is faithful to the model’s internal decision process. Like other trace-based methods, CLER works with generated reasoning steps as inspectable artefacts. This is still valuable for thesis purposes because financial reasoning systems must provide explanations that can be examined by users, but it means that process-level correction should be complemented by verification and human oversight in high-stakes settings.

5.13.2 Transferability of Error Memory

The results support the transferability of error memory across MLLMs, but this transferability is conditional. Errors are reusable when the source and target reasoning steps share a common financial structure, such as the same formula, variable relation, visual comparison, or reporting-period constraint. They are less reusable when similarity is only lexical. This explains why contrastive learning is central to CLER: the retriever must identify transferable reasoning failures rather than merely similar wording.

Cross-model transfer is useful because different MLLMs make different mistakes. A weaker open-source model may fail on a step that a stronger closed-source model usually handles correctly, but that failure can still provide a warning when the stronger model faces a more difficult variant of the same reasoning pattern. Conversely, source-model diversity increases coverage of potential failure modes. The error database therefore acts as a shared external memory of what can go wrong in multimodal financial reasoning.

At the same time, error memory should be maintained dynamically. As model families, prompts, benchmarks, and financial documents change, new error patterns will appear. A static error database may gradually become less representative. Future systems could update `FinErrorSet` with newly observed failures, attach richer meta-

data about error category and evidence source, and combine retrieval confidence with uncertainty-aware verification before deciding whether to revise a reasoning step.

5.13.3 Relationship to SEA

SEA and CLER improve different parts of the reasoning process. SEA asks what evidence is needed and how it should be acquired. CLER asks whether the reasoning steps produced from available evidence resemble known mistakes. These functions are complementary. SEA can reduce errors caused by missing visual, textual, or knowledge evidence; CLER can reduce errors caused by invalid use of evidence or flawed reasoning operations. In a future end-to-end system, SEA could provide an evidence state for each reasoning step, while CLER could inspect each step against an error memory before the final answer is emitted.

The distinction is important because it prevents over-claiming. Evidence acquisition alone cannot guarantee correct reasoning, and error reflection alone cannot fix a trace that lacks the necessary evidence. A retrieved error may warn the model that a formula is often misapplied, but the model still needs the correct inputs. Conversely, a complete evidence state may still be used incorrectly. Reliable multimodal financial reasoning therefore requires both evidence-centered and error-centered interventions.

5.13.4 Threats to Validity

Several threats to validity should be considered when interpreting CLER. The first is error database coverage. `FinErrorSet` reflects the datasets, models, prompts, and reasoning traces used to construct it. If a future task involves a financial concept or visual form not represented in the database, retrieval may return only weakly related errors. The second is retrieval relevance. Even with contrastive learning, retrieved mistakes can be superficially similar but logically different, which may cause negative transfer. The third is model dependence. Different MLLMs may respond differently to the same retrieved error; some may use it productively, while others may over-correct or ignore it.

The fourth threat is reasoning trace faithfulness. CLER operates on generated reasoning steps, but generated traces may not always reveal the model’s true internal computation. This limitation is shared by many process-level evaluation and reflection methods. Nevertheless, step-level traces remain useful because they provide an inspectable artefact for identifying and revising errors. The fifth threat is evaluation scope. The experiments use benchmark-style financial reasoning tasks. Real profes-

sional use would require additional validation under longer documents, real-time data, multilingual settings, and human oversight.

5.13.5 Implications for Error Reflection

The findings suggest that error reflection should be treated as a design space rather than a single prompt. Key design choices include the granularity of errors, the diversity of source models, the retrieval objective, the number of retrieved cases, and the way the target model is asked to revise. The results in this chapter support step-level granularity and cross-model diversity. They also suggest that reflection is most useful when it is concrete: the model benefits from seeing a specific wrong step and its correction, rather than only a broad principle such as “check the formula” or “be careful with numbers”.

For financial reasoning, this concreteness matters because many errors are local but consequential. A sign error, a wrong denominator, a missed reporting period, or a mismatched unit can change the final answer. CLER provides a way to make such local errors visible before they propagate. In the integrated framework of Chapter 6, this role corresponds to the reflection and verification stage of reliable reasoning.

5.13.6 Answer to RQ3

CLER answers RQ3 by treating errors produced by different MLLMs as reusable reasoning signals. The chapter constructs FinErrorSet, a step-level error resource containing wrong reasoning steps and corrected versions generated from multiple open-source MLLMs. It then trains a contrastive retriever to match current reasoning steps with structurally and semantically relevant cross-model errors, allowing a target model to reflect on mistakes made by other models before revising its reasoning.

The results show that cross-model, step-level error reflection can improve multimodal financial reasoning across FinMR, FAMMA, and MMMU-Finance. The ablation studies further indicate that step-level retrieval, contrastive learning, and cross-model error diversity are important to the effectiveness of CLER. At the same time, the chapter identifies limits: irrelevant retrieval can cause negative transfer, novel financial concepts may be absent from the error database, and multiple interacting errors remain difficult to diagnose. These findings position CLER as an error reflection component within a broader reliability framework rather than as a guarantee of correctness.

Chapters 3–5 have examined evaluation, evidence acquisition, and error reflection as complementary components of reliable multimodal financial reasoning. Chapter 6 integrates these findings.

5.14 Chapter Summary

This chapter introduced FinErrorSet and CLER for cross-MLLM error reflection. It showed how structured step-level errors can support reflection and revision in multimodal financial reasoning, and it demonstrated that learning from other models' mistakes can complement evaluation and evidence acquisition. Together, Chapters 3, 4, and 5 address evaluation, evidence acquisition, and error reflection. Chapter 6 integrates these findings into a thesis-level account of reliable multimodal financial reasoning.

General Discussion: Towards Reliable Multimodal Financial Reasoning

6.1 Purpose of the General Discussion

The preceding chapters examined financial reasoning from four perspectives. Chapter 2 defined and reviewed the field. Chapter 3 evaluated current MLLMs through FinMR [39]. Chapter 4 improved evidence acquisition through step-wise multi-agent coordination [?]. Chapter 5 improved reasoning revision through cross-model error reflection [?]. This chapter integrates these findings. It answers the research questions at thesis level, proposes an integrated framework for reliable multimodal financial reasoning, and discusses cross-study findings, implications, limitations, and future research.

The purpose of this chapter is not to repeat the empirical results reported in Chapters 3–5. Instead, it asks what the three studies collectively show about the reliability problem. Each technical chapter addresses a particular gap: FinMR addresses evaluation, SEA addresses evidence acquisition, and CLER addresses error reflection. A thesis-level discussion is needed because reliable multimodal financial reasoning is not reducible to any one of these components. A system can be well evaluated but not improved; it can acquire relevant evidence but still reason incorrectly; it can reflect on errors but still lack the evidence needed for a valid answer. Reliability emerges from the interaction among these components.

This chapter therefore performs four functions. First, it gives integrated answers to the research questions. Second, it presents a unified framework for reliable financial reasoning. Third, it synthesizes findings that appear across the studies rather than

inside any single study alone. Fourth, it discusses implications and limitations at thesis level, including issues that cannot be resolved by a benchmark, an agent framework, or an error reflection method in isolation.

The organizing argument is that reliability is a system-level property. In this thesis, reliability means the ability of a financial reasoning system to consistently produce correct, evidence-grounded, robust, and verifiable conclusions. This definition implies that reliability cannot be established by final-answer Accuracy alone. It also cannot be achieved by adding more context, more model calls, or more reflection prompts without understanding how these components affect the reasoning process. The three technical studies therefore form a progression: FinMR makes limitations visible, SEA addresses evidence acquisition as a controllable part of reasoning, and CLER addresses remaining step-level reasoning errors through external error memory.

This integrated perspective is especially important for multimodal finance because the domain contains several interacting sources of difficulty. Financial evidence may appear in text, tables, charts, screenshots, and document images. The relevant operation may depend on domain knowledge, reporting period, unit conventions, and the question’s decision context. A correct final answer must therefore be supported by a chain of choices: what evidence to inspect, how to interpret it, which financial relation to apply, how to compute intermediate values, and how to verify the final conclusion. Chapter 6 uses this chain as the basis for synthesizing the thesis.

6.2 Integrated Answers to Research Questions

6.2.1 RQ1: Evaluating Multimodal Financial Reasoning

RQ1 asked how knowledge-intensive multimodal financial reasoning capabilities can be comprehensively evaluated. Chapter 3 showed that comprehensive evaluation requires a benchmark that jointly tests visual interpretation, mathematical reasoning, financial expertise, topic diversity, and image modality. FinMR addresses this requirement by distinguishing mathematical questions from expertise questions, covering 15 financial topics, including both single-image and multi-image cases, and supporting error analysis across visual perception errors, evidence selection errors, cross-modal alignment errors, financial knowledge errors, formula or operation selection errors, numerical computation errors, logical inference errors, and answer synthesis errors [39].

The thesis-level answer is that evaluation must be diagnostic rather than only comparative. Ranking models by accuracy is useful, but it does not reveal whether

failures arise from perception, evidence selection, financial knowledge, computation, or synthesis. Reliable financial reasoning therefore requires benchmarks that expose capability structure and error mechanisms.

FinMR contributes this diagnostic orientation in three ways. First, it separates financial mathematical reasoning from financial expertise reasoning, extending earlier financial numerical and table-text reasoning benchmarks [25, 203, 192]. This distinction is important because a model may know financial terminology but still fail to apply a formula, or it may execute arithmetic but fail to select the correct financial concept. Second, FinMR covers multiple financial topics, which reveals that performance is not uniform across the domain. Third, FinMR includes diverse visual inputs, showing that multimodal financial reasoning depends not only on language understanding but also on visual perception and cross-modal alignment, complementing broader multimodal financial benchmarks [169, 103, 141].

The answer to RQ1 also includes a methodological caution. A benchmark cannot fully represent professional financial reasoning by itself. Benchmark questions are necessarily simplified compared with real analyst workflows, which may involve long documents, changing data, uncertain assumptions, and human judgement. Nevertheless, a carefully designed benchmark is necessary because it makes capability limitations measurable and reproducible. FinMR therefore serves as a controlled diagnostic environment rather than a complete simulation of professional finance.

At thesis level, FinMR also clarifies what it means for a benchmark to be useful for method development. A benchmark should not only report that a model is wrong; it should indicate what kind of intervention might be needed. Visual perception and cross-modal alignment failures motivate better evidence extraction. Evidence selection failures motivate adaptive acquisition. Formula, computation, and inference failures motivate process-level checking and reflection. In this way, FinMR acts as the diagnostic starting point for SEA and CLER rather than as an isolated evaluation contribution.

6.2.2 RQ2: Acquiring and Integrating Evidence

RQ2 asked how specialized agents can dynamically acquire and integrate heterogeneous evidence. Chapter 4 showed that evidence acquisition can be formulated as a step-wise process in which a planner identifies missing evidence, generates targeted queries, routes them to modality-specific evidence agents, and updates an evidence state for downstream reasoning [?]. SEA demonstrates that query-based communi-

cation is more effective than fixed pipeline, controller, or debate paradigms when the task requires targeted evidence from visual, textual, and knowledge sources.

The thesis-level answer is that evidence acquisition is part of reasoning. A model cannot reliably reason over financial evidence if it cannot determine which evidence is needed, where that evidence should come from, and whether the current evidence state is sufficient. Agent specialization becomes useful when it is linked to adaptive, evidence-driven communication.

This answer refines how agent-based systems should be understood in financial reasoning. The value of multiple agents does not come merely from assigning names or roles to separate model calls. General agentic reasoning frameworks and financial role-aware systems show the promise of tool use, debate, and role decomposition [175, 157, 47, 201], but role decomposition is useful only when it changes the evidence process. In SEA, the planner, image evidence agent, context evidence agent, knowledge evidence agent, and reasoning agent correspond to different information needs. The planner decides what is missing; the evidence agents supply modality-specific evidence; the reasoning agent integrates the current evidence state. The communication protocol is therefore not ornamental. It determines whether the system can adapt to the task.

The results also show why fixed workflows are limited. A fixed pipeline assumes that each question needs the same sequence of operations, but financial questions vary widely. Some require image inspection, some require contextual retrieval, some require domain knowledge, and some require several of these in sequence. Query-based communication allows the system to ask for evidence only when it is needed and to stop when evidence sufficiency is reached. This is especially important in financial settings, where unnecessary context can introduce distraction, cost, latency, or evidence mismatch.

The integrated answer to RQ2 is therefore not simply that more agents improve performance. The more precise conclusion is that agent specialization helps when it is tied to evidence responsibility and adaptive communication. The planner must maintain a state of what has been established and what remains missing; evidence agents must return targeted information rather than broad context; and the reasoning agent must integrate evidence without treating all retrieved content as equally relevant. SEA’s contribution is to make these dependencies explicit. This matters because reliable financial reasoning often fails at the boundary between information access and reasoning, where the system has enough raw material somewhere in the input but has not transformed it into usable evidence.

6.2.3 RQ3: Learning from Cross-Model Errors

RQ3 asked how step-level errors produced by different MLLMs can be exploited to improve financial reasoning. Chapter 5 showed that errors can serve as external reflection signals when they are structured at the reasoning-step level and retrieved according to semantic and structural relevance [?]. FinErrorSet and CLER demonstrate that cross-model error retrieval can help target models identify possible mistakes in their own reasoning and revise their answers.

The thesis-level answer is that error reflection should be process-level and diversity-aware. Self-reflection may reproduce a model’s own biases, while answer-level feedback is too coarse to diagnose intermediate failures [106, 131]. Cross-model step-level errors provide a complementary source of negative knowledge that can improve reflection without requiring fine-tuning of closed-source models, building on broader findings that negative or erroneous examples can guide reasoning revision [13, 146, 145, 6].

This answer also changes how model errors are interpreted. In ordinary benchmark evaluation, errors are counted as failures. In CLER, errors are also treated as data. A wrong reasoning step becomes useful when it is paired with a correction and retrieved in a relevant context. This does not mean that all errors are beneficial; irrelevant retrieved errors can mislead the target model. The contribution is more specific: curated and structured cross-model errors can become a reusable error memory for future reasoning.

The process-level nature of this answer is essential. Final-answer feedback cannot distinguish a wrong answer caused by a visual perception error from one caused by a formula or operation selection error, numerical computation error, or answer synthesis error. Step-level reflection makes the target of correction explicit. This is particularly valuable in finance because small local mistakes can have large consequences. A wrong sign, denominator, unit, or reporting period can change the final answer while leaving the explanation superficially plausible.

The integrated answer to RQ3 also highlights the conditional nature of error reuse. Cross-model errors are useful when they are relevant, corrected, and presented at the right granularity. They can be harmful when retrieval is weak or when the target model over-applies a correction from a different financial context. CLER therefore does not treat error memory as an oracle. It treats error memory as an external reflection signal that must be interpreted by the target reasoner. This distinction is important for reliable deployment: error examples should support revision, but they should not replace evidence grounding or independent verification.

6.3 An Integrated Framework for Reliable Financial Reasoning

The studies support an integrated framework:

Task Understanding → Evidence Identification → Evidence Acquisition
→ Multimodal Reasoning → Error Reflection → Verification.

Each component corresponds to a part of the thesis. The literature review establishes field and task understanding. FinMR measures capability and diagnoses failure. SEA improves evidence identification and acquisition. CLER improves error reflection and reasoning revision. The general discussion connects these components to end-to-end reliability.

The framework begins with task understanding. A financial reasoning system must identify what kind of problem it is solving: a numerical calculation, a table lookup, a chart interpretation, a financial expertise question, a document-grounded question, or a multi-turn interaction. This is why Chapter 2 matters. It establishes a taxonomy of financial reasoning tasks and methods, drawing on benchmarks for numerical, tabular, document, multimodal, and conversational reasoning [25, 203, 67, 128, 169, 27], and it distinguishes financial reasoning from adjacent financial NLP tasks.

The next two components are evidence identification and evidence acquisition. Evidence identification asks what information is required; evidence acquisition obtains or extracts that information. These components are separated because a system may know that a chart value is needed but still fail to read it correctly, or it may retrieve a relevant document but fail to identify the relevant period. SEA operationalizes both components through planner-guided, query-based interaction among specialized agents.

Multimodal reasoning then integrates evidence into a reasoning trace. This is where financial knowledge, visual interpretation, structural understanding, numerical calculation, and logical inference interact. Error reflection inspects this trace for likely failures, and verification asks whether the answer follows from evidence and reasoning. CLER focuses on error reflection by retrieving relevant step-level errors from FinErrorSet. Verification remains broader than CLER: it may include human review, statistical testing, calibration, citation checking, or independent tool execution.

Although the framework is written as a sequence, it should be understood as an iterative loop. Evidence identification may change after an initial reasoning attempt.

Component	Thesis Work
Field and task understanding	Literature Review
Capability measurement	FinMR
Evidence acquisition	SEA
Error correction	CLER
End-to-end reliability	Integrated discussion

Table 6.1: Mapping between the integrated reliability framework and thesis components.

Error reflection may reveal that the system used the wrong variable, which sends the process back to evidence acquisition. Verification may identify uncertainty or inconsistency, which triggers further evidence search or human review. This iterative character is central to reliable reasoning. A one-pass answer can be accurate in simple cases, but high-risk financial settings require mechanisms for detecting when the reasoning state is incomplete or unstable.

The framework also distinguishes between *available information* and *usable evidence*. A financial report may contain all information required to answer a question, but the system still needs to locate, interpret, and align the relevant parts. SEA addresses this distinction by allowing the planner to ask targeted questions of image, context, and knowledge agents. CLER then addresses a second distinction: usable evidence is not the same as valid reasoning. Even after evidence has been acquired, the system must avoid known reasoning failures and verify whether the final answer follows.

Table 6.1 summarizes how the thesis components map onto this framework. The table should not be read as assigning each chapter to a completely separate module. The chapters are complementary. Chapter 2 defines the task space that makes FinMR meaningful; FinMR exposes the evidence and reasoning failures that motivate SEA and CLER; SEA improves the evidence state but leaves step-level reasoning errors; and CLER improves reflection but still depends on evidence quality. Chapter 6 therefore treats reliability as the coordination of these components rather than as the output of any single method.

6.4 Cross-Study Findings

6.4.1 Finding 1: Financial Reasoning Is a Composite Capability

The studies show that financial reasoning is not a single ability. It combines domain knowledge, evidence grounding, visual perception, structural interpretation, formula or operation selection, numerical computation, logical inference, and answer synthesis [39? ?]. Strong performance in one component does not guarantee strong overall reasoning.

This finding is visible in all three technical studies. FinMR shows that model performance varies across mathematical and expertise questions, visual forms, and financial topics. SEA shows that evidence acquisition itself has multiple components: planning, routing, evidence extraction, and reasoning integration. CLER shows that reasoning errors can occur at different steps even after evidence is available. Together, these findings suggest that evaluating a model as simply “good at finance” or “good at multimodal reasoning” is too coarse. The relevant question is which component of the reasoning process is strong and which component remains fragile.

6.4.2 Finding 2: General Strength Is Not Enough

Strong general multimodal performance does not guarantee strong financial reasoning. Models that perform well on general multimodal tasks may still fail in finance because financial reasoning requires specialized knowledge, precise units and periods, and economically meaningful interpretation. FinMR makes this mismatch visible through topic-level and modality-level evaluation [39], echoing limitations reported in other multimodal financial benchmarks [169, 103, 141].

This matters for both research and deployment. General-purpose multimodal benchmarks often emphasize object recognition, spatial understanding, chart reading, or broad visual question answering. Financial reasoning may require those abilities, but it also requires domain-specific interpretation. A model must know, for example, whether a ratio should increase or decrease, how a numerator and denominator should be selected, why a cash-flow item is treated differently from an income-statement item, or how risk concepts change the interpretation of visual evidence. Strong visual ability is therefore necessary but not sufficient.

6.4.3 Finding 3: Evidence Acquisition Is Part of Reasoning

SEA demonstrates that a major source of failure is not only weak reasoning after evidence is available, but also failure to acquire the right evidence [?]. In multimodal financial reasoning, deciding what to inspect is itself a reasoning step.

This finding challenges a common separation between retrieval and reasoning. In many systems, retrieval is treated as preprocessing: the system first gathers context and then reasons over it. The thesis suggests a more interactive view. Reasoning determines what evidence is needed, and newly acquired evidence changes the reasoning path. This feedback loop is especially important when evidence is heterogeneous. A visual chart, a textual context, and a financial formula may each answer different parts of the problem.

6.4.4 Finding 4: Adaptive Communication Matters

Agent specialization is effective only when communication is adaptive and evidence-driven. The comparison among communication paradigms shows that specialized agents are most useful when they exchange targeted evidence. Static role decomposition or unstructured debate can be less effective than concise query-based communication [? 157, 47].

The implication is that agent systems should be evaluated by their communication behavior, not only by their final accuracy. A multi-agent system may appear sophisticated while simply passing the same context among several model calls. SEA shows that the important design question is how agents decide what to ask, when to ask, and when to stop. Adaptive communication also has cost implications: every additional call and every additional piece of context has a token and latency cost. Reliable systems must therefore balance accuracy gains against communication efficiency.

6.4.5 Finding 5: Correct Evidence Does Not Guarantee Correct Reasoning

SEA’s remaining errors and CLER’s motivation show that relevant evidence can still be followed by invalid reasoning [? ?]. Reliable systems therefore need both evidence acquisition and reasoning-step correction.

This finding is central to the thesis argument. It prevents the conclusion that better retrieval or better evidence acquisition alone will solve financial reasoning. Evidence is necessary, but it must be interpreted through valid financial logic. A model that has

the right values can still use the wrong formula; a model that has the right chart can still compare the wrong periods; a model that has the right definition can still choose an answer that does not follow. CLER addresses this gap by making the reasoning trace itself a target of reflection.

6.4.6 Finding 6: Errors Produced by One Model Can Help Another

CLER shows that cross-model errors can act as useful supervision signals [?]. This finding is important because it reframes model errors not only as failures to be counted, but also as resources for improving future reasoning.

The value of cross-model errors depends on diversity. If all models fail in the same way, one model’s error may not help another. If models fail differently, however, their mistakes can reveal alternative failure modes. This suggests a broader research direction in which error memories are continuously updated as models are evaluated. Such memories could support future models by storing not only correct examples but also structured records of what went wrong and how it was corrected.

6.4.7 Finding 7: Reliability Must Be Evaluated End to End

Across the thesis, reliability emerges as a pipeline-level property. It cannot be inferred from final-answer Accuracy alone, nor from retrieval quality, execution correctness, or self-reflection in isolation.

End-to-end reliability requires multiple forms of evidence. Accuracy shows whether the final output is correct. Evidence-level evaluation shows whether the system used relevant support. Process-level evaluation shows whether intermediate reasoning steps are valid. Robustness and perturbation tests show whether the system remains stable under changes to evidence. Human validation and statistical testing show whether improvements are credible. The thesis therefore argues for evaluation that follows the reasoning process rather than only its final product.

Taken together, these seven findings explain why the thesis cannot be reduced to three independent studies. FinMR, SEA, and CLER each address a different failure point in the same reasoning chain. FinMR shows what current systems fail to do. SEA changes how systems acquire and coordinate evidence. CLER changes how systems learn from process-level mistakes. The cross-study conclusion is that reliable multimodal financial reasoning requires both capability measurement and process intervention.

6.5 Theoretical Implications

The thesis contributes a capability-based account of financial reasoning. It argues that financial reasoning is best understood as an evidence-grounded process in which perception, knowledge, structure, computation, and reflection interact. The studies also show that evidence and reasoning cannot be sharply separated: evidence selection shapes the reasoning path, and reasoning needs determine what evidence should be acquired. Finally, the thesis suggests that model errors exhibit complementarity. Different MLLMs can fail in different ways, and these differences can be converted into useful reflection signals.

This account has implications for how reasoning in specialized domains is conceptualized. Domain reasoning is often described as the application of general reasoning ability to domain knowledge. The findings here suggest a more integrated view. Financial knowledge affects what evidence is considered relevant, how visual information is interpreted, which operations are valid, and what counts as a decision-relevant conclusion. The domain is therefore not merely a database of facts added to a general reasoner. It shapes the reasoning process itself.

The thesis also contributes to theories of multimodal reasoning. Multimodality is not only a matter of processing multiple input channels. In financial reasoning, modalities carry different epistemic roles. A chart may provide quantitative evidence, a textual prompt may specify the question intent, a table may define the variables, and background financial knowledge may determine the operation. Cross-modal reasoning therefore requires more than aligning image tokens and text tokens; it requires understanding how each modality functions as evidence.

Finally, the thesis contributes to a process-level view of reliability. Reliability is often evaluated through outcome correctness, but the studies show that process failures can exist even when outcomes occasionally appear correct. A model may arrive at the right answer through invalid reasoning, and such success is fragile. The theoretical implication is that reliable reasoning systems should be judged by both conclusions and the path by which those conclusions are reached.

The thesis also suggests that perception, knowledge, and logic should be treated as mutually dependent rather than sequentially separate. A visual perception error can cause a formula error because the wrong value is extracted. A knowledge error can cause an evidence selection error because the model does not know which variable matters. A logical inference error can turn correct calculations into the wrong answer. This interaction explains why multimodal financial reasoning is difficult: failures do not remain confined to one module. They propagate through the reasoning trace.

6.6 Methodological Implications

The thesis has several methodological implications. Benchmark design should measure capability structure, not only final accuracy. Evaluation should include topic-level, modality-level, and process-level analyses [39]. Evidence perturbation can reveal whether acquired evidence genuinely contributes to reasoning [?]. Agent communication should be evaluated as a first-class design choice rather than treated as an implementation detail. Step-level error datasets can support more precise reflection than answer-level feedback [?]. Statistical testing and human validation remain essential for distinguishing robust improvements from noise.

For benchmark design, the thesis suggests that task coverage should be tied to an explicit capability taxonomy. FinMR is not only a dataset of financial questions; it is organized around reasoning type, topic, and modality. This organization enables more informative analysis than a single aggregate score. Future benchmarks can extend this approach by adding longer documents, multilingual evidence, real-time data, or open-ended answers, but the same principle applies: the benchmark should make the target capability visible.

For method evaluation, the thesis shows the importance of ablation and perturbation. SEA’s ablation studies test whether each agent and communication choice contributes to performance. Evidence perturbation tests whether visual, textual, and knowledge evidence are genuinely useful. CLER’s ablations test whether contrastive retrieval and reflection both matter. These analyses are important because LLM-based systems can improve for accidental reasons, including prompt length, answer format, or model-specific behavior. Methodological rigor requires isolating which component produces the improvement.

For process-level evaluation, the thesis supports the construction of intermediate artefacts such as evidence states, reasoning traces, retrieved error pairs, and corrected steps. These artefacts make system behavior inspectable. They also create opportunities for human validation and statistical testing. In high-stakes domains such as finance, this inspectability is not merely a research convenience; it is part of responsible system design.

The thesis further suggests that statistical and human validation should be paired. Statistical tests help determine whether observed improvements are likely to be robust across samples or model comparisons. Human validation helps determine whether intermediate artefacts, such as retrieved evidence or wrong-step corrections, are meaningful to domain users. These forms of validation answer different questions. A statistically significant gain may still rely on poor explanations, while a convincing qualitative

example may not generalize. Reliable method development should therefore combine quantitative, process-level, and human-centred evidence.

6.7 Practical Implications

The findings are relevant to financial education, analyst support, financial document analysis, audit preparation, risk assessment, and human–AI collaboration. The thesis does not claim that MLLMs should make autonomous investment decisions. Instead, it supports a view of financial reasoning systems as assistive tools that retrieve evidence, inspect multimodal information, perform calculations, identify possible errors, and present conclusions that humans can verify.

In financial education, systems inspired by this thesis could support students by identifying relevant evidence, explaining formula choices, and showing common reasoning mistakes. FinErrorSet-like resources are particularly relevant for education because students often learn effectively from worked incorrect examples. In analyst support, evidence acquisition mechanisms could help users inspect financial reports, charts, or filings more efficiently, while reflection mechanisms could warn when a reasoning step resembles known errors.

In document analysis and audit preparation, the thesis suggests that systems should preserve evidence trails. A generated answer should be linked to the visual or textual evidence used, the operations performed, and the reasoning steps revised. This is important because financial professionals need to justify conclusions to colleagues, clients, regulators, or auditors. The practical value of an AI system therefore depends not only on answer Accuracy but also on whether the system supports review.

For risk assessment, the findings encourage caution. The thesis does not support using MLLMs as autonomous decision makers in investment, lending, or compliance. Instead, it supports human–AI collaboration in which the system acts as an evidence assistant, calculation helper, reasoning checker, or error warning mechanism. Human oversight remains necessary because financial decisions involve accountability, judgement, and context that extend beyond benchmark-style reasoning.

These practical implications are strongest in settings where the system’s work can be inspected before action is taken. For example, an analyst might use evidence acquisition to locate relevant report figures, then use reflection to check whether a proposed calculation resembles known mistakes. A teacher might use step-level errors to show students why a superficially plausible answer is wrong. An auditor might use evidence trails to verify whether a generated explanation is grounded in the cited

document. In each case, the value of the system lies in supporting human review rather than replacing professional responsibility.

6.8 Overall Limitations

Several limitations remain. First, the studies rely on static benchmark datasets, whereas real financial reasoning often involves real-time data, evolving market conditions, and temporal uncertainty. Second, the datasets and evaluations are primarily English-language and may not fully represent multilingual or jurisdiction-specific financial reasoning. Third, commercial model results may be affected by API version drift and reproducibility constraints. Fourth, benchmark contamination remains difficult to rule out completely. Fifth, the thesis focuses mainly on QA-style settings rather than full professional deployment. Sixth, cost, latency, uncertainty calibration, and user interaction are evaluated only partially. Finally, SEA and CLER are studied as complementary methods, but they are not yet implemented as a single end-to-end production system.

The static nature of the datasets is a particularly important limitation. Many financial questions depend on reporting dates, market conditions, and regulatory context. A model that answers a benchmark question correctly may still fail when the relevant data change over time. Future evaluations should therefore include temporal reasoning and mechanisms for checking whether evidence is current.

Language and jurisdiction are also important. Financial concepts are not always portable across legal systems, accounting standards, or regulatory regimes. The thesis primarily studies English-language resources and common financial reasoning settings. This limits conclusions about multilingual financial reasoning or jurisdiction-specific tasks such as local tax rules, reporting standards, or regulatory filings.

Another limitation concerns model access. Some evaluated models are commercial systems whose versions may change over time. Even when prompts and datasets are preserved, API updates can affect reproducibility. This limitation is common in current LLM research, but it is especially relevant for a thesis that compares multiple model families. Reporting access dates, model names, and experimental settings helps, but it cannot fully eliminate version drift.

Finally, the thesis does not fully integrate SEA and CLER into a single deployed system. The studies show that evidence acquisition and error reflection are complementary, but they are evaluated as separate contributions. An end-to-end implementation would require additional design choices: when to trigger evidence acquisition,

when to trigger error retrieval, how to resolve conflicts between evidence and error warnings, and how to present uncertainty to users.

Cost and latency are also important limitations. SEA can improve reasoning through additional agent calls, and CLER can improve reflection through retrieval and revision, but both introduce computational overhead. In some professional workflows, the additional cost may be acceptable because verifiability matters more than speed. In other settings, such as interactive analysis or large-scale document screening, latency and token cost may constrain deployment. Future systems must therefore include cost-aware control policies, not only accuracy-oriented prompts.

Uncertainty and calibration remain underdeveloped in this thesis. The studies evaluate correctness, evidence use, perturbation behaviour, and human validation, but they do not fully assess whether models know when their answers are uncertain. This matters because a financial reasoning system should be able to state when evidence is incomplete, when retrieved errors are only weakly related, or when a conclusion requires human judgement. Without calibration, even improved reasoning systems may express unreliable confidence.

6.9 Future Research

Future work should investigate real-time financial agents that can reason over changing data streams, temporal reasoning over reporting periods and market events, causal and counterfactual financial reasoning, multi-document and multimodal document reasoning, uncertainty-aware answer generation, multilingual and regulatory settings, and human-in-the-loop verification. A particularly important direction is integrating SEA and CLER into a unified system in which evidence acquisition and error reflection operate within the same reasoning loop. Another direction is dynamic error memory, where systems continuously update their error resources as new model failures are observed and validated.

One direction is real-time and temporal financial reasoning. Future systems should be able to identify the relevant date, retrieve current evidence, distinguish historical from current information, and reason over changes across reporting periods. This would move financial reasoning closer to practical use, where the correctness of an answer may depend on when the question is asked.

A second direction is multi-document multimodal reasoning. Real financial analysis often requires reading several filings, investor presentations, earnings transcripts, and market charts together. This setting raises new evidence acquisition challenges: the

system must decide which document type is relevant, align evidence across documents, resolve conflicting values, and maintain source attribution.

A third direction is uncertainty-aware reasoning. Current benchmark accuracy does not fully measure whether a model knows when it is uncertain. Future systems should provide calibrated confidence, identify missing evidence, ask clarification questions, and defer to human review when the evidence is insufficient. This is especially important for high-stakes financial applications.

A fourth direction is integration of SEA and CLER. A unified system could first use SEA to acquire visual, textual, and knowledge evidence, then use CLER to inspect each reasoning step against a dynamic error memory. Such a system would treat evidence acquisition and error reflection as part of the same loop. It could also learn from its own deployment history by adding validated new errors to the memory.

A fifth direction is human-in-the-loop evaluation. Financial reasoning systems should be tested not only as autonomous question-answering models, but also as collaborators. User studies could examine whether evidence trails, retrieved errors, and reflection warnings help analysts, students, or auditors make better decisions. Such studies would connect benchmark performance to actual human work.

Another direction is agent-level process evaluation. Future benchmarks should evaluate not only whether a multi-agent system answers correctly, but also whether the planner asks appropriate questions, whether evidence agents return relevant and concise evidence, whether the reasoning agent integrates evidence faithfully, and whether reflection modules avoid negative transfer. This would make agent behaviour auditable and would help distinguish genuinely reliable coordination from accidental improvements caused by extra computation.

Finally, future work should develop dynamic error memories. `FinErrorSet` is a static resource constructed for the experiments in this thesis, but real systems will encounter new model failures as models, documents, regulations, and financial practices change. A dynamic error memory could store newly validated mistakes, track which model produced them, record the evidence source and error category, and retrieve them only when they are sufficiently transferable. Such a memory would turn continuous evaluation into continuous reliability improvement.

6.10 Chapter Summary

This chapter integrated the thesis findings and argued that reliable multimodal financial reasoning requires appropriate evaluation, relevant evidence, correct reasoning,

effective reflection, and verifiable conclusions. FinMR, SEA, and CLER address different but complementary parts of this reliability problem. Together, they move financial reasoning research from answer prediction toward evidence-grounded, process-aware, and reflection-capable financial intelligence.

Conclusion

7.1 Thesis Summary

This thesis has investigated reliable multimodal financial reasoning with large language models. The research problem motivating the thesis is that financial reasoning requires more than fluent language generation or general visual understanding. Financial questions often require a system to interpret heterogeneous evidence, apply domain-specific knowledge, perform numerical or logical operations, and produce conclusions that can be inspected, as shown by financial numerical, document, and multimodal reasoning benchmarks [25, 203, 67, 169, 39]. In high-risk financial settings, an answer is not sufficient simply because it is plausible. It must be correct, evidence-grounded, robust, and verifiable.

The aim of the thesis was to advance reliable multimodal financial reasoning with large language models through comprehensive capability evaluation, adaptive multi-agent evidence acquisition, and cross-model error reflection. This aim was pursued through a progression of studies. Chapter 2 first established the conceptual foundation by defining financial reasoning, organizing the literature, and identifying the gaps that motivate the technical chapters. Chapter 3 then addressed the evaluation problem through FinMR, a benchmark designed to diagnose knowledge-intensive multimodal financial reasoning capabilities [39]. Chapter 4 addressed the evidence acquisition problem through SEA, a step-wise multi-agent framework for acquiring and integrating visual, textual, and knowledge evidence [?]. Chapter 5 addressed the error correction problem through FinErrorSet and CLER, which use cross-model step-level errors as reflection signals for reasoning revision [?].

The overall outcome of the thesis is a coherent account of reliable multimodal financial reasoning. FinMR shows where current MLLMs fail and why benchmark design must be diagnostic. SEA shows that evidence acquisition is not merely preprocessing but an adaptive reasoning activity. CLER shows that reasoning errors can be transformed into structured negative knowledge that supports future correction. Together, the studies support the central thesis that reliable financial reasoning emerges from the interaction of evaluation, evidence acquisition, reasoning, reflection, and verification.

This progression also explains why the thesis is not a collection of independent studies. The literature review defines the field and identifies the gaps. FinMR diagnoses model limitations and reveals that many failures involve incomplete evidence use, weak cross-modal alignment, incorrect financial knowledge, or invalid reasoning steps. SEA responds to the evidence-related part of this problem by improving how a system asks for, receives, and integrates heterogeneous evidence. CLER responds to the remaining reasoning-related part of the problem by inspecting step-level errors and supporting model-specific revision. Chapter 6 then integrates these components into a thesis-level framework for reliable multimodal financial reasoning.

Across the thesis, reliability is treated as an end-to-end property rather than a single score. A system may achieve high final-answer Accuracy while using weak evidence, producing an unfaithful explanation, or failing under small perturbations. Conversely, a system may retrieve relevant evidence yet still apply an incorrect operation or draw an invalid conclusion. The thesis therefore argues that reliable multimodal financial reasoning requires attention to the whole reasoning process: task understanding, evidence identification, evidence acquisition, multimodal reasoning, error reflection, and verification.

7.2 Summary of Main Contributions

The first contribution is the structured literature review in Chapter 2. The review synthesizes financial reasoning research by distinguishing five task families: financial numerical reasoning, financial tabular reasoning, financial document reasoning, financial multimodal reasoning, and financial conversational reasoning [25, 191, 128, 169, 27]. It also organizes methods into prompting, program-based reasoning, retrieval augmentation, evidence grounding, agent interaction, training, reflection, and verification approaches [156, 23, 87, 157, 106]. This synthesis provides the conceptual basis for the thesis and clarifies why financial reasoning should be separated from broader financial NLP tasks such as sentiment analysis, information extraction, or market prediction.

The literature review also establishes the three research gaps addressed by the technical chapters. The evaluation gap concerns the limited coverage of knowledge-intensive, multi-topic, and multimodal financial reasoning in existing benchmarks. The evidence acquisition gap concerns the tendency of existing systems to rely on fixed contexts or predetermined workflows rather than dynamically identifying missing evidence. The error reflection gap concerns the dominance of self-reflection and answer-level correction, with limited use of structured step-level errors from diverse MLLMs. These gaps provide the intellectual bridge from the review chapter to FinMR, SEA, and CLER.

The second contribution is FinMR, introduced in Chapter 3 [39]. FinMR is designed to evaluate knowledge-intensive multimodal financial reasoning through 3,200 questions, 3,764 images, 15 financial topics, and both single-image and multi-image settings. It distinguishes mathematical reasoning from financial expertise reasoning, allowing model performance to be analyzed at a finer level than overall Accuracy. By including topic-level, modality-level, and error-level analysis, FinMR turns benchmark evaluation into a diagnostic tool.

FinMR’s contribution is not only that it provides another dataset. Its main role in the thesis is to make capability limitations visible. It shows that current MLLMs can fail through visual perception errors, evidence selection errors, cross-modal alignment errors, financial knowledge errors, formula or operation selection errors, numerical computation errors, logical inference errors, and answer synthesis errors. This error taxonomy becomes a common language for the later chapters. It also motivates the thesis progression from evaluation to intervention: once failures are diagnosed, the next question is how to reduce them.

The third contribution is SEA, introduced in Chapter 4 [?]. SEA develops a step-wise multi-agent evidence acquisition framework for multimodal financial reasoning. It uses a planner, image evidence agent, context evidence agent, knowledge evidence agent, and reasoning agent to acquire heterogeneous evidence through query-based communication. The framework treats evidence acquisition as an adaptive process in which the system identifies evidence gaps, routes targeted queries, integrates agent responses, assesses evidence sufficiency, and terminates when the evidence state is adequate for reasoning.

SEA contributes a methodological view of multi-agent reasoning that is more specific than simply assigning roles to agents. Its core insight is that agent specialization becomes useful when communication is adaptive and evidence-driven. The planner must decide what is missing; evidence agents must provide modality-specific informa-

tion; and the reasoning agent must integrate evidence into a coherent answer. SEA therefore addresses the evidence acquisition gap identified in the literature review and motivated by FinMR’s error analysis.

The fourth contribution is FinErrorSet and CLER, introduced in Chapter 5 [?]. FinErrorSet is a step-level financial reasoning error resource constructed from errors produced by multiple open-source MLLMs. It contains mistake–correction pairs that make local reasoning failures explicit. CLER uses this resource through contrastive error retrieval and model-specific reflection. Given a current reasoning trace, CLER retrieves structurally and semantically relevant cross-model errors and asks the target MLLM to reflect on whether similar mistakes appear in its own reasoning.

The contribution of CLER is to reinterpret model errors as reusable negative knowledge. Instead of treating errors only as failures to be counted, CLER treats curated wrong steps and their corrections as resources for improving future reasoning. This is important because self-reflection can reproduce a model’s own bias, while final-answer feedback is often too coarse to diagnose intermediate failures. CLER addresses this limitation by using cross-model, step-level error reflection to support revision without requiring fine-tuning of closed-source target models.

The fifth contribution is the integrated reliability framework developed in Chapter 6. The framework connects the thesis components through the sequence of task understanding, evidence identification, evidence acquisition, multimodal reasoning, error reflection, and verification. It explains how the literature review, FinMR, SEA, and CLER fit into a unified account of reliable multimodal financial reasoning. The integrated framework is deliberately modest in its claim: it does not say that any single chapter solves reliability, but that reliability depends on coordinating appropriate evaluation, relevant evidence, correct reasoning, effective reflection, and verifiable conclusions.

7.3 Answers to the Overall Research Aim

The overall research aim was to advance reliable multimodal financial reasoning with large language models. The thesis answers this aim in five ways: by clarifying how multimodal financial reasoning should be understood, how it should be evaluated, how evidence acquisition can be improved, how error correction can be improved, and what constitutes reliability in this domain.

First, the thesis clarifies how multimodal financial reasoning should be understood. Financial reasoning is a domain-specific reasoning process in which a system integrates

financial knowledge, heterogeneous evidence, numerical or logical operations, and contextual interpretation to derive justified and decision-relevant conclusions. The multimodal setting adds further complexity because evidence may appear as text, tables, charts, diagrams, screenshots, or document images. A system must therefore reason not only over language, but also across visual and textual evidence. This understanding is developed in Chapter 2 and used throughout the technical studies.

Second, the thesis shows how multimodal financial reasoning should be evaluated. FinMR demonstrates that evaluation should be diagnostic rather than only comparative [39]. A single Accuracy score can rank models, but it cannot explain whether failures arise from visual perception errors, evidence selection errors, cross-modal alignment errors, financial knowledge errors, formula or operation selection errors, numerical computation errors, logical inference errors, or answer synthesis errors. Reliable evaluation therefore requires task coverage, topic diversity, multimodal evidence, reasoning-type distinctions, and fine-grained error analysis. FinMR provides one concrete instantiation of this evaluation philosophy.

Third, the thesis shows how evidence acquisition can be improved. SEA demonstrates that a financial reasoning system should not passively consume all available context or rely on a fixed reasoning pipeline [?]. Instead, the system should identify what evidence is missing, ask targeted modality-specific queries, integrate the resulting evidence, and decide whether the evidence state is sufficient. This makes evidence acquisition an active part of reasoning. In multimodal finance, this is especially important because the relevant information may be distributed across images, textual context, and background financial knowledge.

Fourth, the thesis shows how error correction can be improved. CLER demonstrates that step-level errors produced by one MLLM can help another MLLM reason more carefully when those errors are structured, corrected, and retrieved appropriately [?]. This approach improves error correction by moving beyond self-reflection and answer-level feedback. It gives the target model concrete examples of what went wrong in similar reasoning steps and how those steps should be revised. Error correction therefore becomes process-aware: the model reflects on particular reasoning steps rather than only reconsidering the final answer.

Fifth, the thesis defines what constitutes reliable multimodal financial reasoning. Reliability is the ability of a financial reasoning system to consistently produce correct, evidence-grounded, robust, and verifiable conclusions. This definition requires more than answer Accuracy. A reliable system must understand the task, identify appropriate evidence, acquire and integrate heterogeneous information, perform valid

reasoning, recognize likely errors, revise flawed steps, and support conclusions that can be inspected. Reliability is therefore not a property of a model alone. It is a property of the reasoning process and the system design around that process.

These answers are complementary. Understanding the task gives the system a target capability. Evaluation makes current limitations visible. Evidence acquisition improves the information available for reasoning. Error reflection improves the validity of the reasoning trace. Verification connects the final answer back to the evidence and the reasoning process. If any component is missing, reliability is weakened. A benchmark without intervention diagnoses failure but does not reduce it. Evidence acquisition without correction may still lead to invalid reasoning. Error reflection without evidence grounding may revise a trace that lacks the information needed for a valid answer. The thesis therefore answers the overall aim by showing how these components must work together.

7.4 Final Remarks

Reliable financial intelligence requires more than accurate answer generation. It requires systems that can identify appropriate evidence, integrate heterogeneous information, perform valid reasoning, recognise errors, and support conclusions that can be inspected and verified.

This thesis contributes to that goal by moving from definition and review, to evaluation, to evidence acquisition, to error reflection, and finally to an integrated view of reliability. The progression matters because reliable multimodal financial reasoning cannot begin with a method alone. It first requires a clear understanding of the task, then benchmarks that reveal where systems fail, mechanisms that improve the evidence state before reasoning, and reflection methods that inspect the reasoning process after evidence has been acquired.

The broader message is that financial reasoning systems should be designed for inspectability and human judgement. In finance, a conclusion often has value only when its basis can be checked. Large language models and multimodal large language models provide powerful foundations, but capability alone does not establish trustworthiness. Reliability depends on how models are evaluated, how evidence is acquired, how reasoning is checked, and how conclusions are verified.

By defining and reviewing financial reasoning, constructing FinMR, developing SEA, introducing FinErrorSet and CLER, and integrating these studies into a reliability framework, this thesis offers a step toward financial AI systems that are not only

more capable, but also more transparent, more evidence-aware, and more reflective. The final conclusion is therefore simple but demanding: reliable multimodal financial reasoning requires systems that know what evidence matters, know how to reason with it, know how they may be wrong, and make their conclusions available for human inspection.

Appendix A

SEA Prompt Templates and Notation

This appendix provides supporting notation and prompt templates for the SEA framework introduced in Chapter 4.

A.1 Summary of Notations

Notation	Description
X	A multimodal financial reasoning instance.
C	Textual context of the question.
$I = \{i_1, \dots, i_S\}$	Set of visual inputs, such as charts, tables, or figures.
Q	Financial reasoning question.
$O = \{o_1, \dots, o_M\}$	Set of candidate answer options.
Y^*	Ground-truth answer.
$\hat{Y}$	Predicted answer.
E_n	Evidence state after reasoning step n .
e_n	Evidence retrieved at step n .
$e_{n,\ell}$	Evidence returned for the ℓ -th query at step n .
R_n	Reasoning trajectory after step n .
r_n	Intermediate reasoning output at step n .
t_n	Sub-task generated by the planner at step n .
$q_n = \{q_{n,1}, \dots, q_{n,L_n}\}$	Set of evidence-seeking queries at step n .
s_n	Termination flag indicating whether to continue or stop.
$\mathcal{A}_P$	Planner agent.
$\mathcal{A}_I$	Image evidence-seeking agent.
$\mathcal{A}_C$	Context evidence-seeking agent.
$\mathcal{A}_K$	Knowledge evidence-seeking agent.
$\mathcal{A}_R$	Reasoning agent.
$\mathcal{D}$	External financial knowledge base.
$\rho(\cdot)$	Routing function that assigns each query to an evidence-seeking agent.
$N_{\max}$	Maximum number of reasoning/planning steps.

Table A.1: Summary of the main notations used in the SEA framework.

A.2 Prompt Templates for SEA

To improve reproducibility, this appendix provides the prompt templates used by the agents in SEA. Each agent is instantiated with the same backbone MLLM within an experiment, but is assigned a role-specific instruction. The planner agent generates step-conditioned subtasks and evidence-seeking queries; the evidence-seeking agents return modality-specific evidence; and the reasoning agent performs evidence-grounded inference based on the accumulated evidence state.

A.2.1 Planner Agent Prompt

Planner Agent Prompt

[System Input]:

You are the planner agent in a step-wise evidence acquisition framework for multi-modal financial reasoning. Given the original instance, the accumulated evidence, and the intermediate reasoning trajectory, your task is to decide whether the current evidence is sufficient to answer the question. If the evidence is sufficient, output **stop**. Otherwise, generate the next sub-task and formulate targeted evidence-seeking queries.

The available evidence-seeking agents are: **Image Agent**, which extracts information from charts, tables, figures, and images; **Context Agent**, which retrieves information from the textual context; and **Knowledge Agent**, which provides financial concepts, formulas, definitions, and domain knowledge.

You should generate only the next necessary sub-task, avoid redundant queries, and assign each query to exactly one agent. Do not answer the question directly. Stop only when the accumulated evidence and reasoning trajectory are sufficient for the reasoning agent to derive the final answer. The maximum number of planning steps is $N_{\max} = 6$.

Return only a JSON object in the following format:

```
{
  "status": "continue" or "stop",
  "subtask": "<next reasoning objective, or None>",
  "queries": [
    {
      "agent": "Image Agent / Context Agent / Knowledge Agent",
      "query": "<targeted evidence-seeking query>"
    }
  ]
}
```

If **status** is "stop", set **subtask** to None and **queries** to an empty list.

[User Input]:

Context: {context}

Question: {question}

Options: {options}

Images: {images}

Step ID: {step_id}

Accumulated Evidence: {evidence_state}

Intermediate Reasoning Trajectory: {reasoning_trajectory}

Return only the JSON object.

A.2.2 Evidence-Seeking Agent Prompts

Image Agent Prompt

[System Input]:

You are the Image Agent, a step-wise evidence acquisition framework for multimodal financial reasoning. Given a planner query and the visual inputs, your role is to extract only the visual evidence required by the query. The visual inputs may include charts, tables, figures, plots, or screenshots.

You should identify relevant numerical values, labels, axes, units, table entries, trends, comparisons, and relations from the images. Do not infer unsupported information.

If the requested evidence is not visible, state that it is not available.

Return only the following JSON object:

```
{  
  "agent": "Image Agent",  
  "query": "<planner query>",  
  "evidence": "<extracted visual evidence>"  
}
```

[User Input]:

Images: {images}

Planner Query: {query}

Return only the JSON object.

Context Agent Prompt

[System Input]:

You are the Context Agent in SEA, a step-wise evidence acquisition framework for multimodal financial reasoning. Given a planner query and the textual context, your role is to extract only the textual evidence required by the query.

You should identify relevant facts, variables, assumptions, constraints, numerical quantities, temporal conditions, definitions stated in the context, and relations among entities or variables. If the requested evidence is not stated or implied in the context, state that it is not available.

Return only the following JSON object:

```
{  
  "agent": "Context Agent",  
  "query": "<planner query>",  
  "evidence": "<extracted textual evidence>"  
}
```

[User Input]:

Context: {context}

Planner Query: {query}

Return only the JSON object.

Knowledge Agent Prompt**[System Input]:**

You are the Knowledge Agent in SEA, a step-wise evidence acquisition framework for multimodal financial reasoning. Given a planner query and retrieved entries from the external financial knowledge base $\mathcal{D}$, your role is to provide only the financial knowledge required by the query. You should retrieve relevant financial concepts, mathematical formulas, definitions, and domain-specific reasoning principles. If the requested knowledge is not available in the retrieved entries, state that it is not available.

Return only the following JSON object:

```
{
  "agent": "Knowledge Agent",
  "query": "<planner query>",
  "evidence": "<retrieved financial knowledge>"
}
```

[User Input]:

Retrieved Knowledge Entries: {retrieved_knowledge_entries}

Planner Query: {query}

Return only the JSON object.

A.2.3 Reasoning Agent Prompt

Reasoning Agent Prompt

[System Input]:

You are the Reasoning Agent in SEA, a step-wise evidence acquisition framework for multimodal financial reasoning.

At each reasoning step, you are given the current sub-task, the updated evidence state, and the previous reasoning trajectory. Your role is to perform only the current inference step based on the available evidence. You should produce an intermediate reasoning output that advances the reasoning trajectory.

You must follow these rules:

1. Use only the accumulated evidence provided in the evidence state and the previous reasoning trajectory.
2. Ground the current reasoning step explicitly in the retrieved evidence.
3. Do not introduce unsupported assumptions or external knowledge unless it is already included in the evidence state.
4. Do not answer the final question unless the current sub-task explicitly asks for final decision synthesis.
5. If the evidence is insufficient for the current sub-task, report the missing or unclear information as the reasoning output, so that the planner can generate additional evidence-seeking queries in the next step.
6. Keep the reasoning focused on the current sub-task rather than solving the entire problem at once.

Return only a JSON object in the following format:

```
{
  "subtask": "<current sub-task>",
  "evidence_used": [
    "<relevant evidence item 1>",
    "<relevant evidence item 2>"
  ],
  "reasoning_output": "<intermediate result for the current sub-task>",
  "evidence_issue": "<None, or a description of insufficient evidence>"
}
```

If the available evidence is sufficient for the current sub-task, set `evidence_issue` to `None`. If the available evidence is insufficient, explain what evidence is missing in `evidence_issue` and do not infer beyond the evidence.

[User Input]:

Current Sub-task: {subtask}

Updated Evidence State: {evidence_state}

Previous Reasoning Trajectory: {reasoning_trajectory}

Return only the JSON object.

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